



Microsoft Power Automate for End Users

Two-day course book

Collect, report, remind, approve and brief: automate a training coordinator's routine with automated, instant and scheduled cloud flows, variables, parallel branches and Copilot.

Microsoft Power Automate for End Users

Two-day course book · Participant edition, version 2.0

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Online resources: github.com/Asraf-JS/Microsoft-Power-Automate

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Screenshots show the product at the time of writing. Microsoft updates its apps often, so labels and layouts in your tenant may differ slightly.

About the author

Asraf is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

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Before you begin

How to use this book

This book collects the notes and copy-paste values from the two-day Microsoft Power Automate for End Users course in one place, so you can follow along offline and keep it at your desk afterwards.

Each chapter matches one chapter of the course and has two parts:

- **Notes** explain the concepts and walk you through each build step by step, mostly point and click, with screenshots, checkpoints and troubleshooting.
- **Hands-on exercises** give you every flow name and text value you type, ready to copy.

You'll write just a few short expressions on the main path, one each in Chapters 4, 5 and 7, each with a step-by-step explanation. More are waiting in optional *Going further* boxes for when you're ready.

Look out for the coloured boxes. Blue boxes are notes, teal boxes are tips, purple boxes are things to try, and orange boxes are key points and warnings.

Prefer to copy values rather than retype them? Everything in this book is also online at github.com/Asraf-JS/Microsoft-Power-Automate, where each grey box has a copy button. Copying from there is safer than copying from a PDF, which can add line breaks or change quotation marks.

Course scenario

Throughout the course you automate the work of one training coordinator.

You coordinate training for a small organisation. Course materials arrive by email, session details live in an Excel register, requests need approval, and participants need reminders.

Each chapter turns one of those manual tasks into a flow:

Stage	What happens
Plan	Learn how flows work and write a plan before building anything
Prepare	Set up the OneDrive folders, Excel register, test email, Forms and Teams destinations
Collect	Save PDF attachments from training emails to OneDrive automatically
Report	Send the coordinator a one-click summary table from the Excel register
Remind	Send weekly reminders to every participant on a schedule
Approve	Route a training request form through an approval and post the result to Teams
Refine	Filter reminders to upcoming incomplete sessions, count them and summarise the week

Brief	Build a personal morning briefing with variables and parallel branches
Design	Use Copilot to plan a flow, then review and correct its suggestion
Release	Watch the trainer package the finished flow in a solution for another environment

Ground rules

Important: Use only fictional data. Keep flows turned off until your trainer asks you to test them, and only send test emails to addresses your trainer has approved. Testing a flow can create files, send messages, start approvals or change shared data.

Sample files

Four practice files come with the course. All of them are fictional.

File	Used in	What it is
TrainingRegister.xlsx	Chapters 2 to 7	A ready-made register with the tblTraining table and six records. Use it only if your trainer says so
CourseOutline.pdf	Chapters 2 and 3	The attachment the Chapter 3 flow should save
TrainerPhoto.jpg	Chapters 2 and 3	The attachment the Chapter 3 flow should skip
SessionNotes.docx	Chapter 3	Used in the independent practice

Download them from the course repository: github.com/Asraf-JS/Microsoft-Power-Automate.

Your Power Automate training journey

One coordinator's routine, start to finish: turn each repeated manual task into a cloud flow.

FOUNDATIONS

01

Understand Power Automate

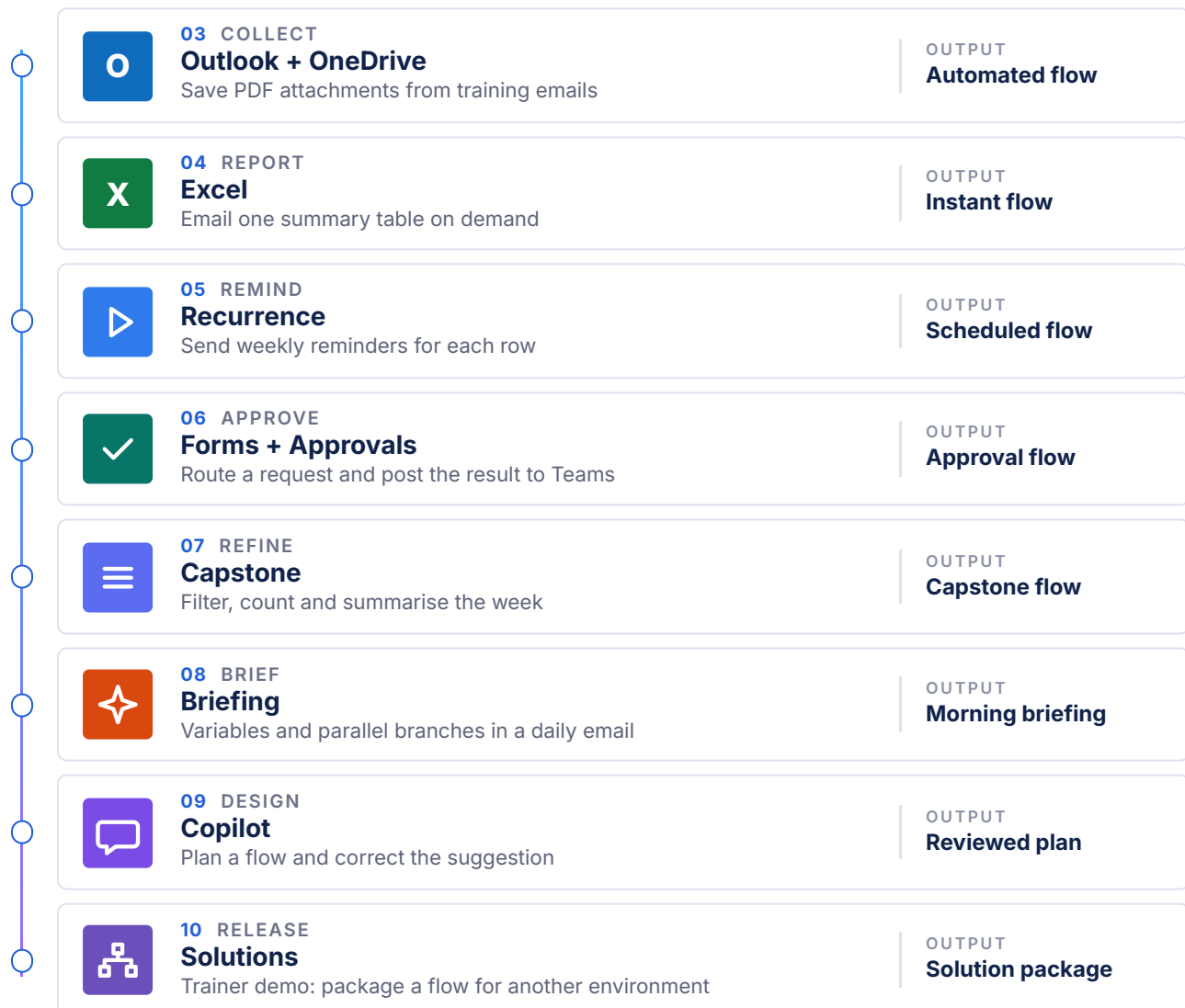
Triggers, actions, flow types and planning

02

Prepare your workspace

OneDrive folders, the Excel register and test data

THE PROJECT, APP BY APP



Collect → Report → Remind → Approve → Refine → Brief → Design → Release

01

CHAPTER 1

Understand Power Automate

Before you build anything, learn the vocabulary of a flow, the three ways a cloud flow can start, and where the main pages live in Power Automate.

Copy-paste values: the planning template for this chapter is on the copy-paste page.

Estimated time: 45 minutes

Your result: A simple automation plan and familiarity with the Power Automate Home and Create pages.

What You Will Learn

- Identify a repetitive business task that could benefit from automation
- Explain the trigger, actions, data, condition and result of a flow
- Choose between an automated, instant and scheduled cloud flow
- Open Power Automate and find Home, Create and My flows
- Write a short plan before building an automation

You need a Windows computer with Microsoft Edge, an internet connection and your training work or school account. You don't need the Excel workbook, a form or any test files yet. You prepare those in Chapter 2.

Your Business Scenario

You coordinate training for a small organisation. Course materials arrive by email, session details are stored in Excel, and participants need reminders. You keep downloading attachments, copying information into messages and checking whether requests were approved.

Each of those tasks has a recognisable starting point and repeatable steps. Over the two days you use Power Automate to do some of those steps for you.

1.1 What Is Automation?

Automation means setting up a repeatable process so software performs the steps when a defined event happens, or when you ask it to run.

Power Automate calls an automation a **flow**. This course builds **cloud flows**, which work with online services such as Outlook and OneDrive for Business.

A flow applies your rule every time, including a wrong one. If you pick the wrong folder or recipient, the flow repeats that mistake. So every flow needs a clear rule and a test.

Tip: Start with a small task whose result you can check easily. A good first flow saves one test attachment to the correct folder.

1.2 Recognise the Parts of a Flow

Take this requirement: *When an email with the subject marker [PA TRAINING] arrives, examine each attachment. If the attachment is a PDF, save it to the training attachments folder.*

Part	Plain-language meaning	Example
Trigger	What starts the flow	A matching email arrives
Action	A step the flow carries out	Create a file in OneDrive for Business
Data	Information a step receives or produces	The attachment's name and contents
Condition	A question that selects a path	Is this attachment a PDF?
Loop	Steps repeated for each item	Check each attachment in the email
Result	The business outcome you verify	Matching files appear in the intended folder

The basic pattern is: something starts the flow, the flow follows its steps, you check the result. A flow only needs a condition or a loop when the process calls for one.

Try it: Imagine a training email with two attachments, `CourseOutline.pdf` and `TrainerPhoto.jpg`. Walk the rule through each one. The PDF is saved, the image is skipped, and the folder should end up with `CourseOutline.pdf` only. If an email has no PDFs at all, the flow still runs successfully and creates nothing.

1.3 Choose How the Flow Starts

Cloud-flow type	When it starts	Training example
Automated	When a specified event occurs	Save attachments when a training email arrives
Instant	When you deliberately start it	Create an Excel summary when you're ready to share it
Scheduled	At a time and frequency you configure	Send training reminders every Monday morning

Pick the type by asking *what should make this process begin?* Sending an email doesn't tell you the type. All three can send email. Only their starting points differ.

Quick check. Pick a type for each request, then check the answers.

1. Whenever a new training request is submitted, ask someone to approve it.
2. At 9:00 AM every Friday, prepare a weekly reminder.
3. Let me generate a summary whenever I need one.

Answers: 1 is Automated, 2 is Scheduled, 3 is Instant.

1.4 Connectors and Connections

A **connector** gives Power Automate a set of triggers and actions for one service. Office 365 Outlook provides email operations. OneDrive for Business provides file operations.

A **connection** links those operations to an account. Whenever a later lesson asks you to sign in to a connector, check that you're using the training account, and that it has permission for the mailbox, folder or resource involved.

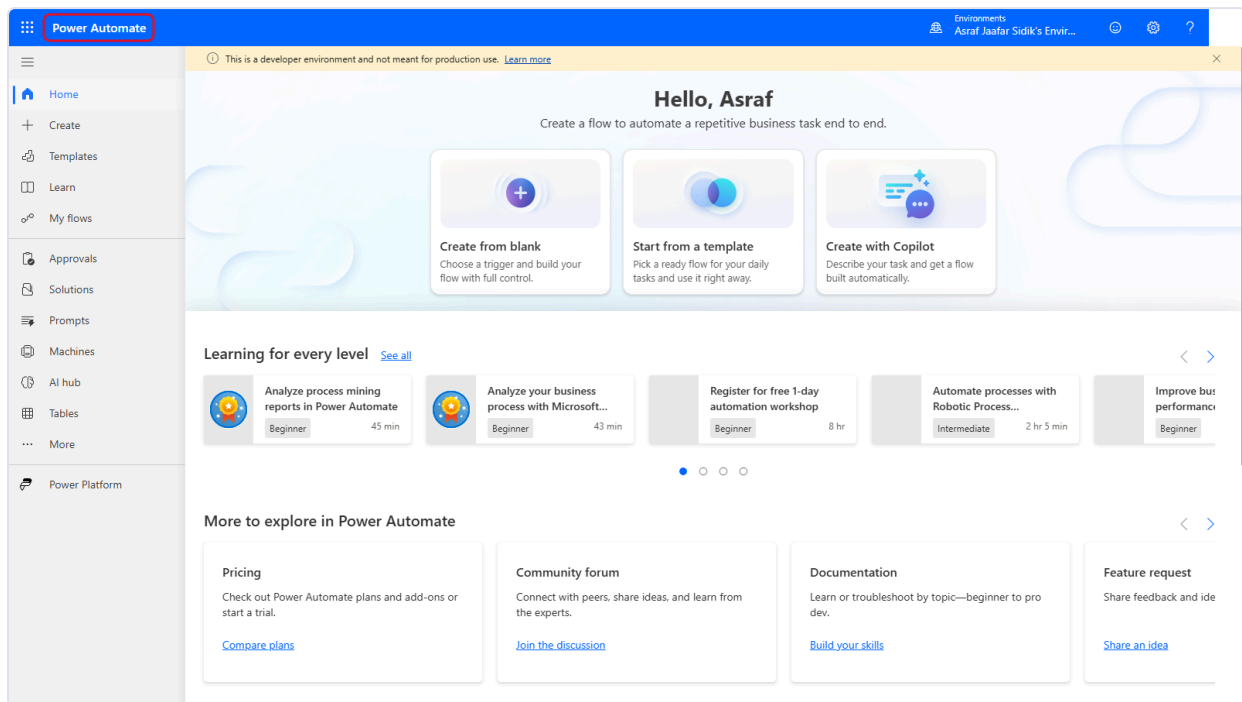
The core exercises use standard connectors only. Power Automate Free can create and run cloud flows with standard connectors but can't share flows, so you build and own your own flows. See [Power Automate license types](#).

Tip: Office 365 Outlook and Outlook.com are different connectors. Go by the connector name in the lesson, not the icon.

1.5 Explore Power Automate

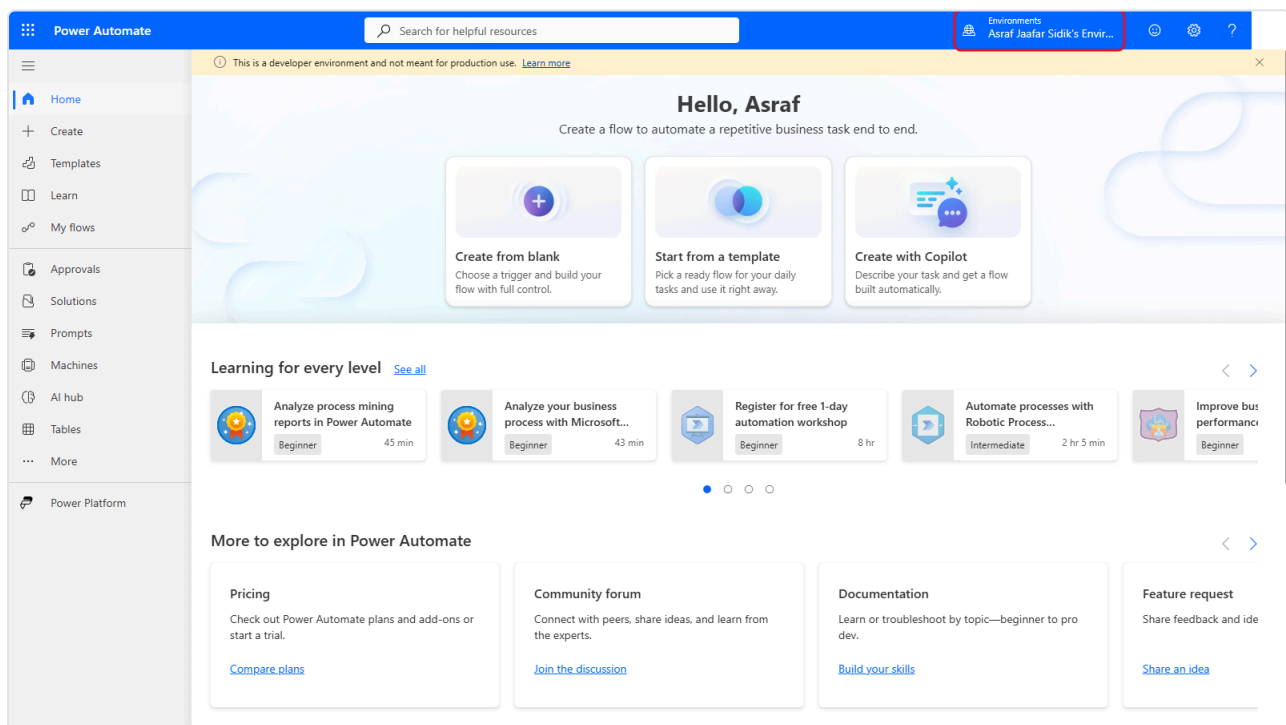
Open the training workspace

1. In Edge, go to make.powerautomate.com and sign in with your training account.
2. Check that the page says **Power Automate** at the upper left.



The Power Automate home page.

- Look at the top right of the blue bar. The small word **Environments** sits above your environment's name. An environment is a workspace that holds Power Platform resources, including flows.
- Compare it with the environment your trainer gave you. If they differ, ask before you build anything.

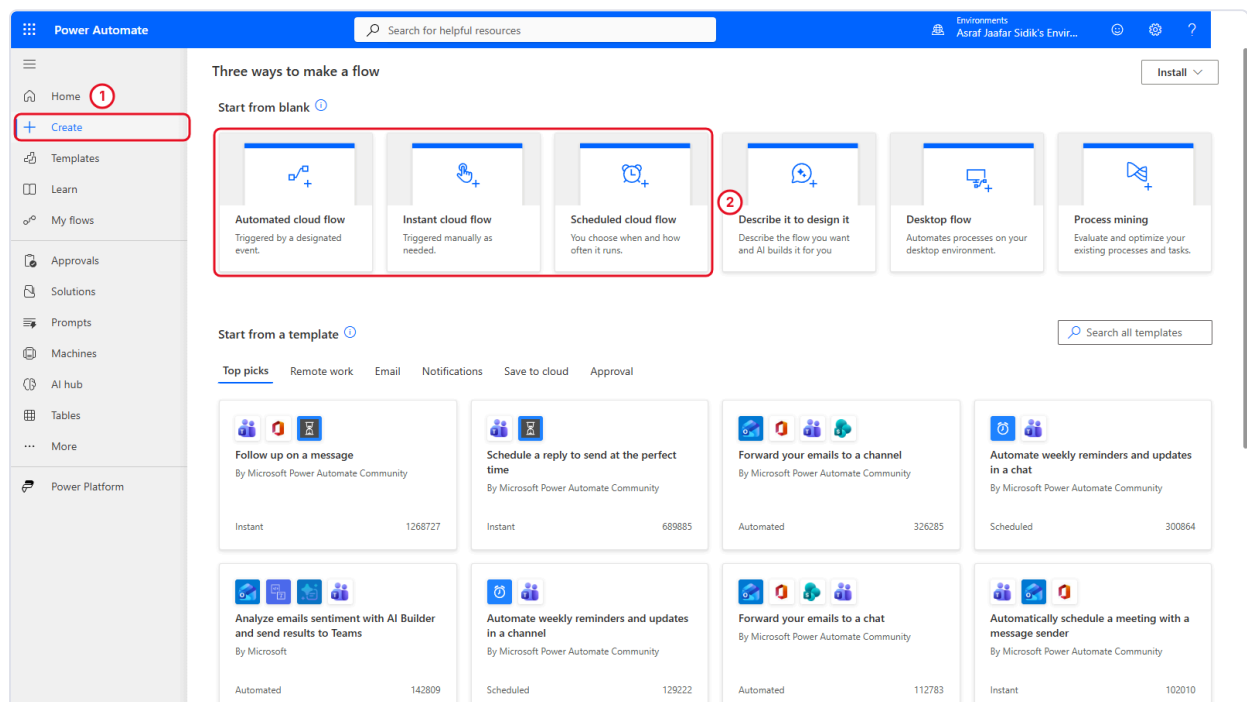


The environment selector shows which workspace you're in. Yours will have a different name.

Find the three cloud-flow choices

1. If you see only icons on the left, select the three-line menu icon at the top left (above **Home**) to show the labels. Then select **Create** in the left navigation.
2. Under **Start from blank** you see six tiles. Three of them create cloud flows from scratch, and those are what this course uses:

Tile	Description on the tile	Used in this course
Automated cloud flow	Triggered by a designated event	Yes
Instant cloud flow	Triggered manually as needed	Yes
Scheduled cloud flow	You choose when and how often it runs	Yes
Describe it to design it	Describe the flow you want and AI builds it for you	Chapter 9
Desktop flow	Automates processes on your desktop environment	No
Process mining	Evaluate and optimize your existing processes and tasks	No



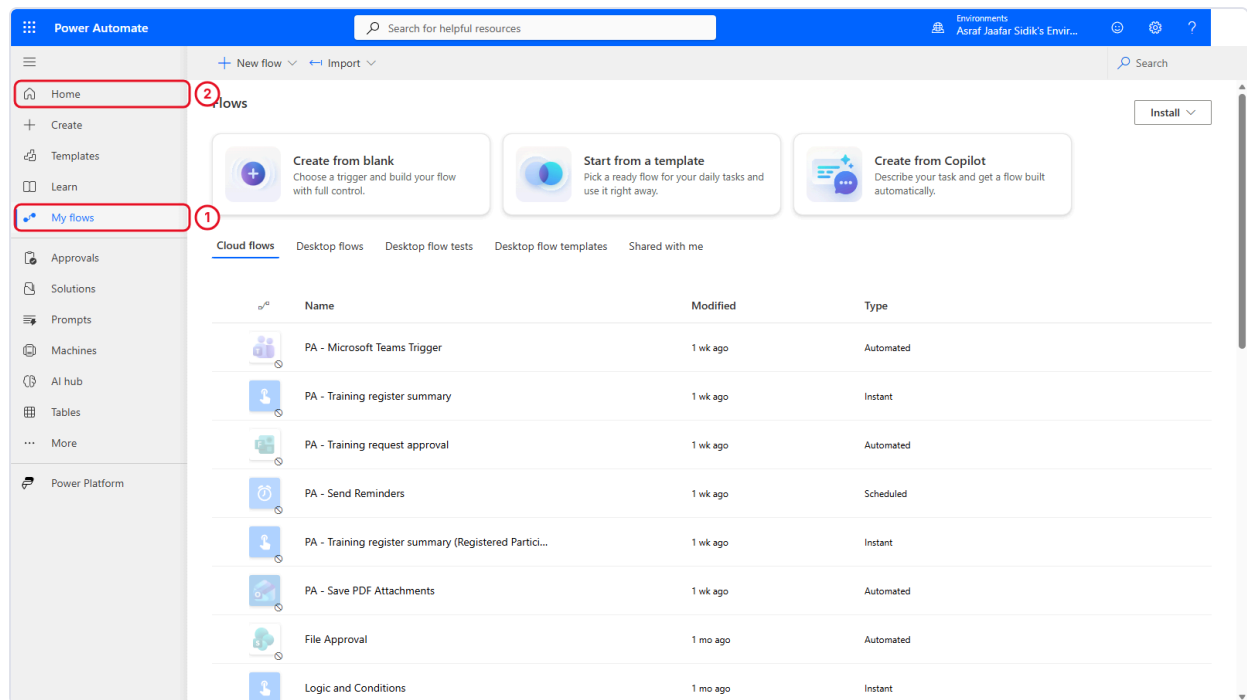
The three cloud-flow tiles come first under Start from blank. Describe it to design it is the Copilot option you'll try in Chapter 9.

Note: The heading **Three ways to make a flow** refers to the three sections of the page: Start from blank, Start from a template and Start from a connector. Those aren't the three cloud-flow types. The types are three of the six tiles inside the first section.

3. Decide which tile you'd use for a flow that starts when someone submits a form. The answer is **Automated cloud flow**: a form submission is an event that happens without you starting it. Instant means *you* start the flow.

Locate your flows

1. Find **My flows** in the left navigation. You'll come back here to find every flow you build.



My flows lists every flow you've built. Your list is probably empty for now. The flows you build in Chapters 3 to 8 appear here, like the ones in this screenshot.

2. Select **Home** at the top of the left navigation to return to the start page.

1.6 Plan Before You Build

A short plan makes the designer much easier to use. Here's the plan for the attachment scenario:

Planning question	Example answer
What task repeats?	Saving PDF training attachments from email
What starts it?	An email with the training subject marker arrives
What information is needed?	Attachment name and contents
What rule applies?	Save PDF attachments; skip other file types
Where should the output go?	The training attachments folder in OneDrive for Business
How will I test it?	Use an email with a PDF and an image, then check the folder

Chapter 2 prepares the resources this plan refers to. Chapter 3 turns it into a working flow.

Independent Practice: Plan a Weekly Reminder

Allow five minutes. Every Monday morning, a training coordinator wants to remind people about upcoming sessions. Session details are in Excel. People whose training is marked Completed should not get a reminder.

1. Choose a cloud-flow type.
2. State the start time and time zone.
3. List three pieces of information the flow needs from Excel.
4. Write one rule that decides whether a reminder is sent.
5. Describe a test with one person who should get a reminder and one who shouldn't.

A good plan uses a scheduled flow with a local time zone, needs values like email address, session date, course title and status, sends only for upcoming incomplete training, and tests both an included and an excluded record.

Tips and Troubleshooting

What you notice	What to do
A different Microsoft account opens automatically	Check the account shown on the page. Use the Edge profile meant for training
You can't open Power Automate	Check the address and connection, then ask the trainer to verify access. Note the error message
The environment name is unfamiliar	Compare it with the class instructions before creating anything
Create isn't visible	Select the three-line menu icon at the top left to show the navigation labels. If it's still missing, ask the trainer to check access
The page looks different from the screenshot	Look for the named command. Layouts vary
A trial or upgrade prompt appears	Check what you selected. The core lessons use the Free baseline
You can't decide which flow type to use	Finish the sentence "This process should begin when..." and match it to a type

Lesson Summary

A trigger starts a flow, actions perform the steps, and data moves between them. Conditions choose a path and loops repeat work for each item. You can now tell automated, instant and scheduled flows apart, find the main Power Automate pages, and describe an automation before you build it.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner.

Automation planning template (section 1.6)

Paste this into your notes and fill in each line before you build a flow. You use the same questions for the independent practice at the end of the chapter.

```
What task repeats?  
What starts it?  
What information is needed?  
What rule applies?  
Where should the output go?  
How will I test it?
```

Power Automate address (section 1.5)

```
https://make.powerautomate.com
```

02

CHAPTER 2

Prepare Your Training Workspace

Every flow in this course reads from or writes to something: a folder, a table, a mailbox, a form, a Teams chat. Set those up first and the later build steps become predictable.

Copy-paste values: the Excel headers, table name and email subject are on the copy-paste page.

Estimated time: 30 minutes

Your result: A training folder, an Excel workbook with a table, and safe test data for the first two build exercises.

What You Will Learn

- Create a dedicated training folder structure in OneDrive for Business
- Build an Excel workbook and format its data as a named table
- Prepare a safe test email with sample attachments
- Confirm that Forms, Teams and your Power Automate environment are ready

Note: Use your own training account and the environment your trainer names. Never put real participant information in the practice resources. Use fictional names and test addresses only.

Forms and Teams preparation in this chapter are trainer-led readiness checks, because those destinations depend on your organisation's policy.

Apps You'll Use

Everything in this course runs in the browser. Start from the Microsoft 365 home page and you can reach every app from it.

App	Web address	What you use it for
Microsoft 365 home	m365.cloud.microsoft	Your starting point. The app launcher (nine dots, top left) opens every app below
OneDrive	Open from the app launcher	The training folder and the Excel register

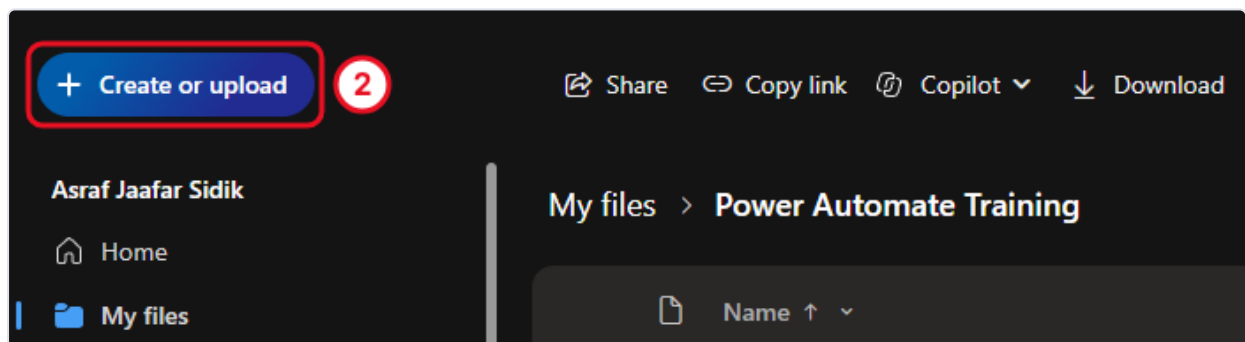
Outlook	outlook.office.com	Test emails and the emails your flows send
Microsoft Forms	forms.office.com	The training request form (Chapter 6)
Teams	teams.microsoft.com	Flow bot messages (Chapters 6 and 11)
Power Automate	make.powerautomate.com	Building your flows

Sign in to each one with your **work or school account**, the same one every time.

Note: Used to OneDrive as a folder in File Explorer? That folder and OneDrive on the web are the same storage, so files appear in both. During class, use the web version so your screen matches the screenshots. Power Automate can only see files inside OneDrive: a workbook saved to your Desktop or Documents outside the OneDrive folder is invisible to it.

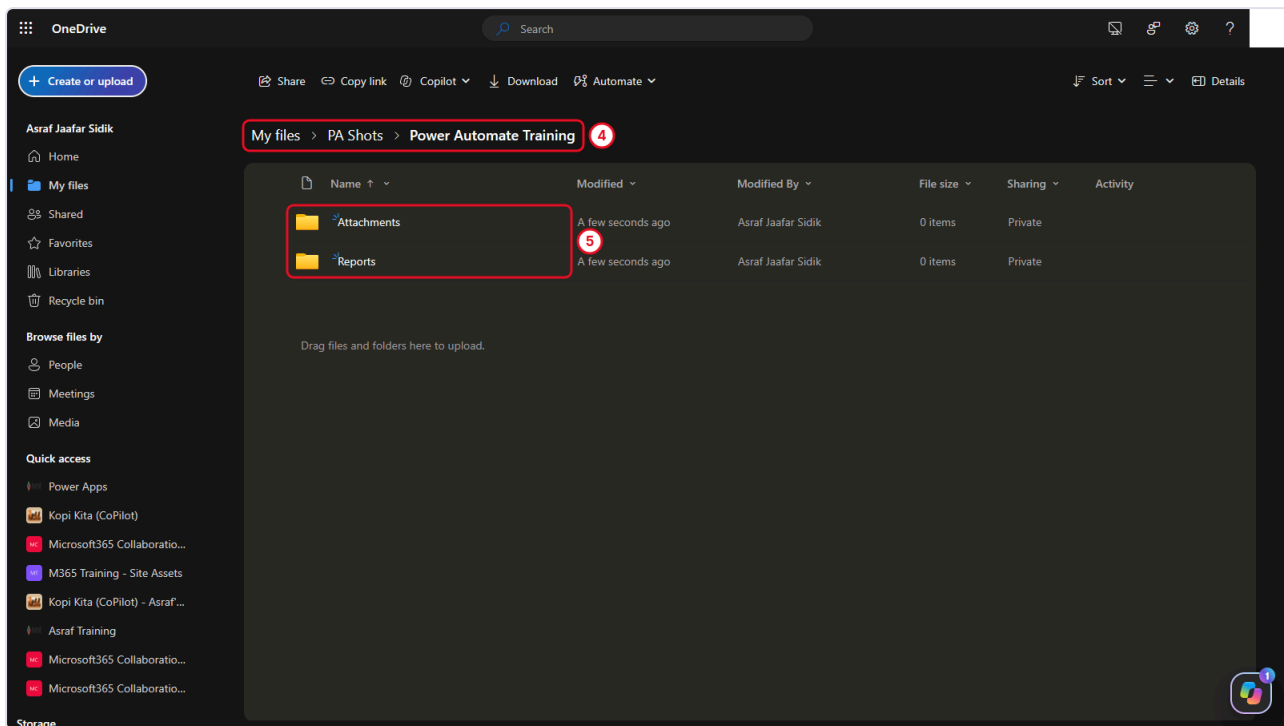
2.1 Create the Training Folders in OneDrive

1. Open m365.cloud.microsoft, select the app launcher (nine dots, top left), then **OneDrive**. Select **My files** on the left.
2. Select the blue **Create or upload** button at the top left, then **Folder**.



Create or upload makes folders, uploads files and creates new Office documents.

3. Name the folder **Power Automate Training** and select **Create**.
4. Select the new folder's name to open it. The path at the top now reads **My files > Power Automate Training**.
5. Inside it, create two subfolders the same way (**Create or upload > Folder**): **Attachments** and **Reports**.



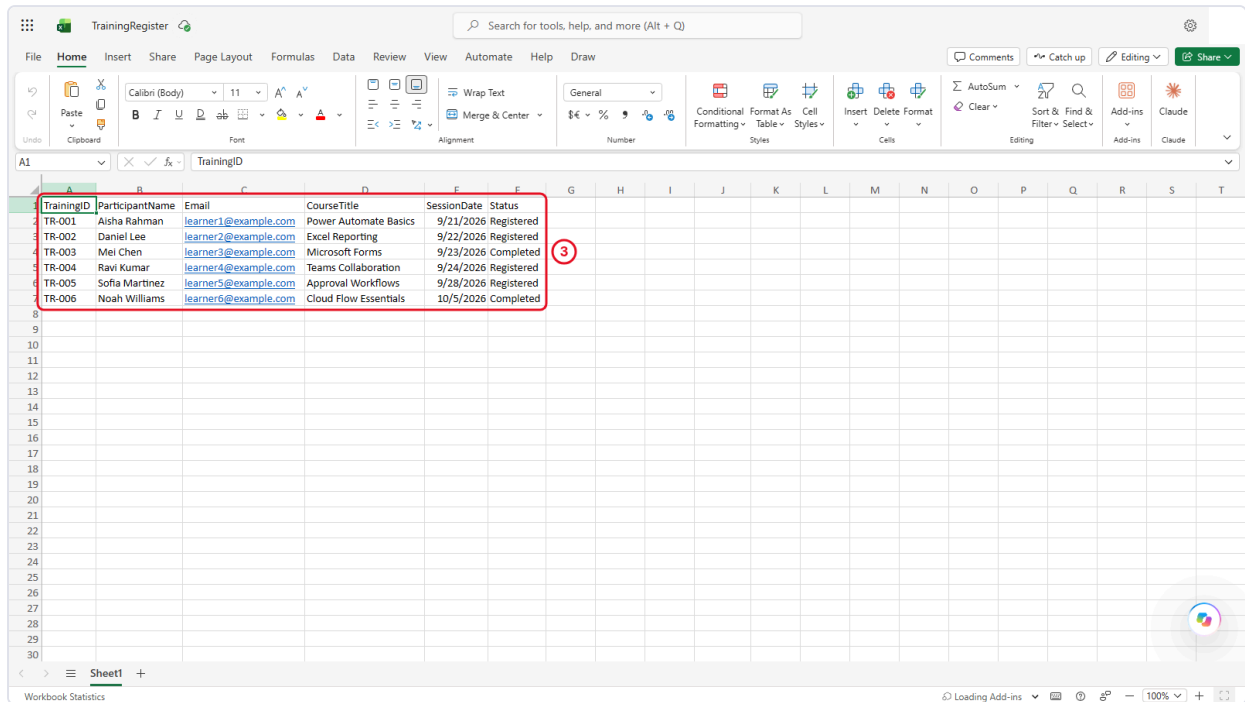
Both subfolders inside Power Automate Training.

Checkpoint: You can see `Attachments` and `Reports` inside `Power Automate Training`. The Chapter 3 flow writes to `Attachments`. Summary outputs can go to `Reports`.

2.2 Create the Excel Workbook

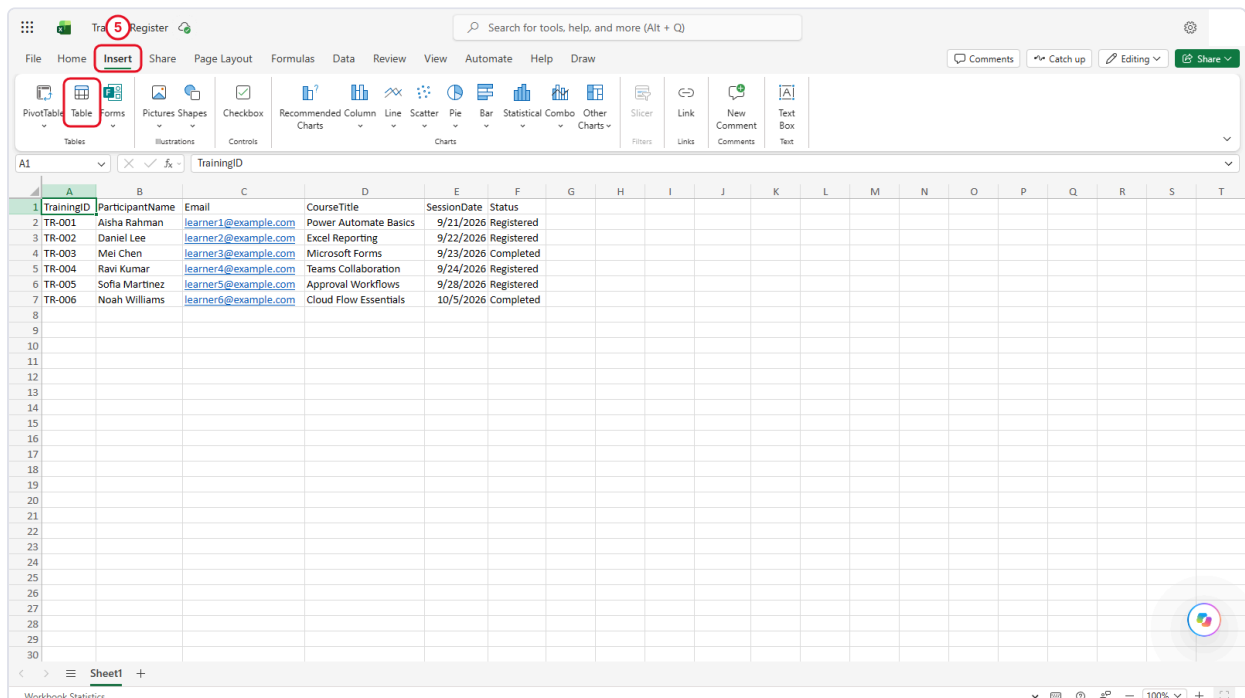
1. Inside `Power Automate Training`, select **Create or upload**, then **Excel workbook**. Check the breadcrumb first: the workbook is created in whichever folder you're viewing.
2. When Excel for the web opens, select the file name in the title bar and rename it `TrainingRegister`. Excel adds `.xlsx` for you.
3. Select cell **A1**, then paste the headers and sample rows from the copy-paste page (section 2.2). Each header lands in its own column (`TrainingID`, `ParticipantName`, `Email`, `CourseTitle`, `SessionDate`, `Status`), and the six fictional rows fill rows 2 to 7.
4. If your trainer gave you approved test addresses, type them over the addresses in the **Email** column. If your trainer asks you to change the session dates, keep at least two upcoming sessions, one completed session and one session outside the training week.

Tip: Short on time? Your trainer may tell you to upload the ready-made `TrainingRegister.xlsx` to `Power Automate Training` instead (**Create or upload > Files upload**). It already has the `tblTraining` table and six fictional records, so you can skip to the checkpoint.



Fictional rows. The example.com addresses can't receive mail. The cells look plain until you format them as a table in the next step.

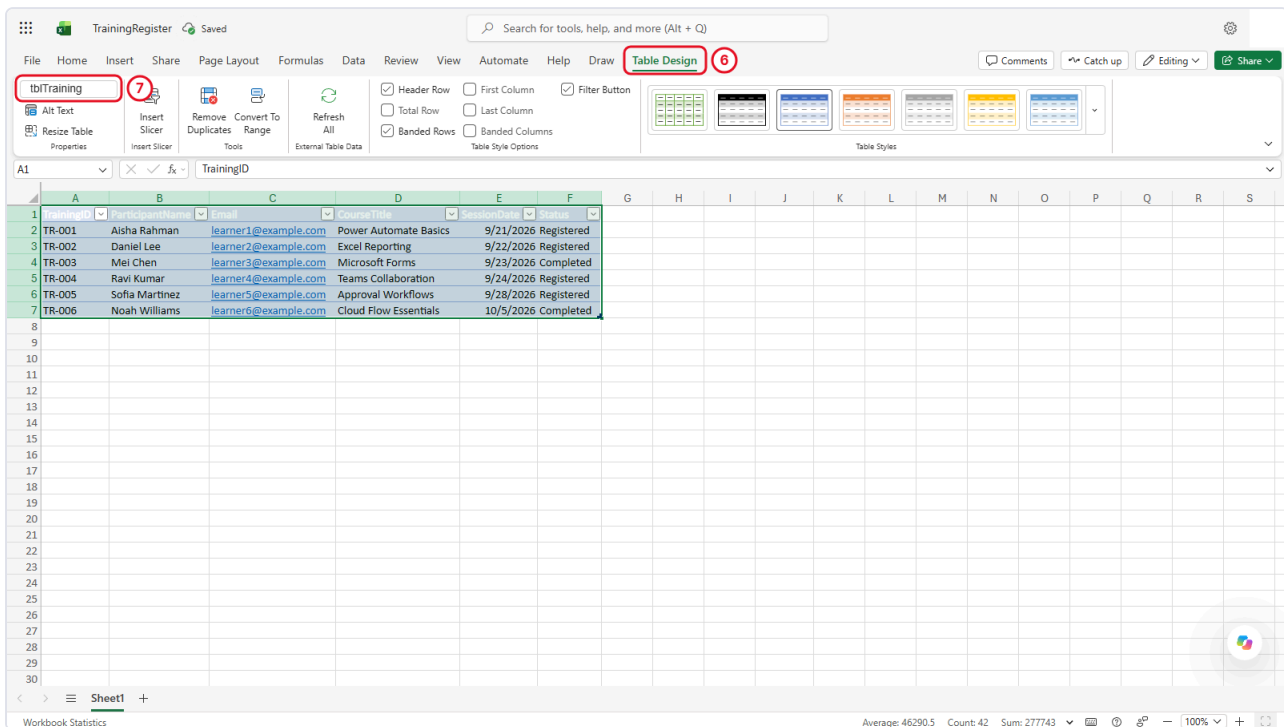
5. Select any cell with data in it, then select the **Insert** tab and **Table**. In the box that opens, make sure **My table has headers** is ticked, and select **OK**. The rows turn into a striped table.



Insert is the third tab, after File and Home. Table is near the left end of the Insert tab.

6. With a cell in the table still selected, select the **Table Design** tab that has appeared at the end of the ribbon.

7. At the far left of that tab, select the name box (it reads something like **Table1**), type `tblTraining` and press Enter.



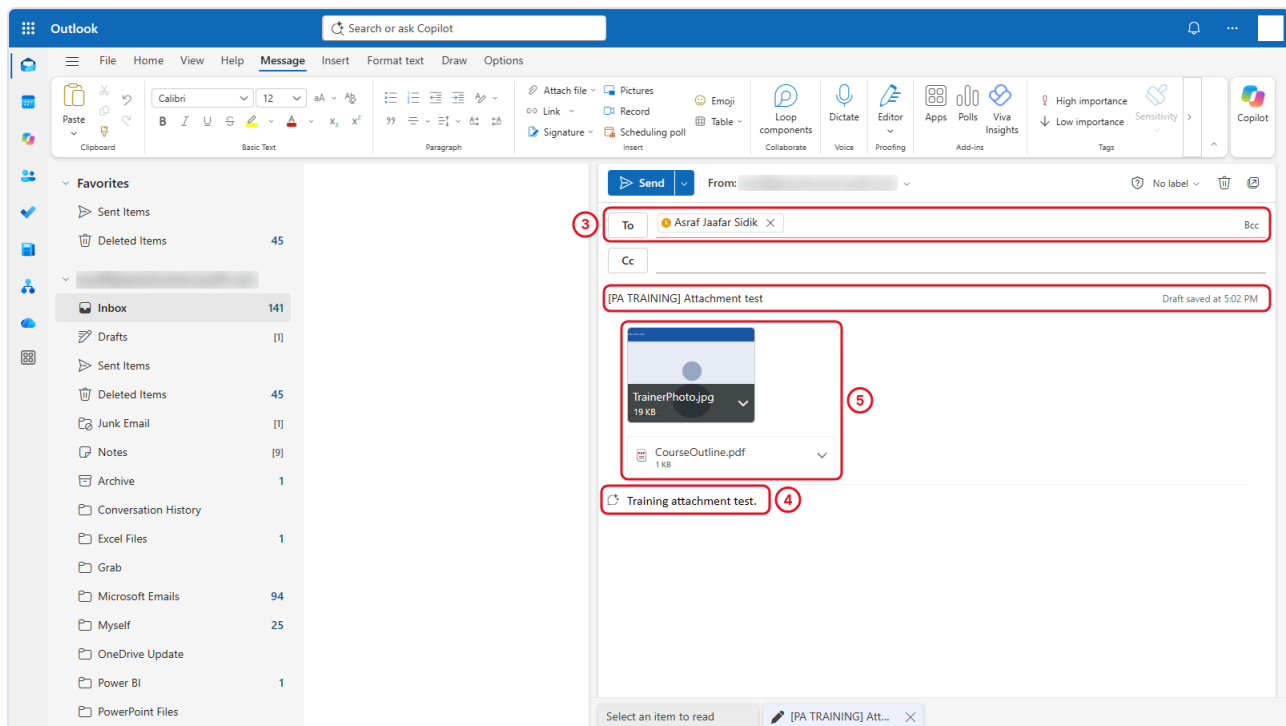
Set the table name on the Table Design tab.

Checkpoint: The workbook is named `TrainingRegister.xlsx`, the data is a formatted table named `tblTraining`, and the table has six fictional records.

If Power Automate can't find the table later, check three things: the workbook is in OneDrive for Business, the range is formatted as a table, and the table name is exactly `tblTraining`.

2.3 Prepare a Test Email

1. Download the test files: [test-files.zip](#) (one click, all three files). Open your Downloads folder, right-click the ZIP and choose **Extract All**.
2. Open Outlook on the web (outlook.office.com) with the training account and select **New mail** (it may just say **New**) at the top left.
3. In **To**, type your own email address, the mailbox you connect to Power Automate in Chapter 3. In the subject line, enter `[PA TRAINING] Attachment test`.
4. Select the large empty area under the subject line (the message body) and type `Training attachment test`.
5. On the **Message** tab of the ribbon, select **Attach file** > **Browse this computer**, go to the extracted test files and pick `CourseOutline.pdf`. Repeat for `TrainerPhoto.jpg`. If Outlook asks whether to attach the picture or insert it inline, choose to attach it.
6. **Don't send it.** Outlook saves the message automatically (you see **Draft saved** at the right of the subject line). Select **Inbox** in the folder list to leave it. The message waits in **Drafts**.



The picture shows as a thumbnail and the PDF as a smaller chip. Both are attachments. The test email waits in Drafts until Chapter 3.

Important: Don't attach real course materials or any file whose name contains a person's name. The filename shows up in run history and in your OneDrive folder.

The Chapter 3 flow only reacts to new messages, so you send this draft during the test in section 3.6.

2.4 Confirm Forms Readiness

Open Microsoft Forms and confirm the training account can create a form. You build the **Training Request** form in Chapter 6, right before the flow that uses it. It will ask for Requester name, Course, Preferred date and Business reason, with fictional answers only.

2.5 Confirm the Teams Training Destination

Ask the trainer for the team and channel, or the training chat, and write its name down. The trainer confirms posting access before the Teams exercises: the optional extension in Chapter 4 and the approval messages in Chapter 6.

The Teams connector is standard, but your organisation may restrict posting or require the Workflows app. If you can't post a test message, tell the trainer before building the summary flow.

2.6 Confirm the Power Automate Environment

1. Return to Power Automate.

2. Look at the top right of the blue bar. The small word **Environments** sits above your environment's name.
3. Confirm it matches the environment your trainer gave you.

Key point: OneDrive files, Excel workbooks, Forms and Teams destinations aren't packaged automatically when a solution is moved. Keep a note of where each resource lives. Your trainer demonstrates moving flows in Chapter 10.

Independent Practice

Find the row in `tblTraining` whose Status is Completed. Which later reminder test should exclude it, and why? Then write a second test email without the `[PA TRAINING]` marker, save it as a draft, and predict whether the attachment flow should process it once it's sent.

Lesson Summary

You created a dedicated OneDrive training area with its folders and Excel table, prepared safe inputs for the attachment and summary exercises, confirmed the environment, and noted where the Forms and Teams work will happen. Next you build the flow that saves qualifying email attachments.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Anything marked **text** is typed or pasted straight into a field.

Folder names (text, section 2.1)

Main folder:

Power Automate Training

Subfolders:

Attachments

Reports

Workbook name (text, section 2.2)

TrainingRegister

Excel headers and sample rows (section 2.2)

Click cell A1 in Sheet1, then paste. Each value lands in its own column: the headers fill row 1 and the six fictional rows fill rows 2 to 7.

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	9/21/2026	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	9/22/2026	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	9/23/2026	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	9/24/2026	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	9/28/2026	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	10/5/2026	Completed

Table name (text, section 2.2)

tblTraining

Test email subject (text, section 2.3)

[PA TRAINING] Attachment test

Test email body (text, section 2.3)

Training attachment test.

03

CHAPTER 3

Save Email Attachments Automatically

Your first build: an automated flow that watches for training emails and saves only the PDF attachments to OneDrive. Everything in this chapter is point and click.

Copy-paste values: the flow name and subject filter are on the copy-paste page.

Estimated time: 50 minutes

Your result: An automated cloud flow that examines new training emails and saves only PDF attachments to OneDrive for Business.

Connectors: Office 365 Outlook and OneDrive for Business, both standard.

What You Will Learn

- Create an automated cloud flow from a business requirement
- Configure an Outlook trigger with a subject filter
- Process several attachments with **Apply to each**
- Use a condition to select PDF files
- Insert values with the **dynamic content** picker
- Test a matching and a non-matching case, then inspect run history

Before You Begin

Complete Chapter 2. Check that the `Attachments` folder exists and the test email from section 2.3 is saved in Drafts. That draft uses the subject `[PA TRAINING] Attachment test` and has two fictional attachments from the Chapter 2 `test-files` folder:

File	Purpose
CourseOutline.pdf	The attachment that should be saved
TrainerPhoto.jpg	The attachment that should be skipped
SessionNotes.docx	Used only in the independent practice

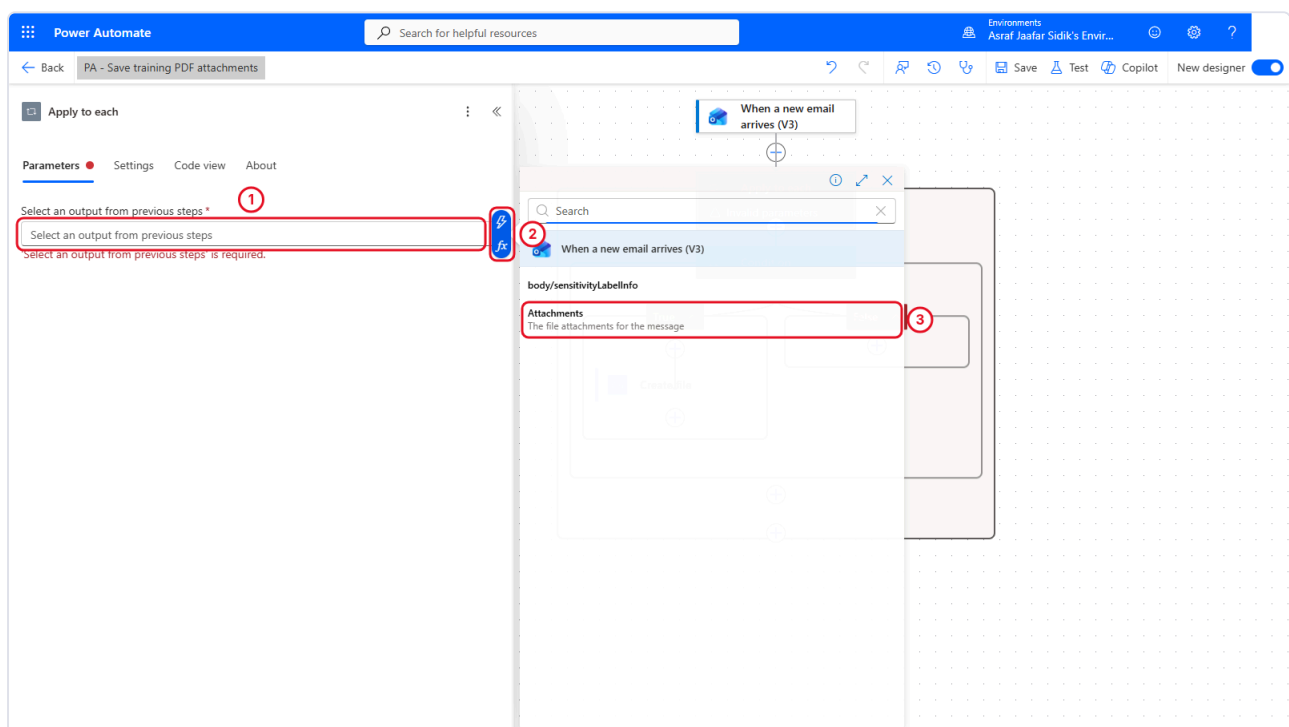
Note: OneDrive never silently overwrites a file. If you run the test twice with the same attachment, Create file either fails with a duplicate-name error or OneDrive saves CourseOutline (1).pdf. Both are correct behaviour. Delete the file from Attachments before a repeat test.

The rule: when a new email arrives with [PA TRAINING] in the subject, examine each attachment. If its name ends in .pdf, create a file in the Attachments folder.

How to Insert Dynamic Content

You'll do this in every chapter, so learn it once. **Dynamic content** is a value produced by an earlier step, such as an email's subject or an attachment's name.

1. Click into the field you want to fill.
2. Select the **lightning bolt** icon that appears beside the field (or type `/` and choose **Insert dynamic content**).
3. Pick the value from the list. Values are grouped under the step that produces them. Use the search box if the list is long.



The lightning bolt opens the dynamic content picker. Values are grouped by the step that produced them.

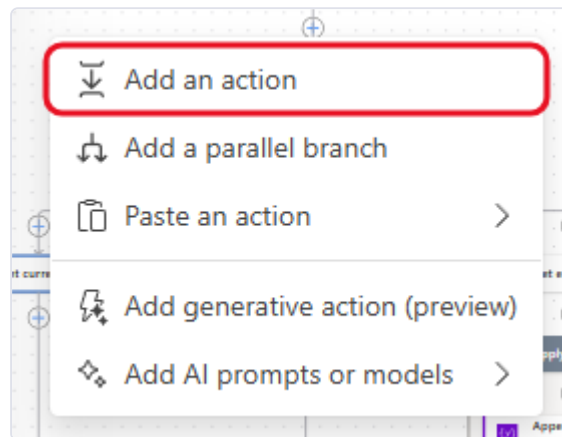
The value appears in the field as a coloured token. To remove it, select the **x** on the token.

Note: After you save and come back, some tokens show their technical name instead of the friendly one: **Attachments Name** shows as `name`, **Attachments Content** as `contentBytes`, **value** as `body/value`, and Forms answers as codes like `body/r96281c9...`. That's normal, and the screenshots in this course show them that way too. They still point at the right value.

How to Add an Action and Check Your Flow

You'll also do these in every chapter.

- **Add an action:** select the **+** on the line where the new step should go (below the last step, or between two steps), then **Add an action**. A panel opens on the left with a search box. Type the action's name; results are grouped by connector, such as **Office 365 Outlook** or **Control**. Select the action you want.
- **Save:** the **Save** button is at the top right of the designer.
- **Flow checker:** the small stethoscope icon just left of **Save**. It opens a panel listing errors and warnings. Close it with the **X**.
- **Test:** the **Test** button next to **Save**. A panel opens on the right; choose **Manually**, then select **Test**.



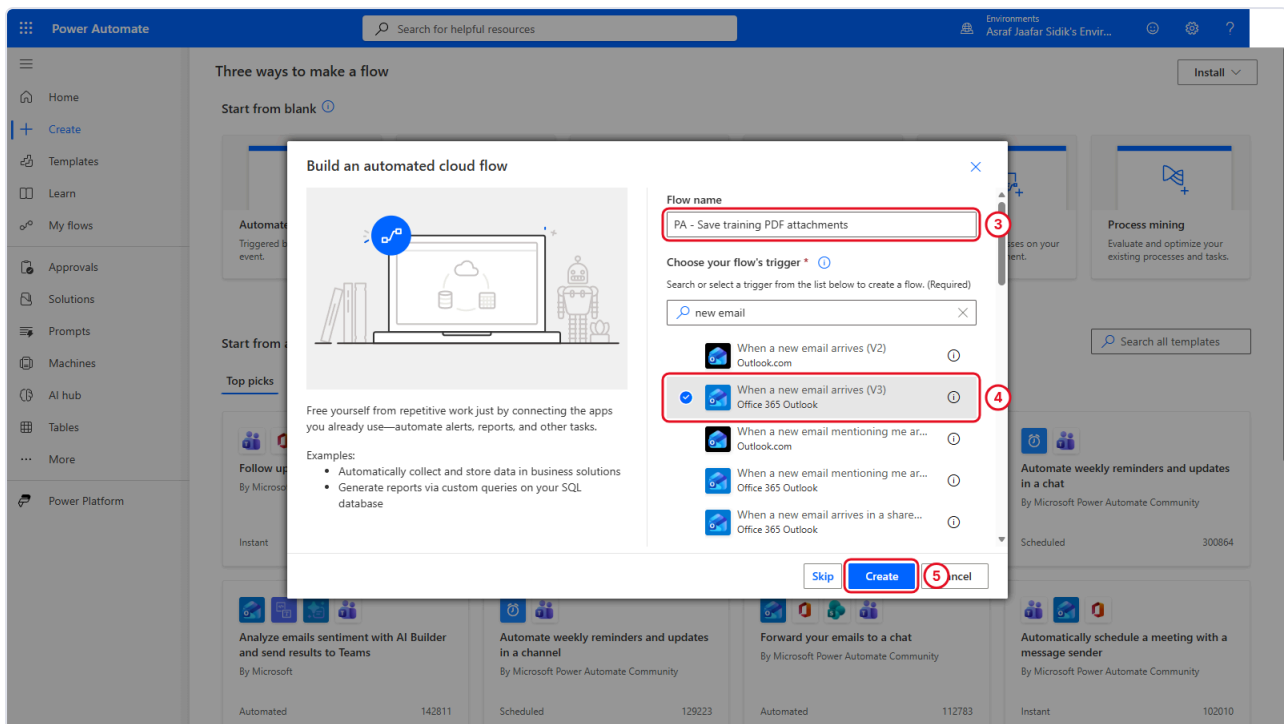
Select a **+** on the canvas and this menu opens. **Add an action** is at the top.



Top right of the designer: Flow checker (the stethoscope), Save and Test.

3.1 Create the Automated Cloud Flow

1. Open Power Automate and confirm the correct environment.
2. Select **Create** on the left, then the **Automated cloud flow** tile (shown in Chapter 1, section 1.5).
3. In **Flow name**, enter **PA - Save training PDF attachments**.
4. In the trigger search box, type **new email**, then select **When a new email arrives (V3)** under **Office 365 Outlook**.
5. Select **Create**. The designer opens with only the trigger on the canvas.



Name the flow and pick the Office 365 Outlook trigger before selecting Create.

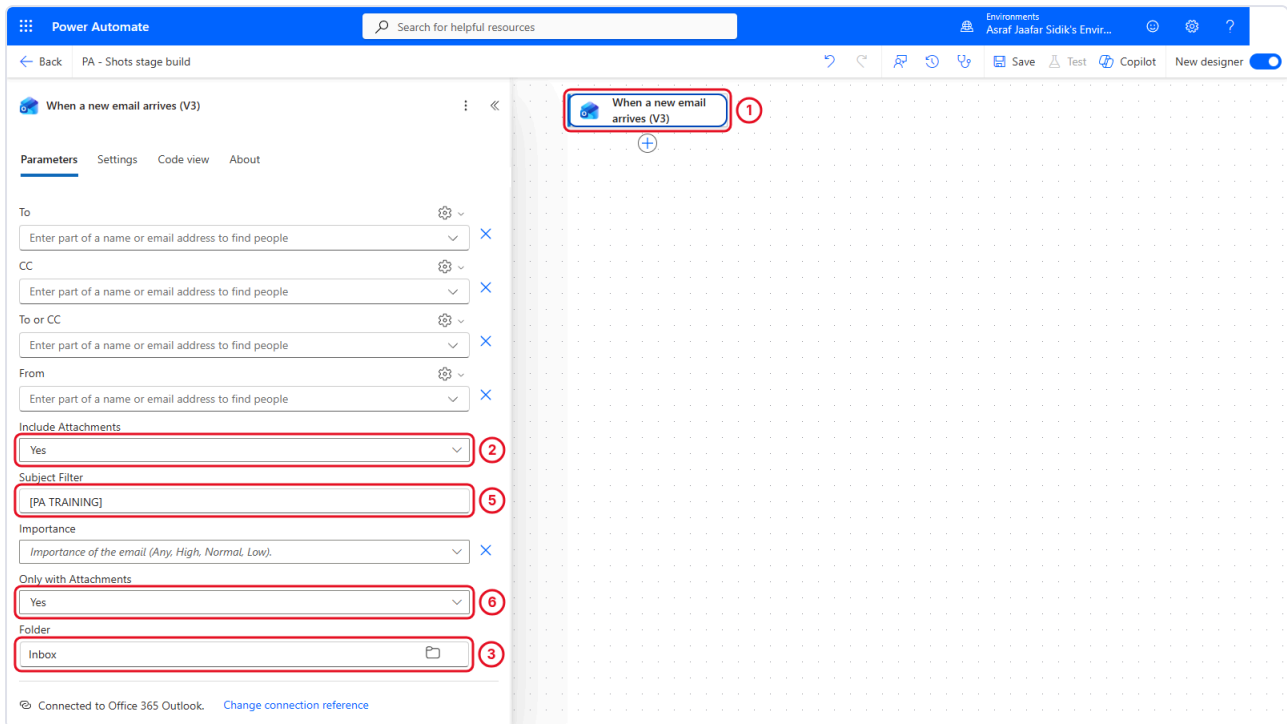
3.2 Configure the Trigger

1. Select the trigger (**When a new email arrives (V3)**) on the canvas. Its settings open in a panel on the left.

The fields below may sit in a different order on your screen. Find each one by its name.

2. Set **Include Attachments** to **Yes**.
3. Set **Folder** to **Inbox**.
4. Select **Show all** under **Advanced parameters**.
5. In **Subject Filter**, enter `[PA TRAINING]`.
6. Set **Only with Attachments** to **Yes**.

Don't save yet. Power Automate won't save a flow that has only a trigger, so you'll save once the first action is in place.

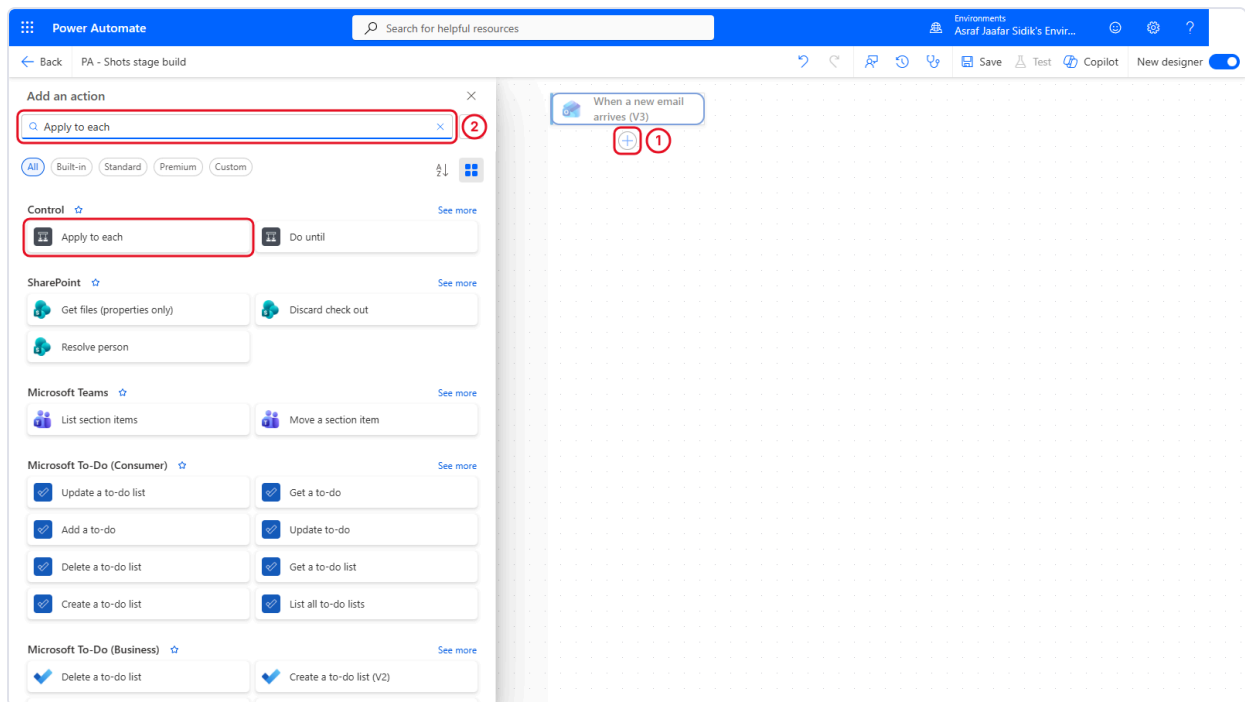


All four trigger fields set. Use Show all if a field is hidden.

Checkpoint: No field shows a red validation message. The flow only starts for *new* matching messages. It doesn't go back and process mail already in the inbox.

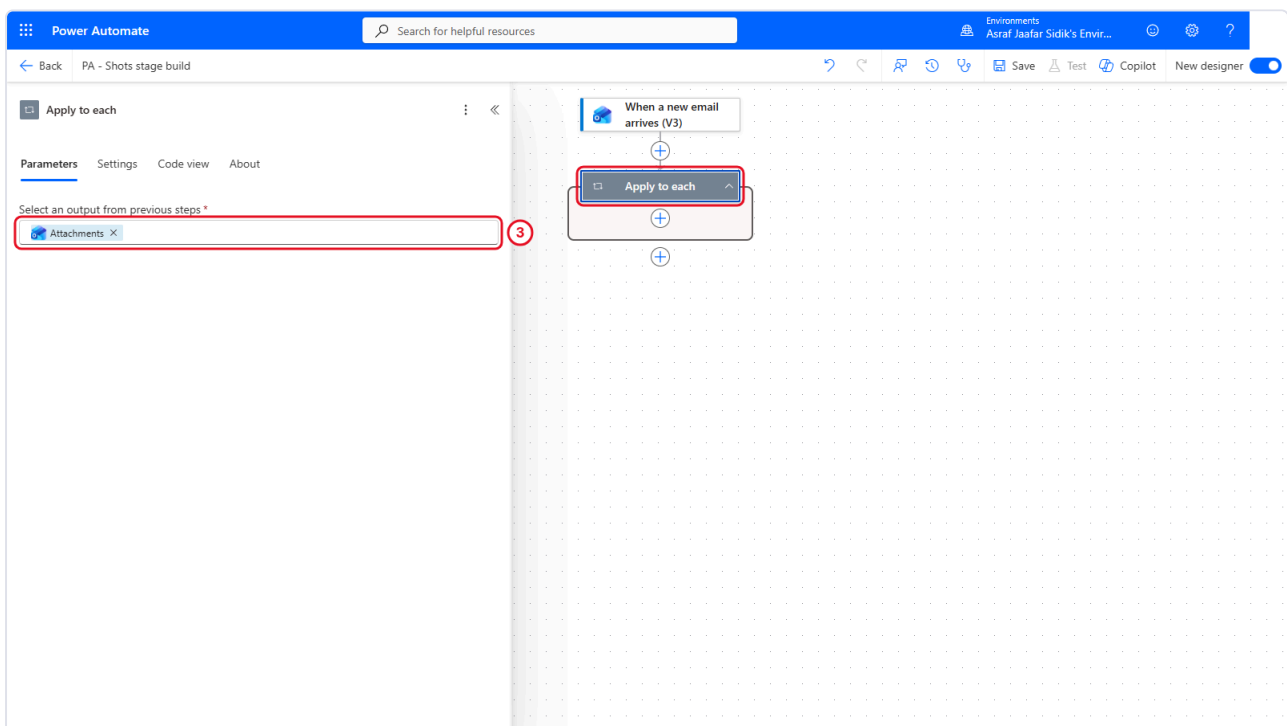
3.3 Add Apply to Each

1. Select the **+** below the trigger, then **Add an action**.
2. Search for **Apply to each** and select it (it's under **Control**).



Search by name, then pick the result under Control.

3. The loop's settings open on the left. Click into **Select an output from previous steps**, open the dynamic content picker (the lightning bolt), and select **Attachments** under *When a new email arrives (V3)*.



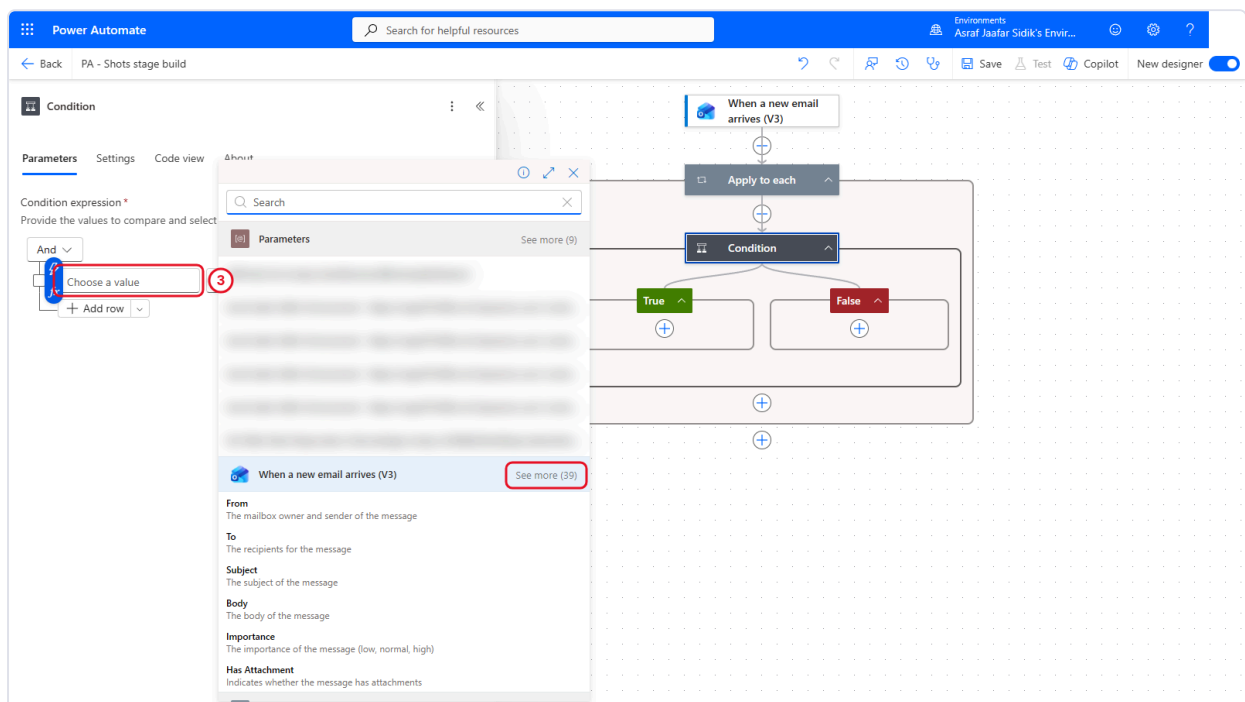
The loop runs once for every attachment in the email.

If **Attachments** isn't in the list, go back to the trigger and confirm Include Attachments is set to Yes.

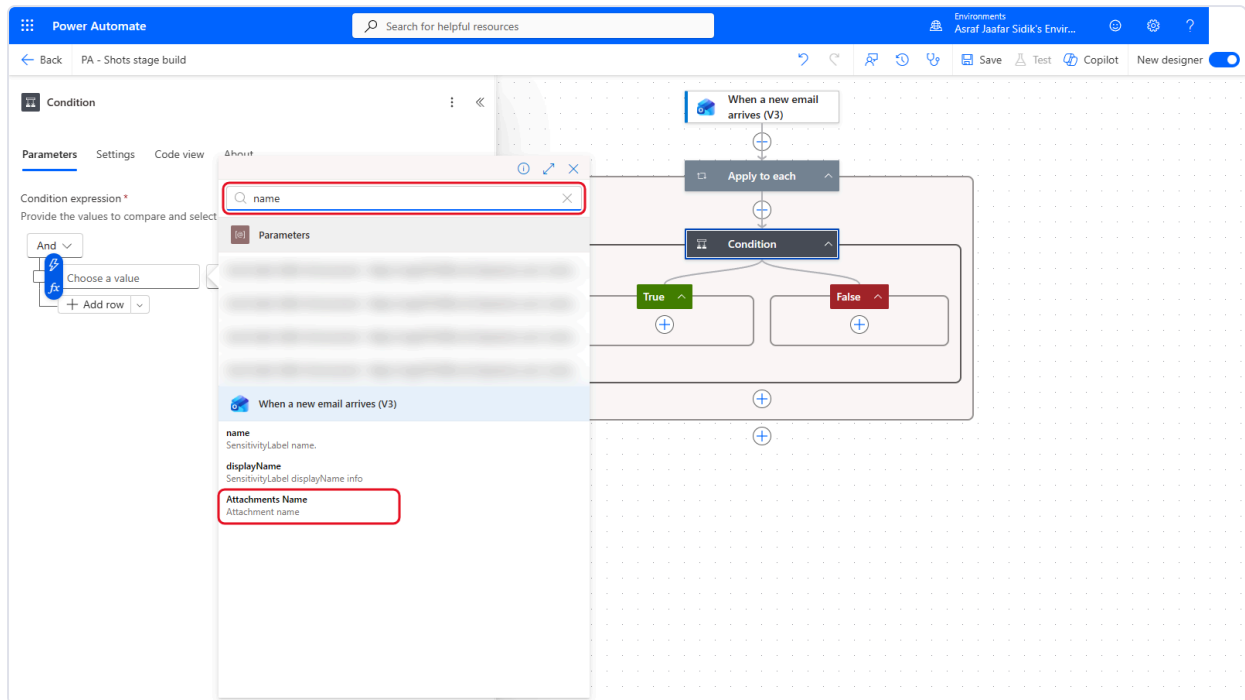
3.4 Add the PDF Condition

1. Select the **+** inside the **Apply to each** box (not the one below it), then **Add an action**.
2. Search for **Condition** and select it (under **Control**).
3. Click into the left **Choose a value** box. Open the dynamic content picker and select **Attachments Name**. (After saving it shows as `name`.)

Can't see Attachments Name? The picker only shows a few values per step at first. Select **See more** beside *When a new email arrives (V3)*, or type `name` in the picker's search box.



See more lists every value from the trigger.

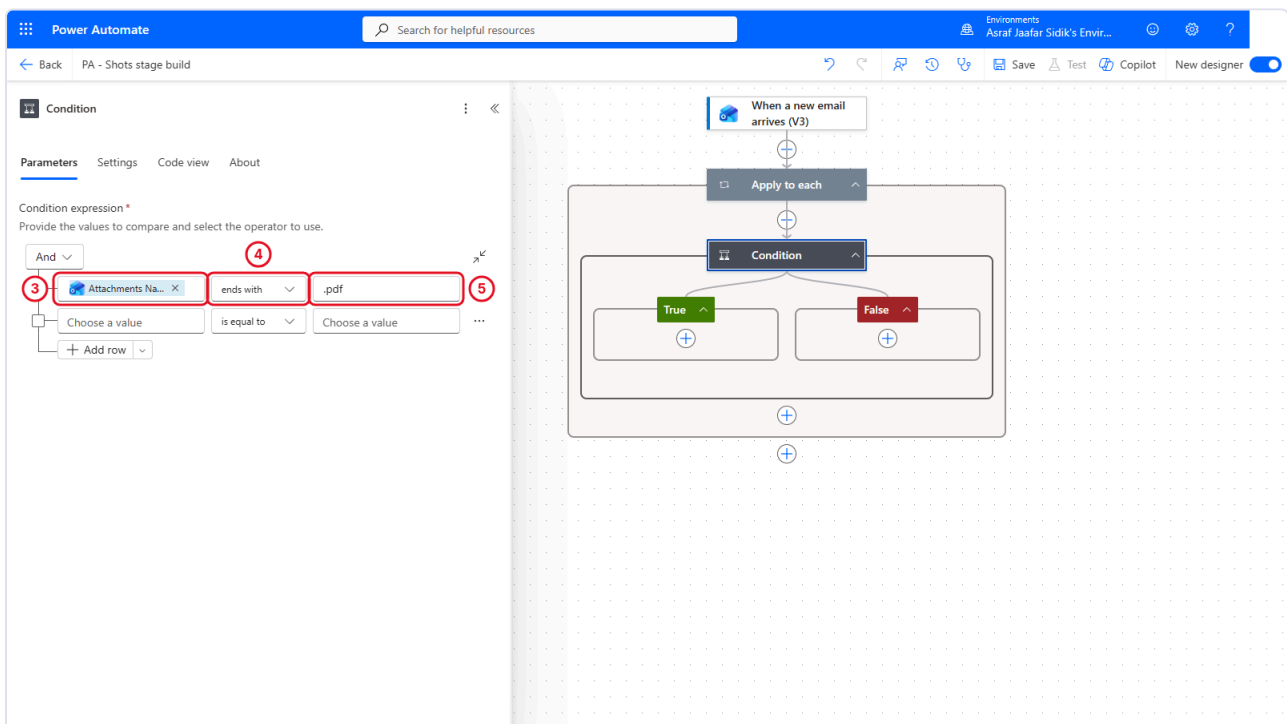


Or search: typing name brings Attachments Name straight up.

4. Set the middle box to **ends with**.

5. In the right box, type **.pdf**.

If a second, empty row appears below yours, select its ... and then **Delete**.



The condition asks one question about the current attachment: does its name end in .pdf?

Tip: Make sure you picked **Attachments Name** (the current attachment's name), not **Subject** or another text value. Inside a loop, the "Attachments ..." values always refer to the item the loop is currently on.

3.5 Create the OneDrive File

1. Select the **+** inside the green **True** box, then **Add an action**.
2. Search for **OneDrive for Business**, then select **Create file** from that connector's group.

Tip: Search by connector name, not action name. Several connectors have an action called Create file, and it's easy to grab the wrong one.

3. In **Folder Path**, select the folder icon at the right end of the box. A list of your OneDrive folders opens: select the **>** arrow beside **Power Automate Training** to open it, then select **Attachments**.

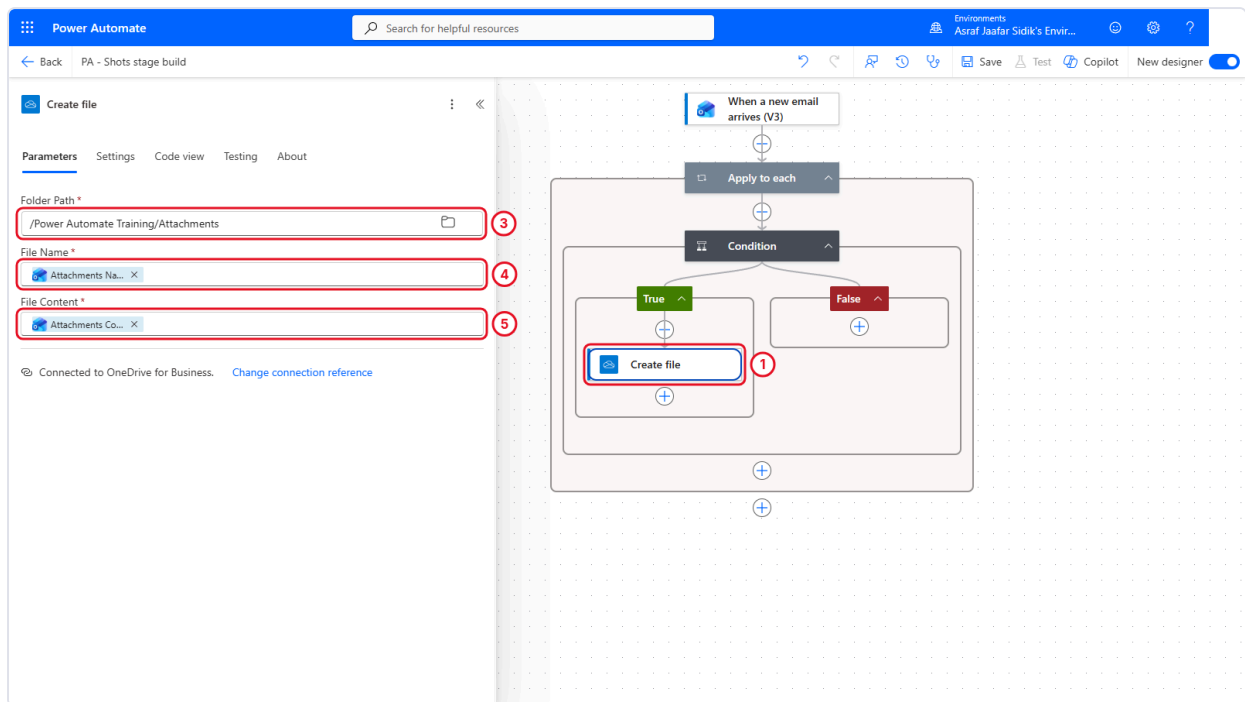
Folder Path *

/Power Automate Training/Attachments

 **3**

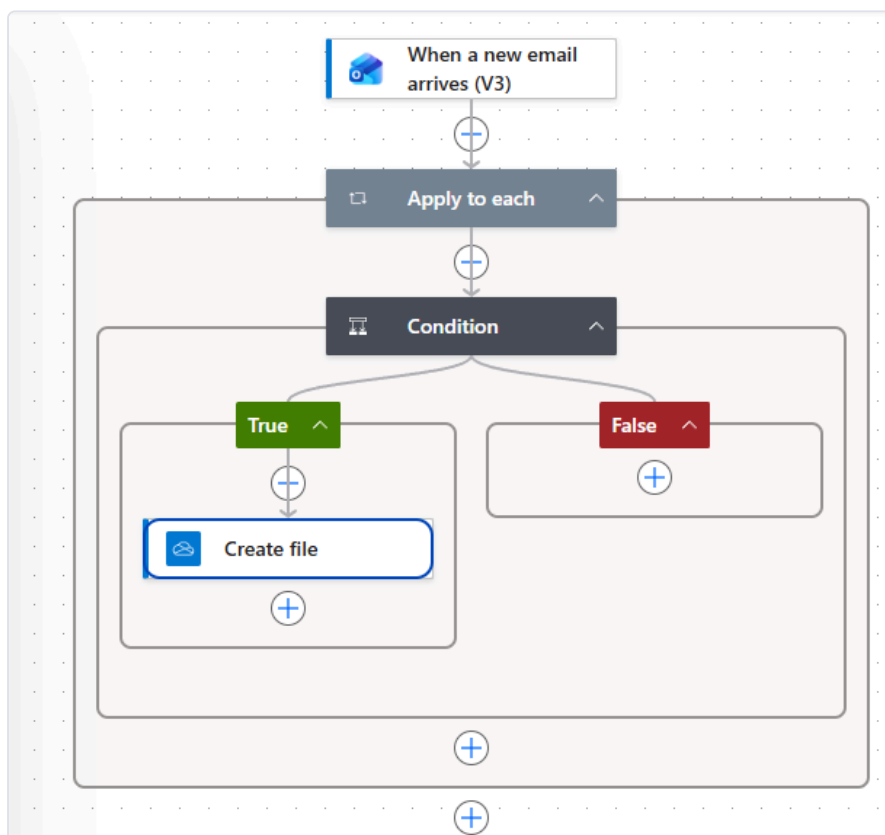
The folder icon opens your OneDrive folders.

4. In **File Name**, insert **Attachments Name** from the dynamic content picker (use **See more** or search **name** if it isn't listed).
5. In **File Content**, insert **Attachments Content** (search **content** if it isn't listed).
6. Leave the **False** branch empty. That's the deliberate decision to skip non-PDF files.



Create file sits inside True. False stays empty.

7. Select **Save** (top right), then the **Flow checker** icon (the stethoscope just left of **Save**). Confirm **Errors (0)** and **Warnings (0)**, then close the panel.



The finished flow: trigger, loop, condition, and one action in True.

3.6 Test a Positive Case

1. Select **Test** (top right). In the panel that opens, choose **Manually**, then select **Test**.
2. Wait for the banner asking you to send a new email. The flow is now listening.
3. Open the draft from section 2.3 in Outlook and send it.
4. Back in Power Automate, wait for the run to finish. Every step shows a green tick.
5. Select **Apply to each**. The counter reads **2 of 2** (or 1 of 2). Select **Create file** to see that it succeeded.

The screenshot shows the Power Automate interface. On the left, the 'Run results' panel for the 'Create file' step is visible, showing inputs and outputs. The main canvas on the right displays the flow steps: 'When a new email arrives (V3)', 'Apply to each' (with a counter of 1 of 2), and a 'Condition' step. The 'Create file' step is highlighted with a green tick, indicating success.

The run processed both attachments.

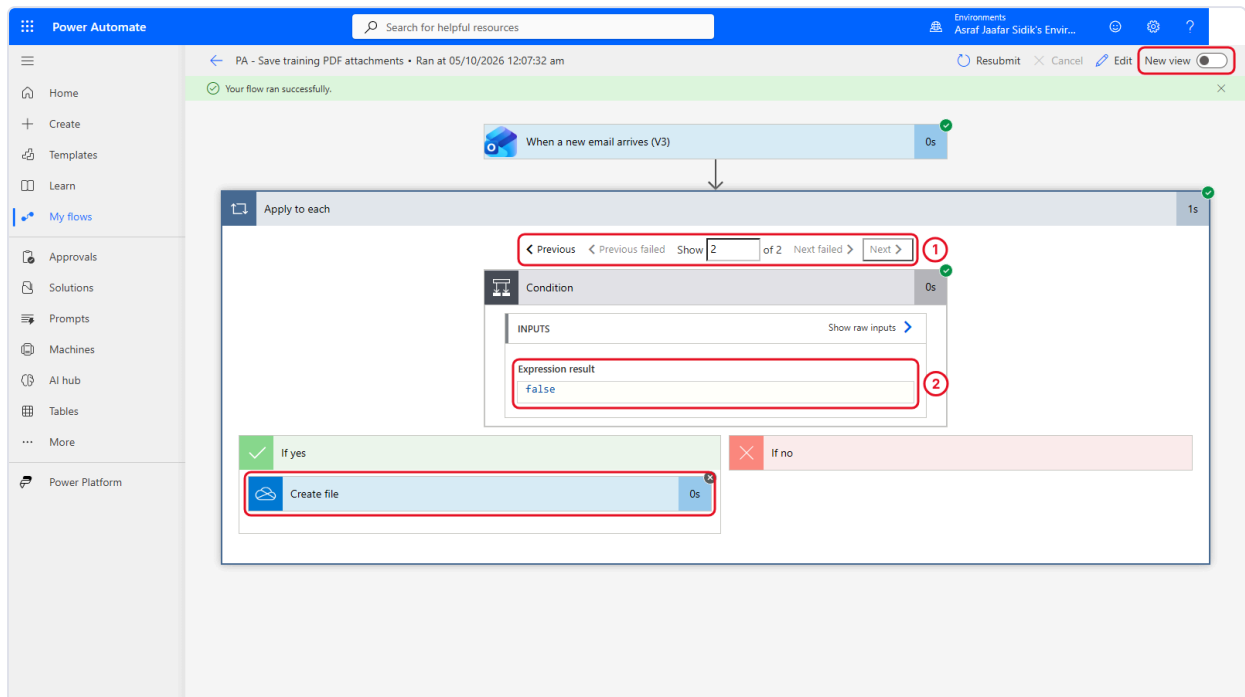
6. Open the Attachments folder in OneDrive. The PDF is there. No image file was created.

3.7 Test a Negative Case

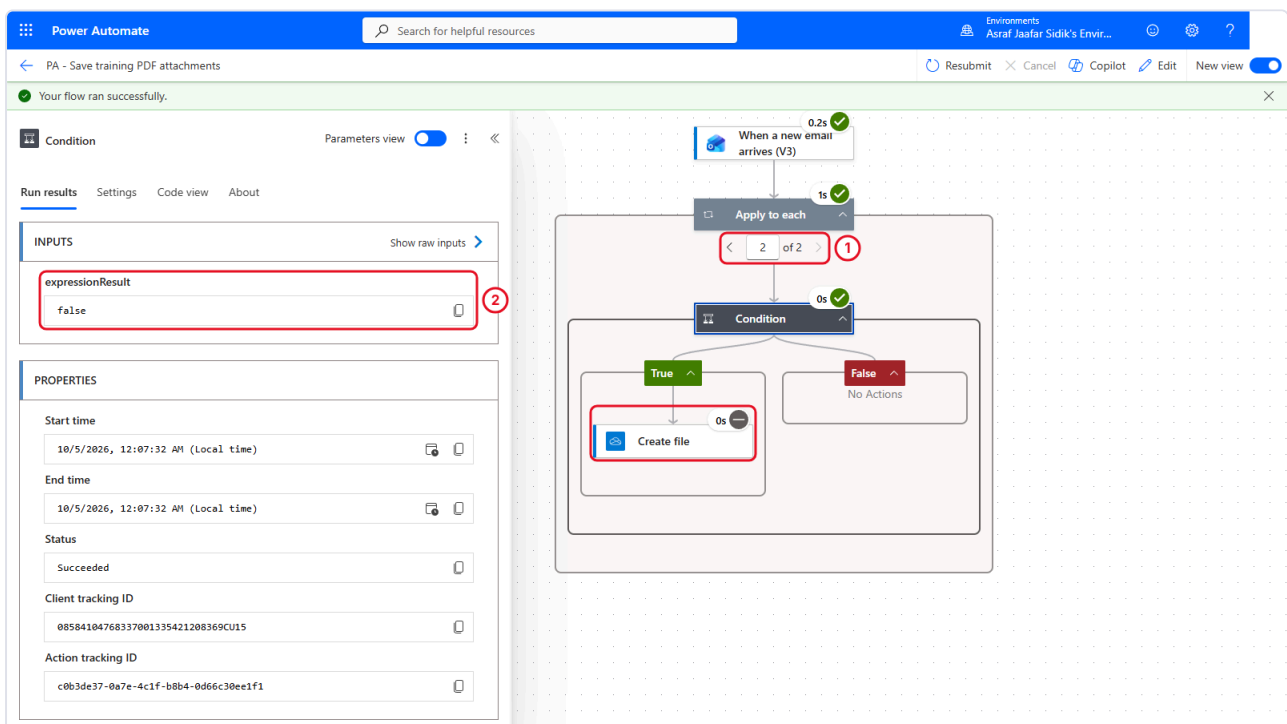
You don't need another email. The mixed test already produced both outcomes.

1. In the same run, use the arrows on **Apply to each** to switch to the other iteration.
2. Select **Condition**. The result is **false**, Create file shows as skipped, and the False branch has nothing to run.

The panel still shows the old result? The run view sometimes doesn't refresh the details panel when you change the iteration. Select a different step (such as **Apply to each**), then select **Condition** again. You can also trust the canvas: on this iteration **Create file** shows a grey skip icon instead of a green tick. If it still won't update, switch **New view** off at the top right of the run page to use the classic run view, where each iteration opens inline.



The classic view: True and False show as If yes and If no, and the skipped Create file has a grey cross.



The image attachment took the False branch, so nothing was created.

Key point: A run can succeed while deliberately producing nothing. *Succeeded* means the process completed, not that every branch created an output.

Independent Practice

Change the condition to save `.docx` attachments instead of PDFs. Send one test email with both `SessionNotes.docx` and `CourseOutline.pdf`. Predict which file appears, run it, then change the condition back to `.pdf`.

Going further: your first look at an expression. Send the same PDF twice and OneDrive numbers the copy or refuses it. To give every file a unique name, replace the **File Name** token with an expression that adds the date and time in front: type `/`, choose **Insert expression**, and paste the timestamp expression from the copy-paste page. It produces names like `20260913-094512-CourseOutline.pdf`. You'll meet expressions properly in Chapters 4 and 5. Optional.

Tips and Troubleshooting

Problem	Check
The flow doesn't run	The message must be new, contain <code>[PA TRAINING]</code> in the subject, and reach the mailbox used by the Outlook connection
Attachments isn't in the dynamic content list	Turn on Include Attachments in the trigger, save, and reopen the loop
The loop runs but creates nothing	Check the condition: left is Attachments Name , middle is <i>ends with</i> , right is <code>.pdf</code>
Create file fails	OneDrive connection, folder path, file name and file content
Duplicate file error	Delete the old file from Attachments, or try the Going further box
The JPG is saved	Create file is outside the condition. Drag it into True

Lesson Summary

This flow combines an event trigger, a loop over attachments, a condition for the business rule and a OneDrive action for the result, all built by clicking and picking dynamic content. You tested both a file-producing path and a successful no-output path. Next, an instant flow reads the Excel table and sends a summary.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything in the main steps is **text**, typed or pasted straight into a field.

Flow name (text, section 3.1)

```
PA - Save training PDF attachments
```

Trigger subject filter (text, section 3.2)

```
[PA TRAINING]
```

Condition right value (text, section 3.4)

```
.pdf
```

Going further (optional)

Unique file name with a timestamp

Paste into the expression editor (type `/` in File Name, then choose **Insert expression**).

```
concat(formatDateTime(utcNow(),'yyyyMMdd-HH:mm:ss'), '-', item()?['name'])
```

`item()` means "the attachment the loop is on right now", the same thing the **Attachments Name** token points to.

04

CHAPTER 4

Create and Distribute an Excel Summary

The coordinator wants a readable list of training sessions on demand, without copying cells into an email by hand. An instant flow reads the register and sends one summary table. Along the way you write your first expression.

Copy-paste values: the flow name, subject and your first expression are on the copy-paste page.

Estimated time: 60 minutes

Your result: An instant cloud flow that reads the training register, picks useful columns, counts the rows and sends one summary email. A Teams update is an optional extension.

What You Will Learn

- Start a flow on demand with an instant trigger
- Read rows from an Excel table with **List rows present in a table**
- Choose report columns with the **Select** action
- Turn the rows into one HTML table and send it in a single email
- Write your first expression to count the rows
- Inspect run history to confirm what each action produced

How the Flow Works

Manually trigger a flow > List rows present in a table > Select > Create HTML table > Send an email (V2)

The Excel action returns a list of rows. Select keeps only the columns readers need. Create HTML table turns the whole list into one table. The email action runs once, after the table is complete.

Before You Begin

You need `Power Automate Training/TrainingRegister.xlsx` in OneDrive for Business with a formatted table named `tblTraining`:

Column	Contents	Used in this chapter for
--------	----------	--------------------------

TrainingID	TR-001 to TR-006	The independent practice
ParticipantName	Fictional names	A report column
Email	Placeholder addresses	Read but deliberately left out of the report
CourseTitle	Fictional course names	A report column
SessionDate	Real Excel dates	A report column
Status	Registered or Completed	A report column

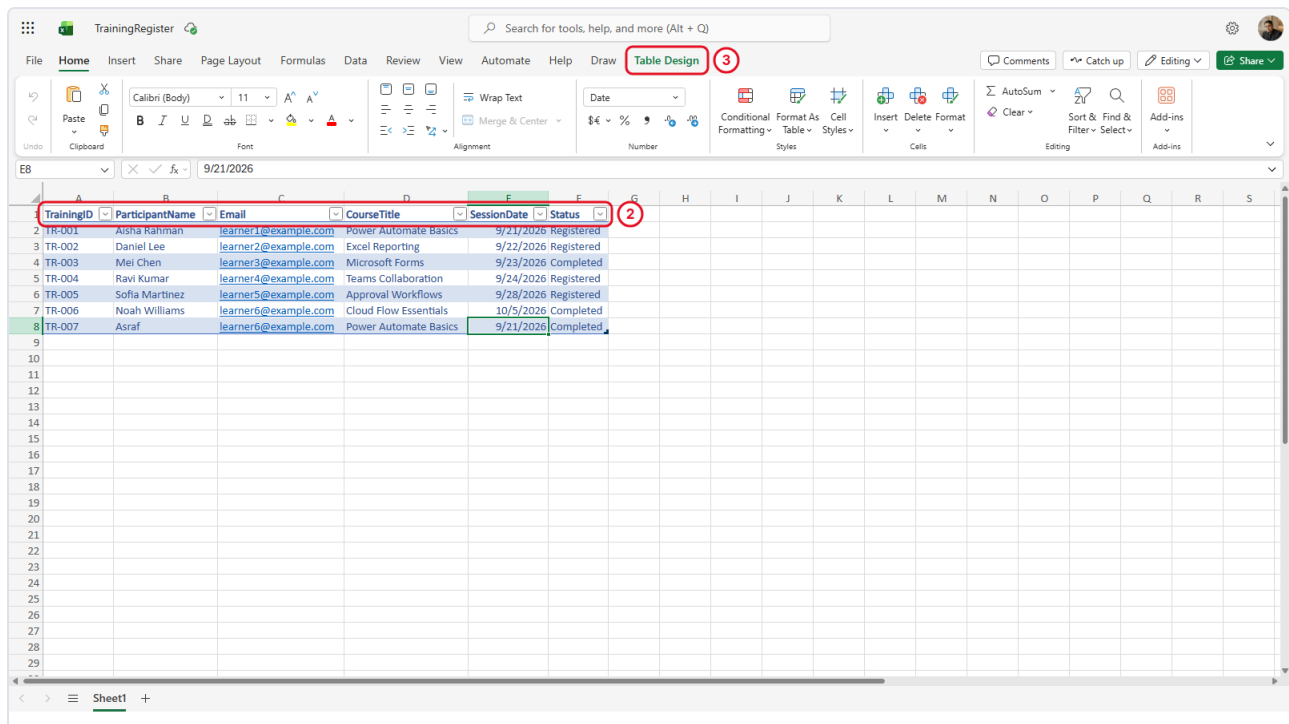
Note: Addresses like `learner1@example.com` are just data in this chapter. `example.com` can't receive mail. The summary goes to your own training mailbox, not to the people in the table.

Tip: Before each run, finish any cell edit and let Excel save. The connector reads the saved cloud file, so an open edit gives you a stale report.

Important: Keep the workbook in Excel for the web during class. If it's open in the Excel desktop app while a flow runs, the flow can read old data or fail because the file is locked. Close the desktop app before testing.

4.1 Confirm the Source Table

1. Open `TrainingRegister.xlsx` from the training folder. You built it in section 2.2, so you're only checking it here.
2. Check the header row. All six columns must be present and spelled exactly as above.
3. Select a cell in the data, open **Table Design**, and confirm the table name is exactly `tblTraining`.

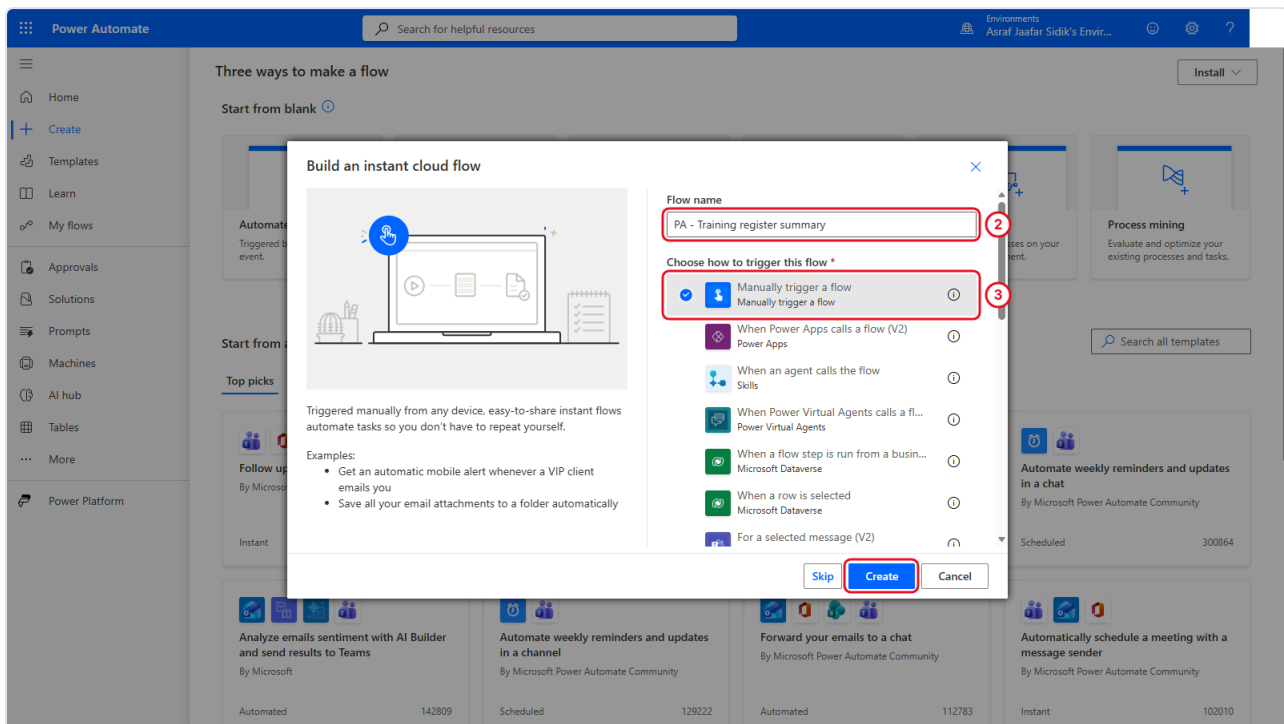


Select any cell in the table, then open the Table Design tab to see its name. The table name, not the sheet name, is what Power Automate looks for.

Key point: A worksheet name and a table name are different things. If the Table dropdown is empty later, a missing or misnamed table is almost always why.

4.2 Create an Instant Cloud Flow

1. Confirm the training environment, select **Create** on the left, then the **Instant cloud flow** tile.
2. In **Flow name**, enter PA - Training register summary .
3. Select **Manually trigger a flow**, then **Create**.



This trigger starts only when you run it.

4.3 Read the Excel Records

1. On the canvas, select the **+** below **Manually trigger a flow**, then **Add an action**.
2. In the search box at the top of the panel, type `List rows present in a table`. Under the **Excel Online (Business)** heading, select **List rows present in a table**. Its settings open on the left.
3. Open the **Location** dropdown and select **OneDrive for Business**.
4. Open the **Document Library** dropdown and select **OneDrive**.
5. In **File**, select the folder icon at the right end of the box. Select the **>** beside **Power Automate Training**, then select **TrainingRegister.xlsx**.
6. Open the **Table** dropdown and select **tblTraining**.
7. Under **Advanced parameters**, open the dropdown (it reads **Showing 0 of 6**) and tick **DateTime Format**. A new **DateTime Format** box appears: choose **ISO 8601**. This returns dates as dates instead of serial numbers like `46286`.

Pick the file by browsing, then the table from the dropdown. Once saved, Location, Document Library and Table show internal IDs like `me` or `{4672...}` instead of the names you picked. That's normal.

If **Table** shows *No items*, stop. The workbook has no formatted table. Go back to section 2.2.

4.4 Choose the Report Columns

1. On the canvas, select the **+** below **List rows present in a table**, then **Add an action**.
2. Search for `Select`. Under the **Data Operation** heading, select **Select**.
3. Click into the **From** box. Select the lightning bolt icon that appears beside it to open the dynamic content picker. Under *List rows present in a table*, select **body/value**. This is the list of all rows. (It may be labelled **value**. If you can't see it, select **See more** or type `value` in the picker's search box.)
4. In the **Map** area, click the first **Enter key** box (left) and type `Participant`.
5. Click the **Enter value** box beside it (right), open the lightning bolt, and select **ParticipantName** under *List rows present in a table*. Type `Participant` in the picker's search box if it isn't listed. Don't type the value yourself: it must be the green token from the picker.
6. A new empty row appears underneath. Fill three more rows the same way, typing the key and picking the value. Leave the last empty row blank:

Enter key	Enter value (dynamic content)
Participant	ParticipantName
Course	CourseTitle

Session date	SessionDate
Status	Status

Select

Parameters Settings Code view About

From *

body/value X

Map *

Participant	ParticipantName X	4
Course	CourseTitle X	
Session date	SessionDate X	6
Status	Status X	

Each key becomes a column heading. Every value is a green Excel token you picked, not typed. Email and the internal row data are left out.

The Session date comes through as 2026-09-21T00:00:00.000Z . That's fine for now: the Going further box at the end of the chapter shows how to tidy it.

Select handles every row for you, so you don't need an Apply to each here.

4.5 Build One HTML Table

1. On the canvas, select the + below **Select**, then **Add an action**.
2. Search for **Create HTML table** and select it under **Data Operation**.
3. Click into **From**, open the lightning bolt, and select **Output** under **Select**.
4. Leave everything else as it is. Select already defined the headings.

Create HTML table

Parameters Settings Code view About

From *

Output X

Advanced parameters

Showing 0 of 1 Show all Clear all

The whole list goes in, one table comes out.

Checkpoint: If an Apply to each suddenly wraps this action, you picked a single column instead of the whole Output. Delete the loop and pick **Output** again.

4.6 Send the Coordinator Summary

1. On the canvas, select the + below **Create HTML table**, then **Add an action**. Search for **Send an email**, then select **Send an email (V2)** under **Office 365 Outlook**.
2. In **To**, start typing your name or email address, then select yourself from the list that appears.
3. In **Subject**, enter **[PA TRAINING] Training register summary**.
4. Click into **Body** and type **Here is the current training register**. then press Enter.
5. On the new line, type **/** and choose **Insert dynamic content** (or select the lightning bolt). Select **Output** under the **Create HTML table** heading. The picker lists two values called **Output**: pick the one under **Create HTML table**, not the one under **Select**.

Leave the body as these two lines for now. You add the count sentence in 4.7.

Send an email (V2)

Parameters Settings Code view Testing About

To *

Enter part of a name or email address to find more people

Subject *

[PA TRAINING] Training register summary

Body *

Here is the current training register.

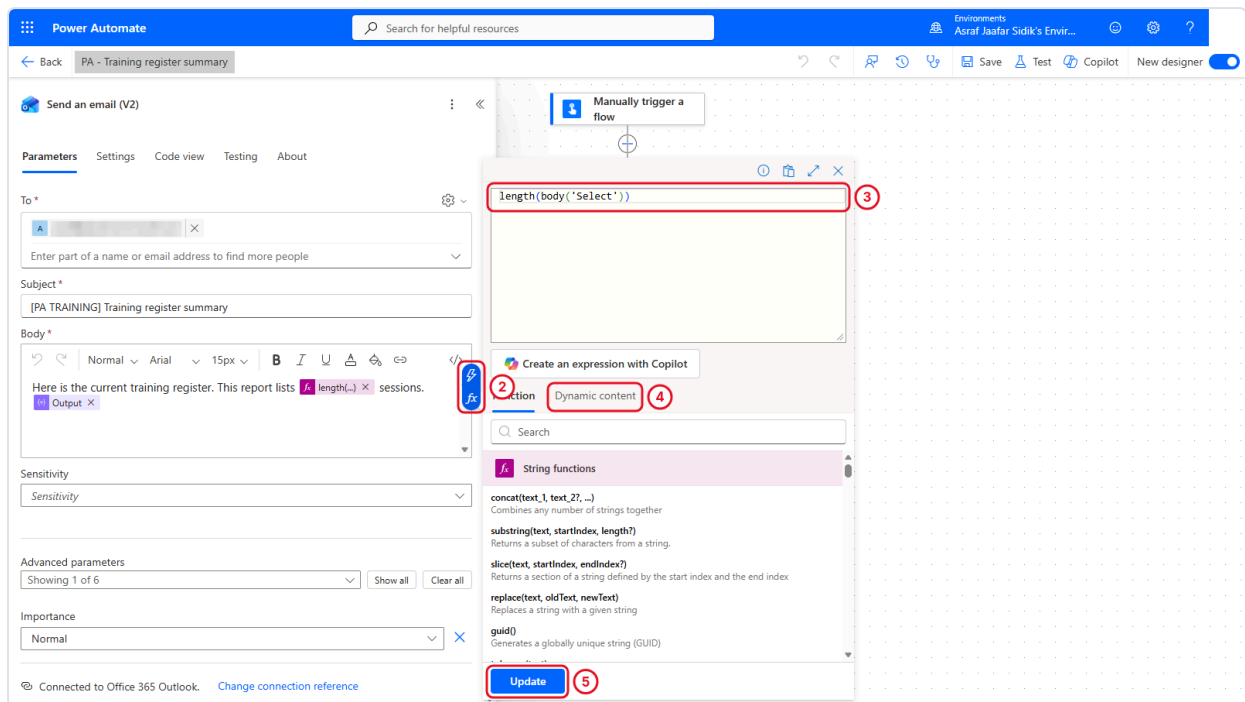
Output X

The table token goes in the body below a short introduction.

4.7 Your First Expression: Count the Rows

The coordinator wants a line like *"This report lists 6 sessions."* None of the dynamic content values is a count. Dynamic content gives you values that already exist. When you need a value *calculated*, you write an **expression**.

1. On the canvas, select **Send an email (V2)** to open it again. In **Body**, click right after `register.` at the end of the first line and type `This report lists` (with a space before and after).
2. Type `/` and choose **Insert expression**.
3. In the expression box, type `length(`
4. At the top of the same panel, switch from the **Function** tab to the **Dynamic content** tab and select **Output** under *Select*. It drops into the expression.
5. Type `)` to close the bracket, then select **Add** (it reads **Update** if you're editing an expression that's already there).
6. After the new token, type `sessions.`

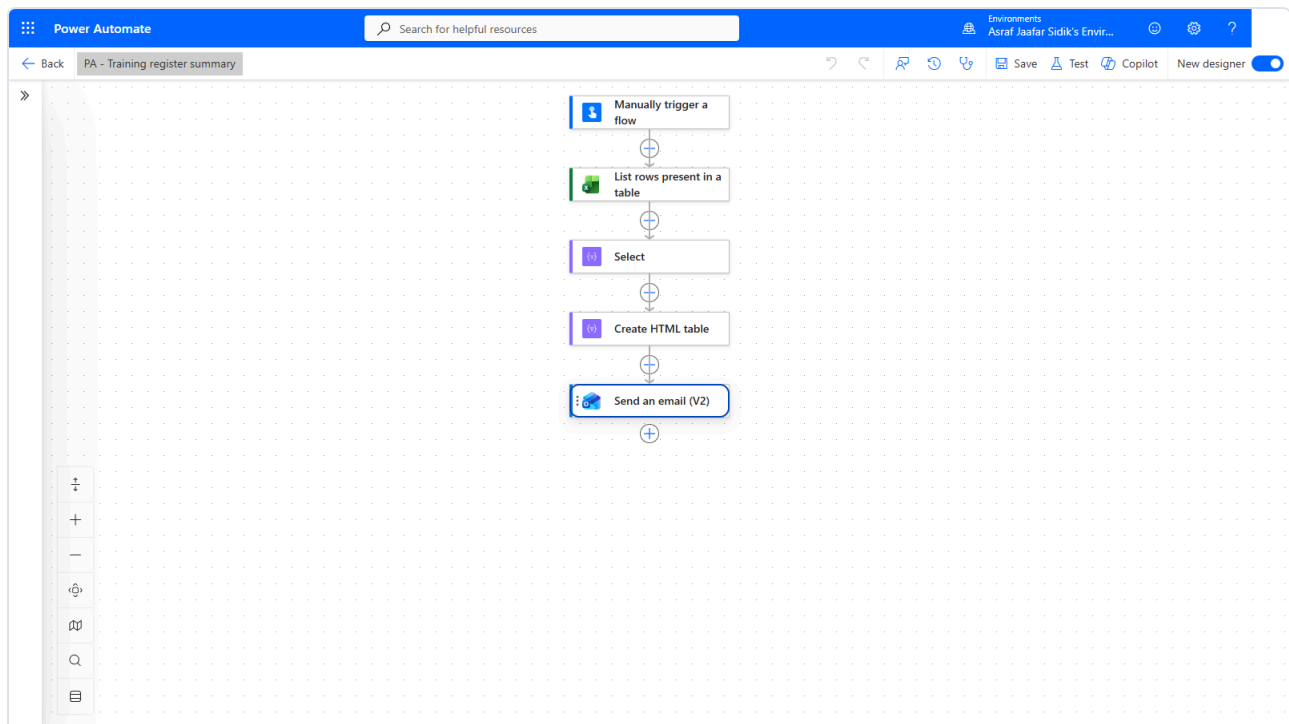


length() counts the items in a list. Here it counts the rows Select produced.

The expression reads `length(body('Select'))`. You can also paste it from the copy-paste page. `length()` is a **function**: you give it something inside the brackets and it gives back an answer.

Tip: If you renamed the Select action, the name inside the quotes changes too. Picking Output from the Dynamic content tab handles that for you, which is why step 4 picks it instead of typing it.

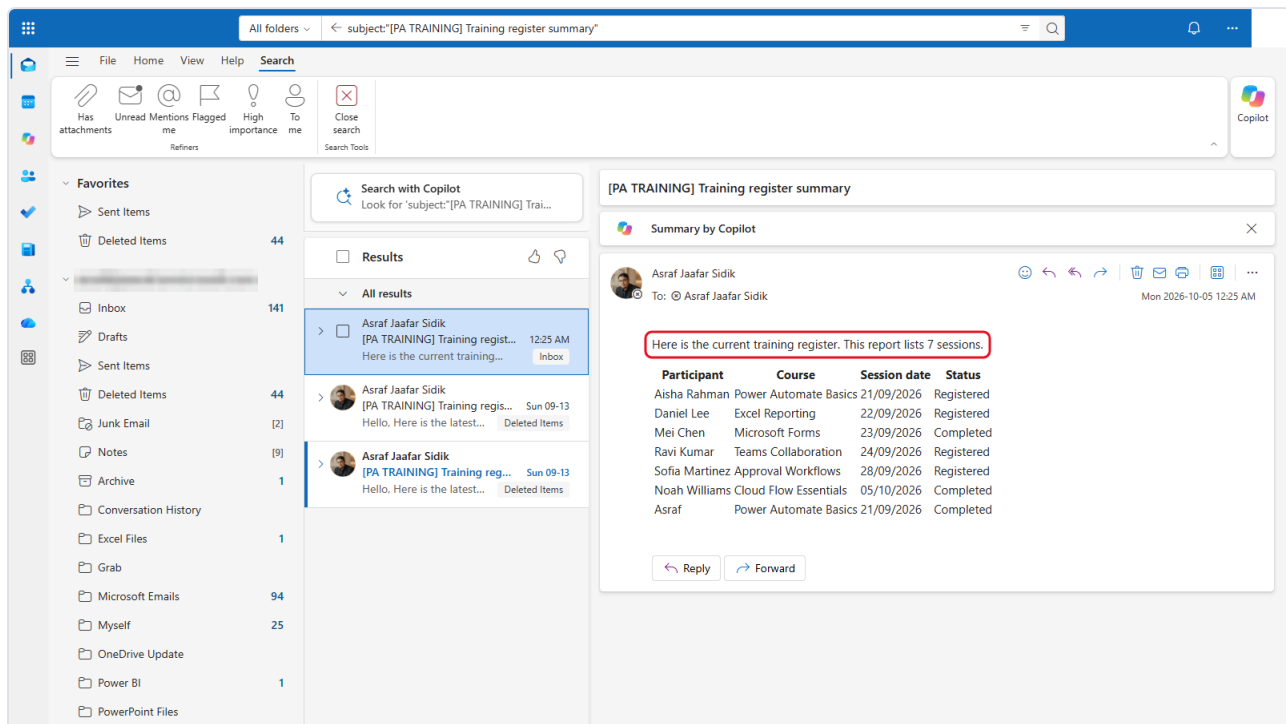
7. Select **Save**, then **Flow checker**, and fix anything it reports.



Five steps. One email.

4.8 Test and Inspect

1. Select **Test** > **Manually** > **Test**. A **Run flow** panel opens; select **Continue** if it asks about connections, then **Run flow**, then **Done**. This sends a real email to you, so follow your trainer's instruction first.
2. Open the completed run. Every action has a green tick, and Send an email (V2) ran once.
3. Select the **Select** step on the canvas. In the panel on the left, scroll to **Outputs** and select **Show raw outputs**. You see six records with real names and courses.
4. Open the email. You should see the sentence *This report lists 6 sessions.*, four headings, and six rows. The number always matches the rows in your table, so if you added a row it goes up too.



One email, one table, counted by your first expression. This workbook had a seventh test row, so it reads 7.

Each manual run sends one current report. Earlier emails stay as historical snapshots.

4.9 Optional Teams Update

Trainer-led only, once a permitted destination is confirmed. After the email, add **Microsoft Teams > Post message in a chat or channel**, choose **Flow bot** and **Chat with Flow bot**, pick yourself as the recipient, and post a short message such as *Training register summary sent*. Keep it short. Teams doesn't show the email's HTML table the same way.

Independent Practice

Add Training ID to the report without adding Email. Predict where the change belongs, make it, and check the received message. Then explain why a six-record table still produces only one email.

Going further: format the date. The Session date column shows values like 2026-09-21T00:00:00.000Z. In Chapter 5 you'll learn the `formatDateTime()` expression that turns it into 21 Sep 2026. Come back afterwards and use it in Select's **Session date** value. Optional.

Troubleshooting

Symptom	What to check
No Excel table is listed	Workbook location, saved table, table name and connector account

One email arrives for every row	The email ended up inside a loop. Use Output from Create HTML table
The report includes internal columns	Use Select to define headings and values
The email shows HTML tags	Check you inserted Output from Create HTML table, not from Select
A date appears as a number	Set DateTime Format to ISO 8601 in List rows present in a table
The expression shows an error	Brackets must match: <code>length(...)</code> . Pick Output from the Dynamic content tab instead of typing it
The report is stale	Finish the Excel edit, wait for save, and run again

Lesson Summary

An instant trigger lets the coordinator decide when to run. Excel supplies the records, Select defines the report columns, Create HTML table produces one report, and your first expression counted the rows. Next, a recurrence trigger starts a reminder process on a schedule.

For reference, see the [Excel Online \(Business\) connector](#) and [data operations in Power Automate](#).

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Anything marked **text** is typed or pasted straight into a field.

Flow name (text, section 4.2)

```
PA - Training register summary
```

Select keys (text, section 4.4)

Type each key, then insert the matching column from dynamic content as its value.

```
Participant
```

```
Course
```

```
Session date
```

```
Status
```

Email subject (text, section 4.6)

```
[PA TRAINING] Training register summary
```

Email introduction (text, section 4.6)

```
Here is the current training register.
```

Your first expression: count the rows (section 4.7)

Type `/` in the Body, choose **Insert expression**, paste, then select **Add**. Works if the Select action still has its default name.

```
length(body('Select'))
```

05

CHAPTER 5

Send Weekly Training Reminders

Every week the coordinator opens the register and writes the same reminder for each participant. A scheduled flow can do that at a predictable local time.

Copy-paste values: the flow name, email text and the date expression are on the copy-paste page.

Estimated time: 75 minutes

Your result: A scheduled cloud flow that runs every Monday at 9:00 AM Malaysia time, reads the training register, and prepares one personalised reminder email for each row.

What You Will Learn

- Schedule a flow with the **Recurrence** trigger and a local time zone
- Convert the current time into readable local time
- Loop through Excel rows with **Apply to each**
- Build a personalised email by typing text and inserting dynamic content
- Use a second expression, `formatDateTime()`, to make a date readable

How the Flow Works

Recurrence > Current time > Convert time zone > List rows present in a table > Apply to each > Send an email (V2)

Recurrence decides when the flow starts. Current time and Convert time zone produce a readable local timestamp for the message. The Excel action returns the rows, and Apply to each handles them one at a time.

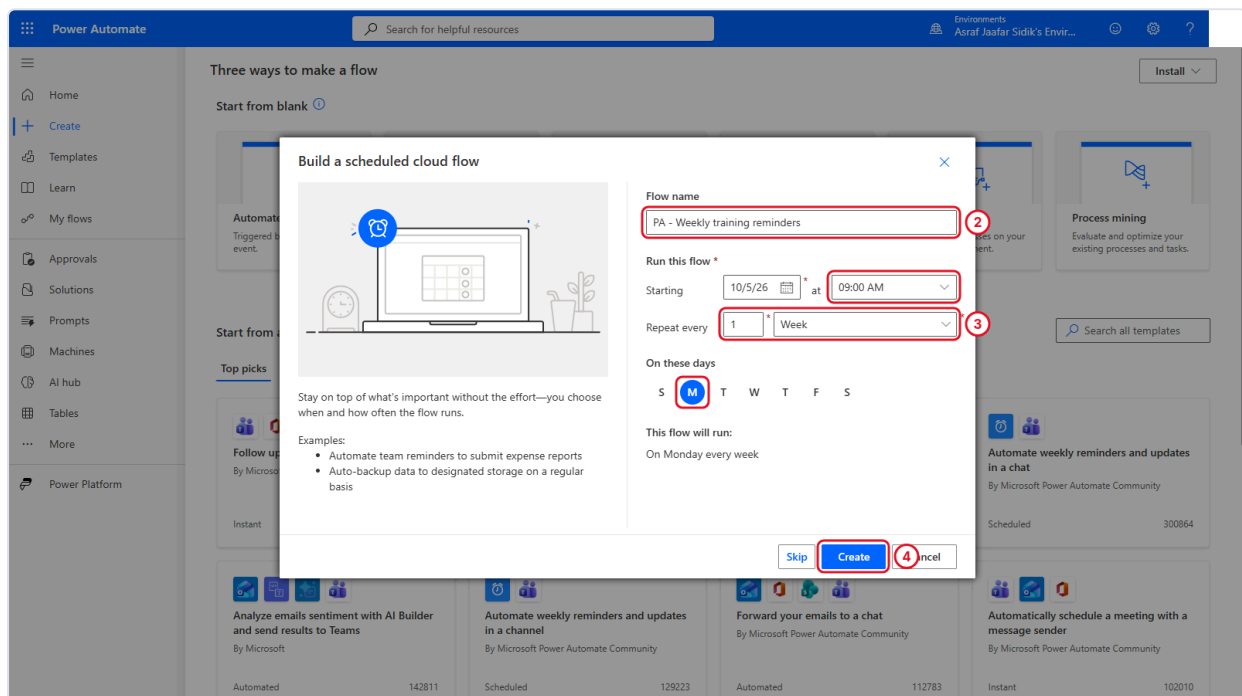
Before You Begin

You need `Power Automate Training/TrainingRegister.xlsx` with the `tblTraining` table. This chapter uses every column: ParticipantName for the greeting, Email as the recipient, CourseTitle in the subject and body, SessionDate and Status in the body.

Important: The sample register uses addresses like `learner1@example.com`. They can't receive mail, which is what you want while learning. Replace them with trainer-approved addresses only when the trainer authorises a live test.

5.1 Create the Weekly Schedule

1. Select **Create** on the left, then the **Scheduled cloud flow** tile.
2. In **Flow name**, enter `PA - Weekly training reminders`.
3. Set **Repeat every** to **1 Week**, tick **M** (Monday), and set the start time to **9:00 AM**. Your starting date will differ.
4. Select **Create**.



The dialog creates the Recurrence trigger for you.

5. The designer opens. On the canvas, select the **Recurrence** card to open its settings on the left, and check these values. If you can't see the time zone fields, open **Advanced parameters** and select **Show all**.

Setting	Value
Interval	1
Frequency	Week
Time zone	(UTC+08:00) Kuala Lumpur, Singapore
Start time	Filled in by the dialog. Leave it.

On these days	Monday
At these hours	9
At these minutes	0

PA - Weekly training reminders

Recurrence *

Interval * 1 Frequency * Week

Time zone (UTC+08:00) Kuala Lumpur, Singapore

Start time 2026-10-12T01:00:00.000Z

On these days Monday

At these hours 9

At these minutes 0

Preview
Runs at 9:00 on Monday every week

Read the preview line at the bottom. It should say 9:00 on Monday every week.

5.2 Produce a Readable Local Timestamp

1. On the canvas, select the **+** below **Recurrence**, then **Add an action**. In the search box, type **Current time**. Under the **Date Time** heading, select **Current time**. It needs no settings.
2. Select the **+** below **Current time**, then **Add an action**. Search for **Convert time zone** and select it under **Date Time**.
3. Click into **Base time**, select the lightning bolt icon beside it, and pick **Current time** under **Current time**.
4. Set the other boxes from their dropdowns:

Setting	Value
Base time	Current time (the token from step 3)
Source time zone	(UTC) Coordinated Universal Time
Destination time zone	(UTC+08:00) Kuala Lumpur, Singapore
Format string (may be labelled Time unit)	Full date/time pattern (short time)

Pick the format from the dropdown. No expression needed.

Key point: The two time zone settings do different jobs. Recurrence decides *when the flow starts*. Convert time zone only changes a timestamp shown in the message.

5.3 Read the Excel Rows

The settings are the same as Chapter 4, section 4.3.

1. On the canvas, select the **+** below **Convert time zone**, then **Add an action**. Search for **List rows present in a table** and select it under **Excel Online (Business)**.
2. Open the **Location** dropdown and select **OneDrive for Business**.
3. Open the **Document Library** dropdown and select **OneDrive**.
4. In **File**, select the folder icon at the right end of the box. Select the **>** beside **Power Automate Training**, then select **TrainingRegister.xlsx**.

Tip: Clicking a folder's name selects it but doesn't open it. Use the **>** arrow on the right to go inside.

5. Open the **Table** dropdown and select **tblTraining**.
6. Under **Advanced parameters**, open the dropdown (it reads **Showing 0 of 6**) and tick **DateTime Format**. In the new **DateTime Format** box, choose **ISO 8601**.

After you pick them, **Location** shows `me`, **Document Library** shows a long code, and **Table** shows an ID in braces like `{46727A35-...}`. That's normal.

Same settings as Chapter 4. The codes in Location, Document Library and Table stand for the names you picked.

5.4 Process One Row at a Time

1. On the canvas, select the **+** below **List rows present in a table**, then **Add an action**. Search for **Apply to each** and select it under **Control**.
2. Click into **Select an output from previous steps**, select the lightning bolt icon beside it, and under **List rows present in a table** pick **body/value** (it may be labelled **value**). If you can't see it, select **See more** or type `value` in the picker's search box.

The loop runs once per row in the table.

Checkpoint: The input must be the whole list (**value**). A single column such as Email isn't a list of rows.

5.5 Prepare the Reminder Email

On the canvas, select the **+** inside the **Apply to each** box (below its header), then **Add an action**. Search for **Send an email**, then select **Send an email (V2)** under **Office 365 Outlook**.

1. **To:** this box opens a people picker, so it needs switching first. Select the small **settings** (gear) icon above the right end of the **To** box, on the same line as the **To** label. Choose **Use dynamic content**. Then click into the box, select the lightning bolt icon, and pick **Email** under *List rows present in a table* (select **See more** or type **Email** in the search box if it isn't listed).



The gear switches To from picking people to accepting dynamic content.

2. **Subject:** click into the box and type **[PA TRAINING] Reminder:** (with the space at the end). Select the lightning bolt icon and pick **CourseTitle** under *List rows present in a table*.
3. **Body:** click into the box and type the message below. Where a value belongs, put the cursor there, type **/** and choose **Insert dynamic content** (or select the lightning bolt), then pick the value:

Hello **[ParticipantName]**,

This is your weekly training reminder.

Course: **[CourseTitle]**

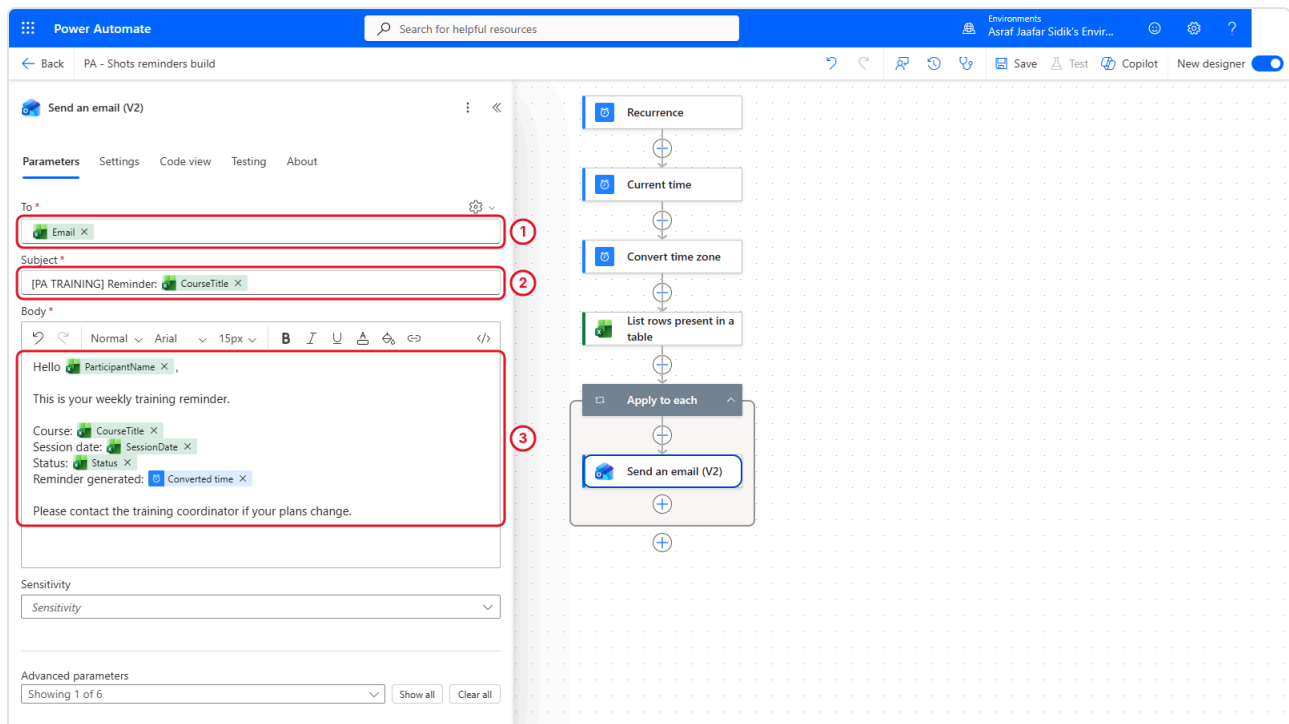
Session date: **[SessionDate]**

Status: **[Status]**

Reminder generated: **[Converted time]**

Please contact the training coordinator if your plans change.

Each **[bold name]** is a dynamic content token. **Converted time** comes from *Convert time zone*. The rest come from *List rows present in a table*. If a value isn't listed, select **See more** or type its name in the picker's search box. You can also paste the whole body from the copy-paste page and replace each **[Name]** with its token.

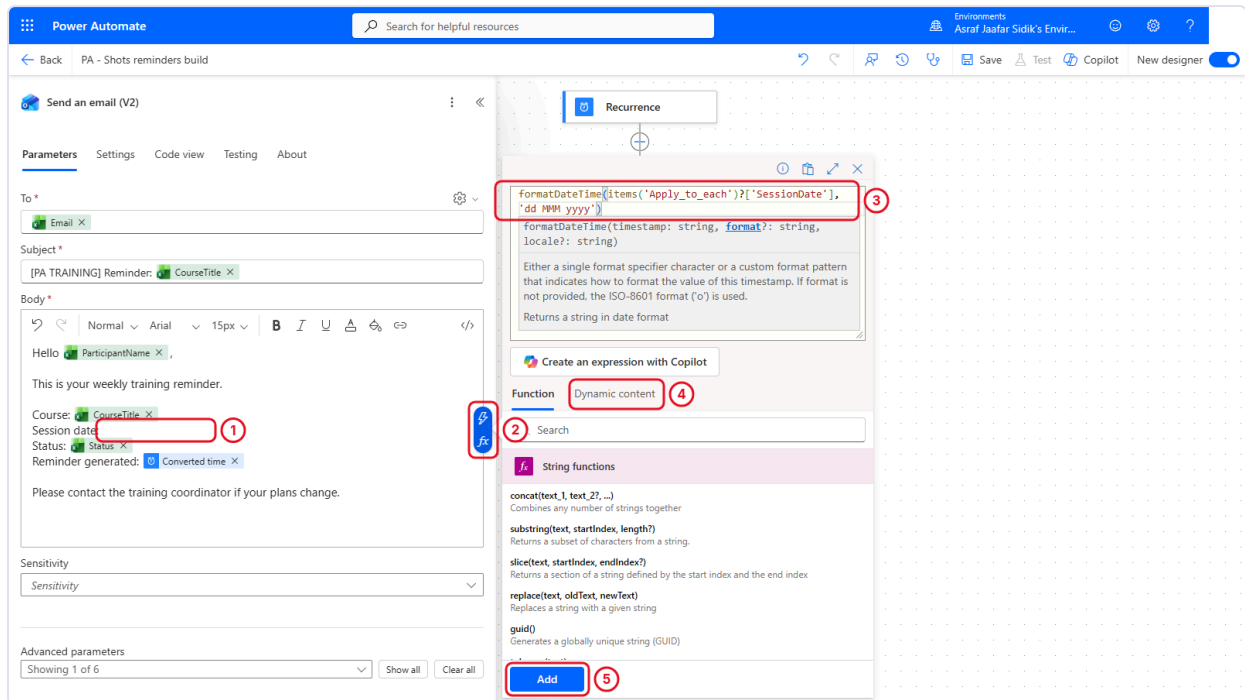


Type the words, insert the tokens. The email reads like a normal message.

5.6 Make the Date Readable with an Expression

Run the flow now and the session date reads `2026-09-21T00:00:00.000Z`. That's correct, but nobody wants to read it. There's no dynamic content value for "the date, written nicely", so this is a job for an expression.

1. In the **Body** of **Send an email (V2)**, select the **x** on the **SessionDate** token to remove it. Make sure the cursor stays right after `Session date:` on that line (click there if it moved).
2. Type `/` and choose **Insert expression** (or select the **fx** button that appears at the right edge of the Body box).
3. In the expression box, type `formatDateTime(`
4. At the top of the same panel, switch from the **Function** tab to the **Dynamic content** tab and select **SessionDate** (type `Session` in its search box if it isn't listed). It drops into the expression.
5. Type `, 'dd MMM yyyy')` and select **Add** (it reads **Update** if you're editing an expression that's already there).

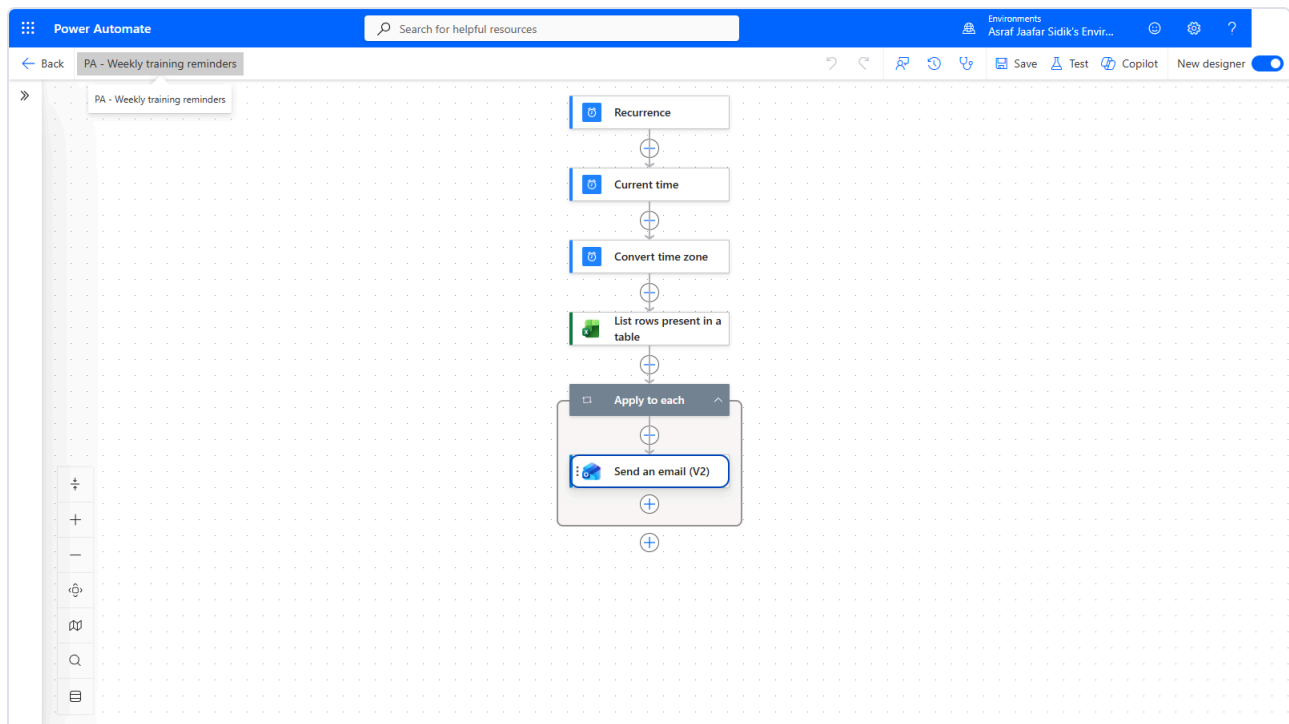


formatDateTime() takes a date and a pattern, and gives back the date written in that pattern.

The finished expression is on the copy-paste page if you'd rather paste it. The pattern `dd MMM yyyy` means two-digit day, short month name, four-digit year, so `21 Sep 2026`.

Tip: Compare this with Chapter 4. `length()` counted a list, `formatDateTime()` reshapes a date. Both follow the same shape: a function name, then what you give it in brackets.

6. On the toolbar at the top, select **Save**, then **Flow checker**, and fix any error it reports.



Recurrence, Current time, Convert time zone, List rows, and the loop with its email.

Trainer-Controlled Test

Important: Don't select Test while the workbook still has `example.com` placeholders. A test sends one email per row, so six messages with the sample data. The trainer confirms the recipients and authorises the run immediately before you select Run flow.

After an authorised test, check the run history:

- Every step before the loop succeeded
- Apply to each shows six iterations
- Each email used that row's Email, CourseTitle and Status, and the date reads like `21 Sep 2026`
- The reminder timestamp is in Malaysia time

In this version every row gets a reminder, including Completed ones. Chapter 7 fixes that.

Independent Practice

Change the scheduled hour to a value your trainer gives you, read the Recurrence preview, then set it back to 9:00. Explain why changing Convert time zone wouldn't change when the flow starts.

Going further: try other date patterns. Change `'dd MMM yyyy'` to `'dddd, d MMMM'` and the date reads *Monday, 21 September*. You can also go back to Chapter 4 and use the same expression in Select's **Session date** value. Optional.

Troubleshooting

Symptom	What to check
The flow starts at the wrong local time	Recurrence time zone, day, hour, minute and preview
The timestamp is eight hours early	Convert time zone source and destination
No Excel rows are returned	OneDrive location, workbook saved, formatted table and table name
Apply to each is empty	Use the Excel action's value list
A recipient or course is blank	Header spelling in the workbook. Re-pick the token
The date shows as a number	Set DateTime Format to ISO 8601 in List rows present in a table
The date expression shows an error	Brackets and quotes: <code>formatDateTime(... , 'dd MMM yyyy')</code>
Completed records get reminders	Expected here. Chapter 7 adds the filter

Lesson Summary

A scheduled trigger starts the flow at a predictable local time. Date Time actions give a readable timestamp, Excel supplies the rows, Apply to each hands the current row to the email, and a second expression turned a raw date into something a person wants to read. Next you add a human approval decision.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Anything marked **text** is typed or pasted straight into a field.

Flow name (text, section 5.1)

```
PA - Weekly training reminders
```

Email subject start (text, section 5.5)

Paste, then insert **CourseTitle** after it.

```
[PA TRAINING] Reminder:
```

Email body (text, section 5.5)

Paste, then replace each [Name] with the matching dynamic content token.

```
Hello [ParticipantName],

This is your weekly training reminder.

Course: [CourseTitle]
Session date: [SessionDate]
Status: [Status]
Reminder generated: [Converted time]

Please contact the training coordinator if your plans change.
```

Readable session date (section 5.6)

With the cursor right after `Session date:` (where the `SessionDate` token was), type `/`, choose **Insert expression**, paste, then select **Add**.

```
formatDateTime(items('Apply_to_each')['SessionDate'], 'dd MMM yyyy')
```

Note: If your loop has a different name (for example `Apply_to_each_1`), build the expression by picking **SessionDate** from the Dynamic content tab as in section 5.6 instead of pasting.

06

CHAPTER 6

Process a Training Approval

A training request needs a person to say yes or no before anything is confirmed. This flow collects the request with Microsoft Forms, asks an approver to decide, and posts a different Teams message for each outcome.

Copy-paste values: the flow name, form text, approval title and messages are on the copy-paste page.

Estimated time: 75 minutes

Your result: An automated cloud flow that reads a Forms response, asks a person to approve or reject it, and prepares a different Teams message for each outcome.

What You Will Learn

- Start a flow when a Microsoft Forms response is submitted
- Retrieve the answers with **Get response details**
- Request a decision with **Start and wait for an approval**
- Branch on the approval outcome and post a different Teams message for each result

How the Flow Works

When a new response is submitted > Get response details > Start and wait for an approval > Condition > Approved or rejected Teams message

The Forms trigger only supplies a response ID. Get response details uses that ID to fetch the answers. The approval action pauses the flow until someone decides. The Condition reads the approval **Outcome** and sends the flow down True or False.

Before You Begin

You need Microsoft Forms, Approvals and Microsoft Teams through your work or school account. Your trainer names an approved test approver and Teams destination before any live test.

Important: The screenshots use `approver@example.com`, a documentation placeholder. Replace it with a trainer-approved account before testing. Don't submit the form or run the flow while the placeholder is still there.

6.1 Prepare the Request Form

1. Go to forms.office.com and sign in with your training account.
2. Select **New Form**.
3. Select **Untitled form** at the top and type `Training Request`. In the description box under it, paste the description from the copy-paste page.
4. Scroll to the bottom of the form and select **+ Add new question**, then **Text**. Type `Requester name` as the question. Switch on **Required**: it's a switch along the bottom edge of the question you're editing.
5. Select **+ Add new question** > **Choice**. Type `Course` as the question, then type the three options from the copy-paste page (Power Automate Basics, Approval Workflows, Excel Reporting), one per option box. Select **Add option** if you need a third box. Switch on **Required**.
6. Select **+ Add new question** > **Date**. Type `Preferred date` and switch on **Required**.
7. Select **+ Add new question** > **Text**. Type `Business reason`. Switch on **Long answer** and **Required**, both along the bottom edge of the question.

Forms saves as you go. There's no Save button.

The screenshot shows the Microsoft Forms editor interface. At the top, the title 'Training Request' is highlighted with a red box. Below the title, there's a description box. The form contains four questions, each marked with a red asterisk to indicate it is required:

- 1. Requester name ***: A text input field with the placeholder 'Enter your answer'.
- 2. Course ***: A choice question with three radio button options: 'Power Automate Basics', 'Approval Workflows', and 'Excel Reporting'.
- 3. Preferred date ***: A date input field with the placeholder 'Please input date (M/d/yyyy)' and a calendar icon.
- 4. Business reason ***: A text input field with the placeholder 'Enter your answer'.

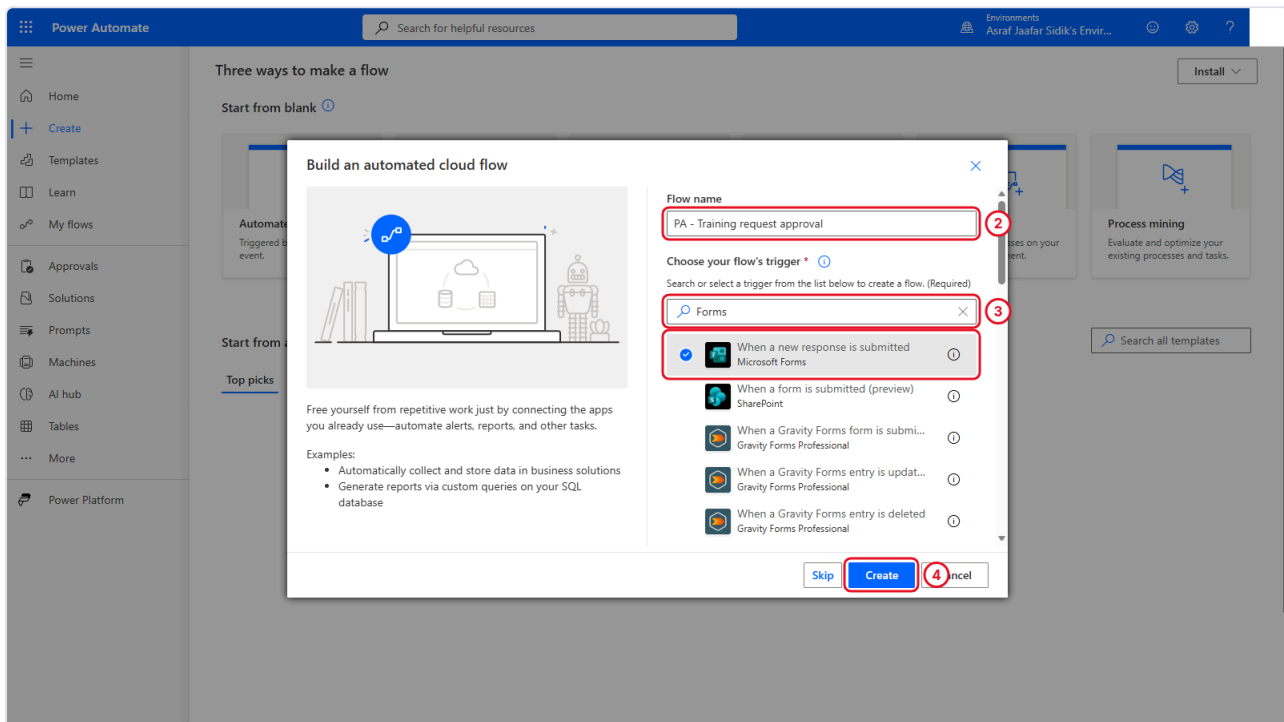
At the bottom of the form, there are buttons for 'Add new question' and 'Add with Copilot'. The background of the form is a decorative image of a modern interior with a green wall and a white sofa.

Four required questions. The red asterisk marks each one as required. Your form uses the plain default theme. The background picture is optional.

Checkpoint: All four questions show the red required marker. A missing answer gives the approver an incomplete card.

6.2 Create the Automated Flow

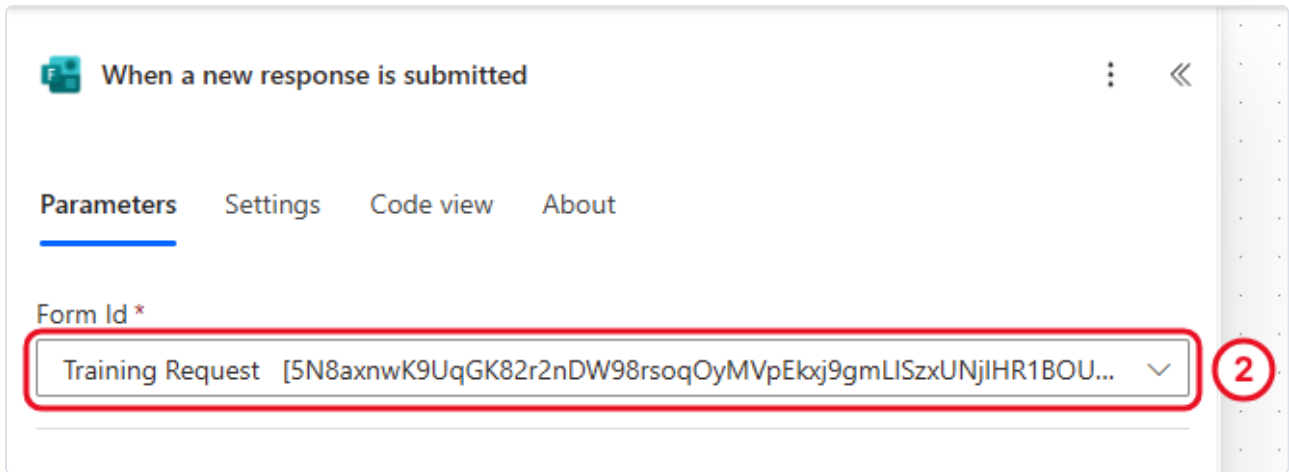
1. In Power Automate, select **Create** on the left, then the **Automated cloud flow** tile.
2. In **Flow name**, enter PA - Training request approval.
3. In the trigger search box, type Forms and select **When a new response is submitted** (Microsoft Forms).
4. Select **Create**.



The Forms trigger starts the flow whenever the form gets a response.

6.3 Select the Form

1. On the canvas, select the **When a new response is submitted** card. Its settings open on the left.
2. Open the **Form Id** dropdown and select **Training Request**.



When a new response is submitted

Parameters Settings Code view About

Form Id *

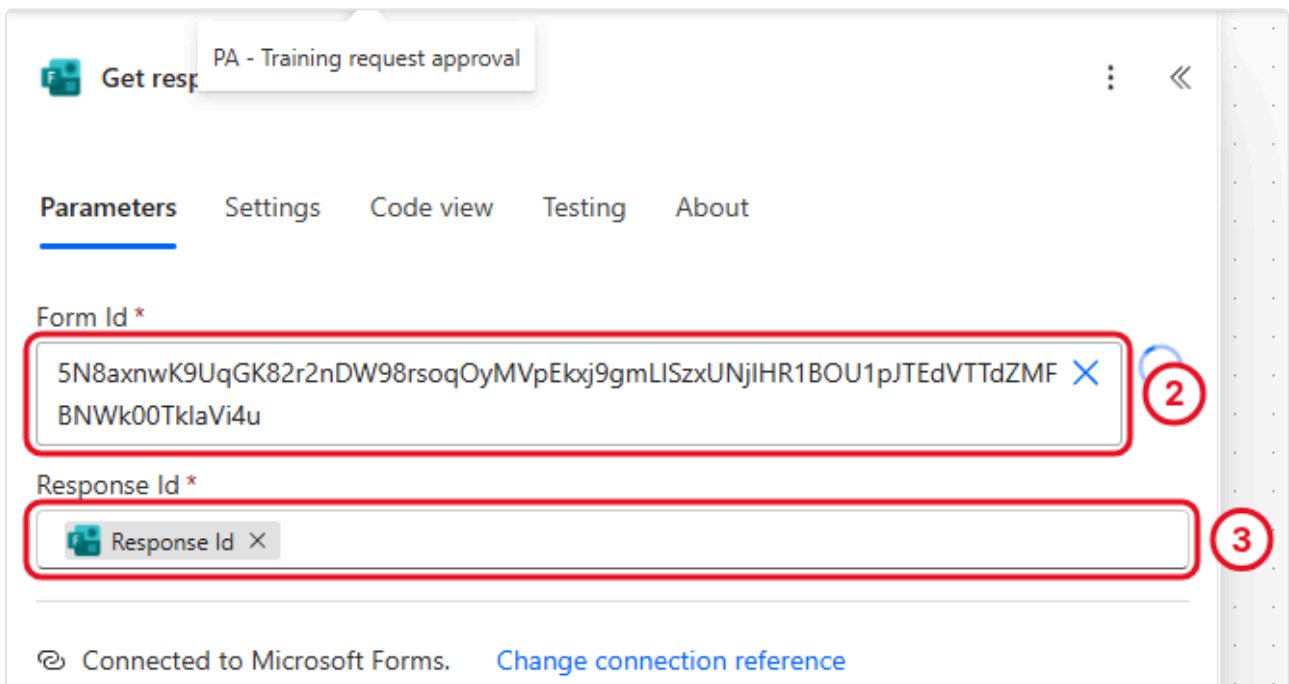
Training Request [5N8axnwK9UqGK82r2nDW98rsoqOyMVpEkxj9gmLISzxUNjIHR1BOU...]

After saving, Form Id shows the form's long internal ID instead of its name. That's normal.

If the form isn't listed, check that you own or can access it, refresh the list, and confirm the Forms connection uses the right account.

6.4 Retrieve the Answers

1. On the canvas, select the **+** below **When a new response is submitted**, then **Add an action**. Search for **Get response details** and select it under **Microsoft Forms**.
2. Open the **Form Id** dropdown and select **Training Request** again.
3. Click into **Response Id**, select the lightning bolt icon beside it, and pick **Response Id** under **When a new response is submitted**.



Get response details

PA - Training request approval

Parameters Settings Code view Testing About

Form Id *

5N8axnwK9UqGK82r2nDW98rsoqOyMVpEkxj9gmLISzxUNjIHR1BOU1pJTEdVTTdZMF BNWk00TklaVi4u

Response Id *

Response Id

Connected to Microsoft Forms. [Change connection reference](#)

The Response Id token links this step to the exact submission that started the flow. As in 6.3, Form Id shows the form's internal ID after saving.

The trigger only says a response exists. Get response details returns the individual answers so later actions can use them.

6.5 Request the Decision

1. On the canvas, select the + below **Get response details**, then **Add an action**. Search for **Start and wait for an approval** and select it under **Approvals**.
2. Set **Approval type** to **Approve/Reject - First to respond**. If your list doesn't have it, choose **Basic**.
3. In **Title**, enter **[PA TRAINING] Training request**.
4. In **Assigned to**, type the trainer-approved approver's name or email and select them from the list.
5. In **Details**, paste the details template from the copy-paste page. Put the cursor after each label, select the lightning bolt icon, and under *Get response details* pick the matching answer: **Requester name**, **Course**, **Preferred date**, **Business reason**. If the list is cut short, select **See more** or type the question name in the picker's search box.

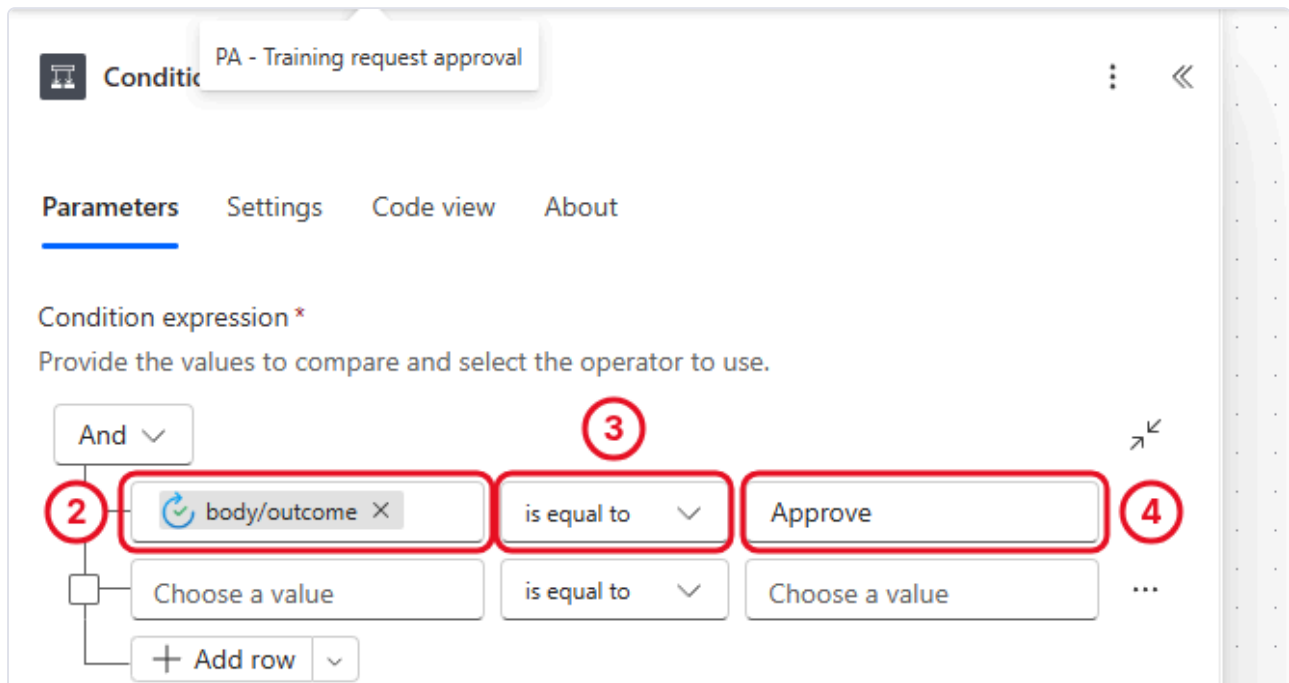
Each label in Details is followed by its answer. In the picker the answers show their question names; after saving they show as codes like `body/r96281c9...`. The screenshot uses Basic.

Checkpoint: Use the values from **Get response details**, not similarly named values from another action.

6.6 Branch on the Outcome

1. On the canvas, select the + below **Start and wait for an approval**, then **Add an action**. Search for **Condition** and select it under **Control**.

2. Click into the left **Choose a value** box, select the lightning bolt icon, and pick **Outcome** under *Start and wait for an approval*. (After saving it shows as `body/outcome`.)
3. Leave the middle box on **is equal to**.
4. In the right box, type `Approve`.
5. If an empty second row appears, select its ... and then **Delete**.



An Approve outcome goes to True. Anything else goes to False.

Key point: The text must be exactly `Approve`. A typo or a trailing space sends every approval down the False branch.

6.7 Prepare the Approved Message

1. On the canvas, select the **+** inside the green **True** box, then **Add an action**. Search for `Post message in a chat or channel` and select it under **Microsoft Teams**.
2. Open the **Post as** dropdown and select **Flow bot**. Open **Post in** and select **Chat with Flow bot**.
3. In **Recipient**, type the trainer-approved Teams account and select it from the list.
4. Click into **Message** and type `Training request approved for` (with the space at the end). Select the lightning bolt icon and pick **Course** under *Get response details*.

The *True* branch posts the approved message. After saving, the *Course* token shows as a code like `body/rc6d429a1...`. That's normal. Replace `approver@example.com` with your trainer-approved account.

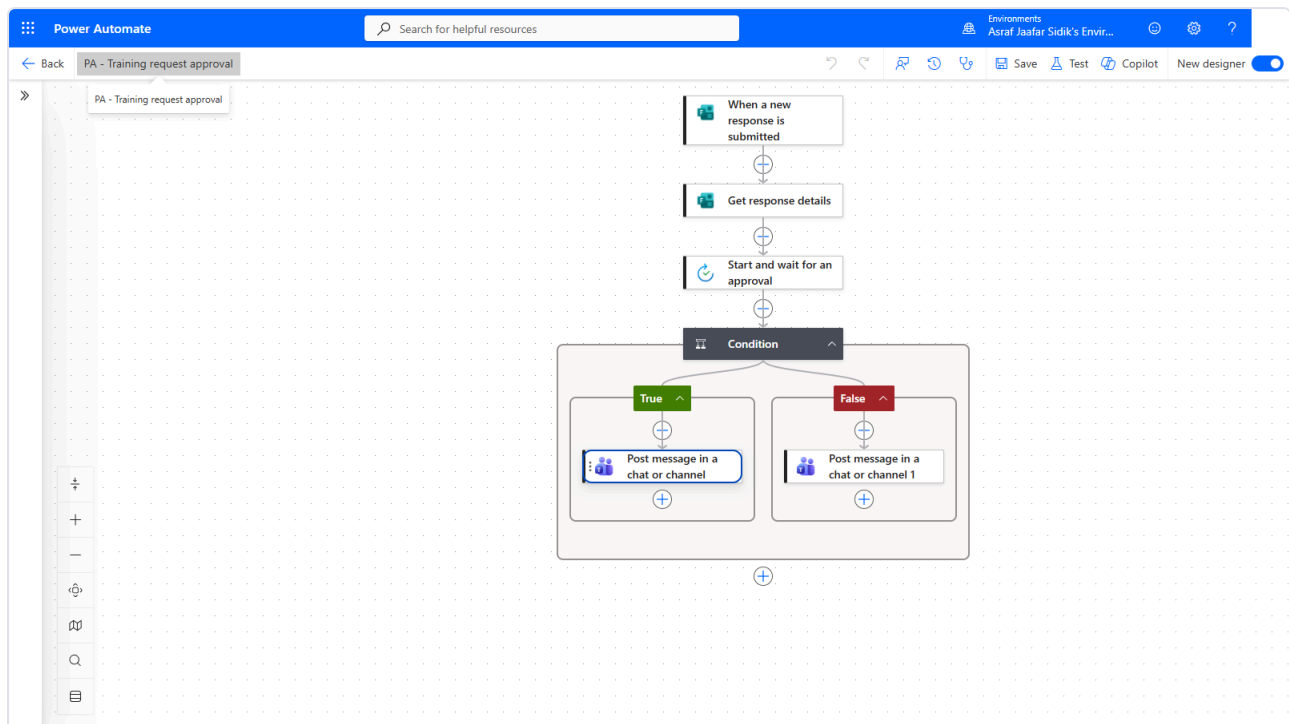
6.8 Prepare the Rejected Message

1. On the canvas, select the **+** inside the red **False** box, then **Add an action**, and add a second **Post message in a chat or channel** (Microsoft Teams).
2. Use the same **Post as**, **Post in** and **Recipient** as section 6.7.
3. In **Message**, type `Training request rejected for` and insert **Course** from *Get response details* with the lightning bolt, as in 6.7.

A production process would also handle outcomes like cancelled or timed-out approvals. This exercise sticks to the two standard Approve and Reject buttons.

6.9 Save and Check the Flow

1. On the toolbar at the top, select **Save**, then **Flow checker**. Fix every error before testing.
2. Confirm the canvas: three steps before the Condition, and one Teams action in each branch.



Three steps, then one Teams message per branch.

Trainer-Controlled Test

Testing creates an approval request and may post to Teams. Replace both `approver@example.com` placeholders first and confirm the recipient is ready. Then, with the trainer's go-ahead:

1. Submit one fictional Training Request response.
2. Approve it, and check the True branch and the approved Teams message.
3. Submit a second fictional response.
4. Reject it, and check the False branch and the rejected Teams message.

After each test, open run history and confirm the right branch succeeded while the other was skipped.

Independent Practice

Without running the flow, change the approval title so it also includes the **Course** value. Which action supplies that value, and why can't the trigger supply it on its own?

Troubleshooting

Symptom	What to check
Training Request isn't in Form Id	Form ownership or access, the Forms connection account, and a refreshed list
The approval has no request details	Get response details Form Id, the Response Id token and the Details tokens

The flow waits forever	The assigned approver, the notification, and whether a decision was made
An approval goes down False	The Outcome token, <i>is equal to</i> , and the exact text Approve
Teams can't resolve the recipient	Replace the placeholder with a trainer-approved account
No Teams message appears	The branch taken, Teams connection, Post in setting, recipient and the action result
Flow checker is clean but you can't test	Flow checker checks structure only, not placeholder recipients or policy

Lesson Summary

A Forms trigger starts the flow with a response ID, Get response details turns it into usable answers, Start and wait for an approval records a human decision, and a Condition routes the request to the matching branch. The flow is ready for a trainer-controlled test once approved recipients replace the placeholders.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

Form name (text, section 6.1)

```
Training Request
```

Form description (text, section 6.1)

```
Submit a fictional training request for the Power Automate approval exercise.
```

Course choices (text, section 6.1)

```
Power Automate Basics  
Approval Workflows  
Excel Reporting
```

Flow name (text, section 6.2)

```
PA - Training request approval
```

Approval title (text, section 6.5)

```
[PA TRAINING] Training request
```

Approval details template (text, section 6.5)

Paste into **Details**. Put the cursor after each label, select the lightning bolt icon, and pick the matching answer under *Get response details*.

```
**Requester:**  
**Course:**  
**Preferred Date:**  
**Business reason:**
```

The ****** marks make the labels bold on the approval card.

Condition right value (text, section 6.6)

```
Approve
```

Teams messages (text, sections 6.7 and 6.8)

Type the text, then insert **Course** from Get response details after it.

Training request approved for

Training request rejected for

07

CHAPTER 7

Capstone: Weekly Reminders and Summary

Chapter 5 sent a reminder to every row, including people who had already finished. The capstone turns that into a process a coordinator would actually use: skip completed training, limit the window to the next 14 days, count what was sent, and handle the week when nothing matches.

Copy-paste values: action names, the variable name and summary text are on the copy-paste page.

Estimated time: 90 minutes

Your result: A copy of the weekly reminder flow that keeps only upcoming incomplete records, counts the reminders, and follows a clear summary or no-records branch.

What You Will Learn

- Filter a list before a loop with **Filter array**
- Work out a future date with **Get future time**
- Use a condition with two rules that must both be true
- Count processed items with an integer variable
- Branch after a loop, including a week with no matching records

How the Flow Works

Recurrence > Current time > Convert time zone > List rows > Filter array > Get future time > Initialize variable > Apply to each (Condition > Send reminder > Increment variable) > Condition > Summary or no-records message

Filter array removes completed rows before the loop starts. Inside the loop, a condition checks that the session falls in the next 14 days. A counter goes up once per reminder. After the loop, a second condition checks the counter and picks what to tell the coordinator.

Before You Begin

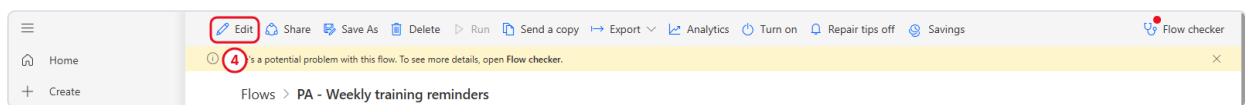
Complete Chapter 5.

Important: The sample register uses `example.com` addresses, which can't receive mail. Keep the capstone turned off and don't test it until the trainer has replaced every recipient with an approved destination.

Note: The sample sessions run from 21 September to 5 October 2026. If your class is on a different date, ask your trainer to shift the SessionDate values so some fall in the next 14 days.

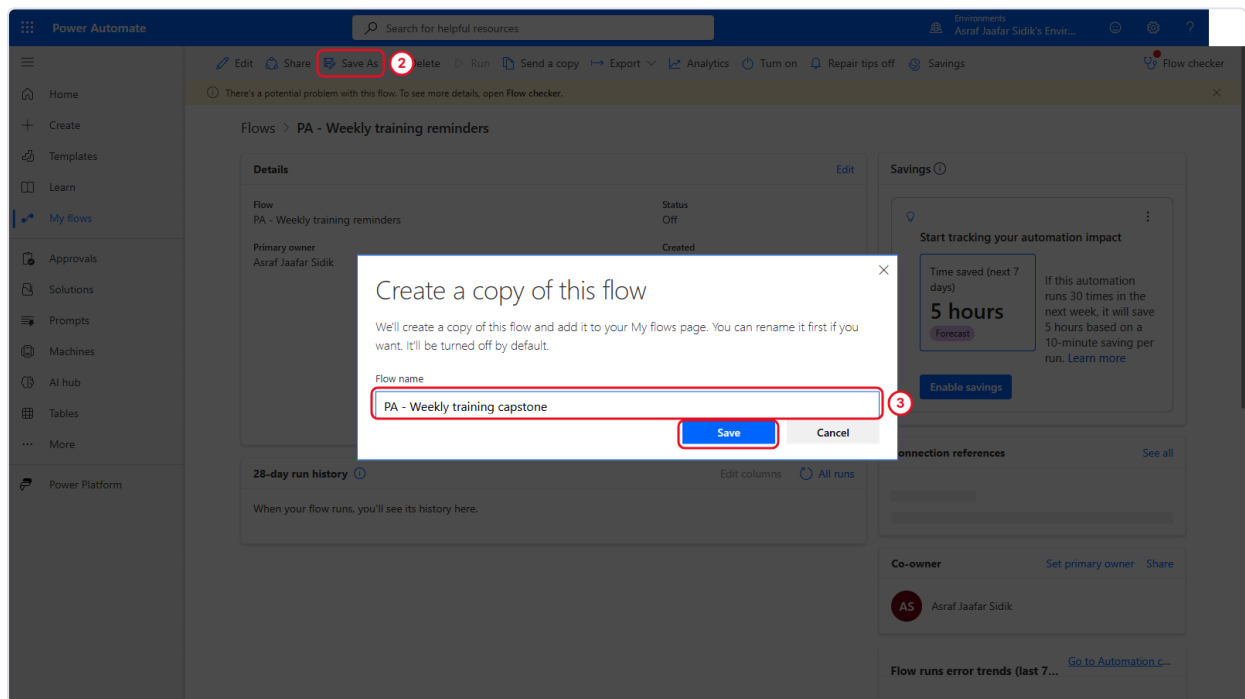
7.1 Make a Safe Capstone Copy

1. In the left navigation, select **My flows**. Select the flow's name `PA - Weekly training reminders` (not the pencil) to open its details page.



The original flow's details page. Save As and Edit are in this toolbar.

2. In the toolbar at the top, select **Save As** (in the ... menu if it isn't visible).
3. In **Flow name**, enter `PA - Weekly training capstone` and select **Save**. The copy starts turned off.



Working on a copy keeps your Chapter 5 flow safe.

4. Go back to **My flows** and select the capstone copy's name (not the pencil). Its details page opens. Select **Edit** at the left of the toolbar to open it in the designer.

Checkpoint: The name at the top of the designer reads `PA - Weekly training capstone` before you change anything.

7.2 Remove Completed Rows with Filter Array

1. On the canvas, select the **+** between **List rows present in a table** and **Apply to each**, then **Add an action**.
2. In the search box at the top of the panel, type `Filter array`. Under the **Data Operation** heading, select **Filter array**. Its settings open on the left.
3. Rename it: select the action's title at the top of the panel and type `Filter out completed training`.
4. Click into the **From** box. Select the lightning bolt icon beside it (or type `/` and choose **Insert dynamic content**), then select **body/value** under *List rows present in a table*. (It may be labelled **value**. If you can't see it, select **See more** or type `value` in the picker's search box.)
5. Under **Filter Query**, click the left box, open the lightning bolt, and select **Status** under *List rows present in a table*. Type `Status` in the picker's search box if it isn't listed.
6. Open the middle dropdown and select **is not equal to**.
7. Click the right box and type `Completed`.

The screenshot shows the configuration panel for the 'Filter array' action. The title is 'Filter out completed training' (3). The 'From' field is set to 'body/value' (4). The 'Filter Query' section is configured with 'Status' (5), 'is not equal to' (6), and 'Completed' (7). There is also an 'Edit in advanced mode' link (5).

Filter array keeps only the rows where the rule is true. Completed rows are dropped before the loop.

Tip: The comparison is exact, including capital letters. `Completed` and `completed` count as different. The sample data uses `Completed`.

7.3 Work Out the End of the Window

1. On the canvas, select the **+** below **Filter out completed training**, then **Add an action**.
2. Search for `Get future time` and select it under **Date Time**.
3. In **Interval**, enter `14`. Open the **Time unit** dropdown and select **Day**.

Get future time

Parameters Settings Code view About

Interval *

14

Time unit *

Day

2

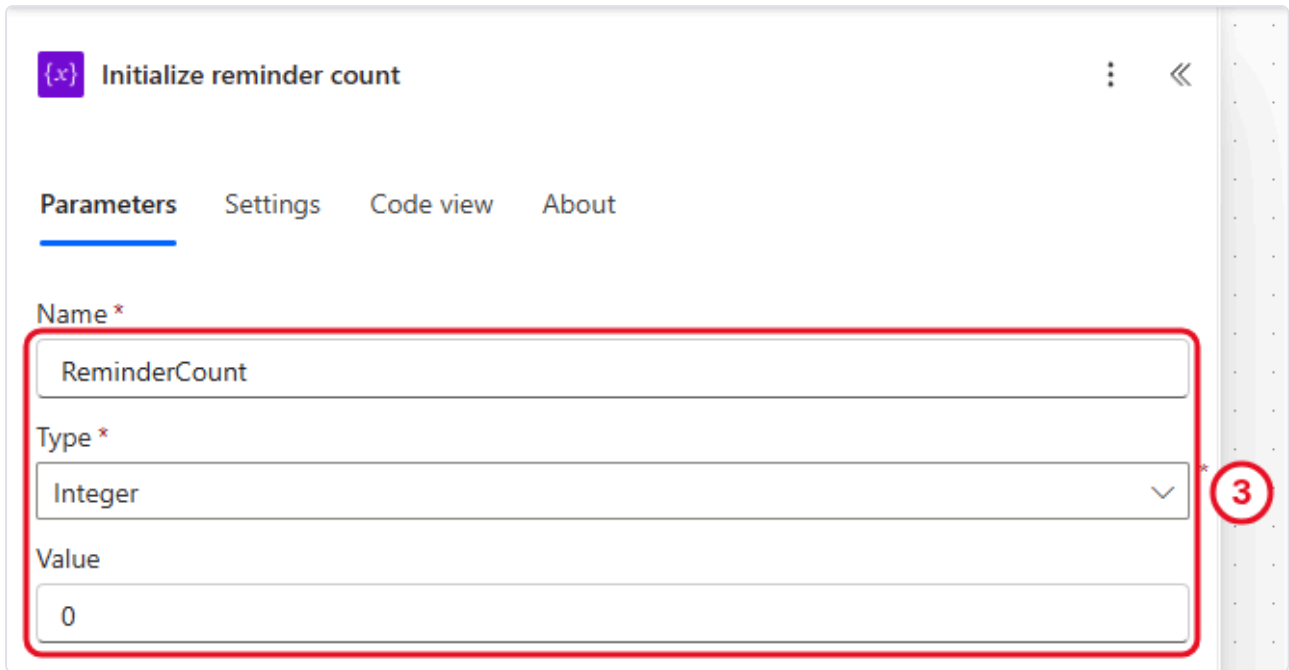
This gives you "14 days from now" without any expression.

You now have both ends of the window: **Current time** (already in the flow from Chapter 5) and **Future time** from this step.

7.4 Start the Reminder Counter

1. On the canvas, select the + below **Get future time**, then **Add an action**. Search for `Initialize variable` and select it under **Variables**.
2. Rename it: select the title at the top of the panel and type `Initialize reminder count`.
3. Fill in the boxes below. For **Type**, open the dropdown and select **Integer**.

Field	Value
Name	ReminderCount
Type	Integer
Value	0



Initialize reminder count

Parameters Settings Code view About

Name *

ReminderCount

Type *

Integer

Value

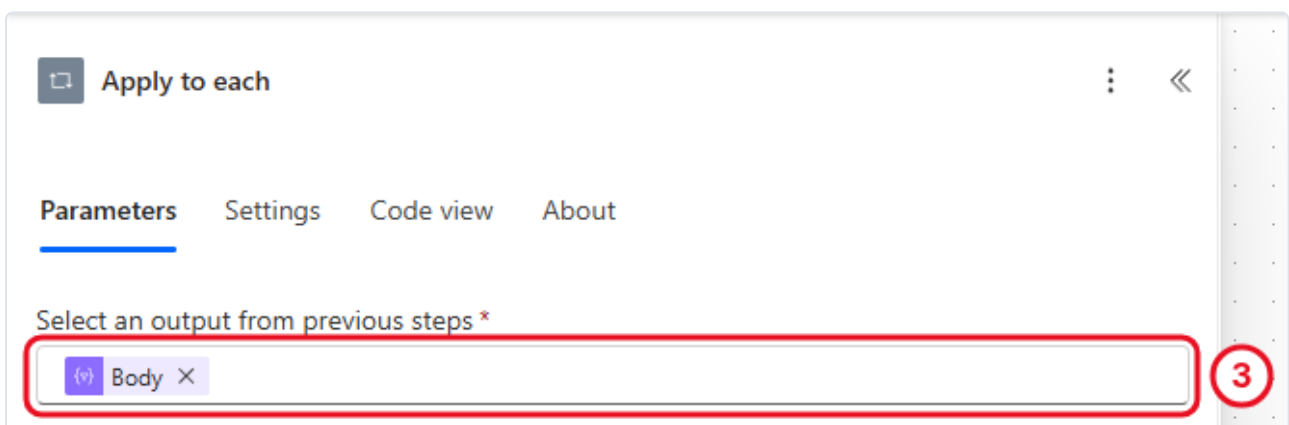
0

A variable is a named box that can change while the flow runs. This one starts at zero.

Key point: Variables must be created at the top level of the flow, never inside a loop or a condition. That's why this step comes before Apply to each.

7.5 Loop Through the Remaining Rows

1. On the canvas, select the **Apply to each** card to open its panel.
2. In **Select an output from previous steps**, select the **x** on the current token to remove it.
3. Click into the empty box, open the lightning bolt, and select **Body** under *Filter out completed training*.



Apply to each

Parameters Settings Code view About

Select an output from previous steps *

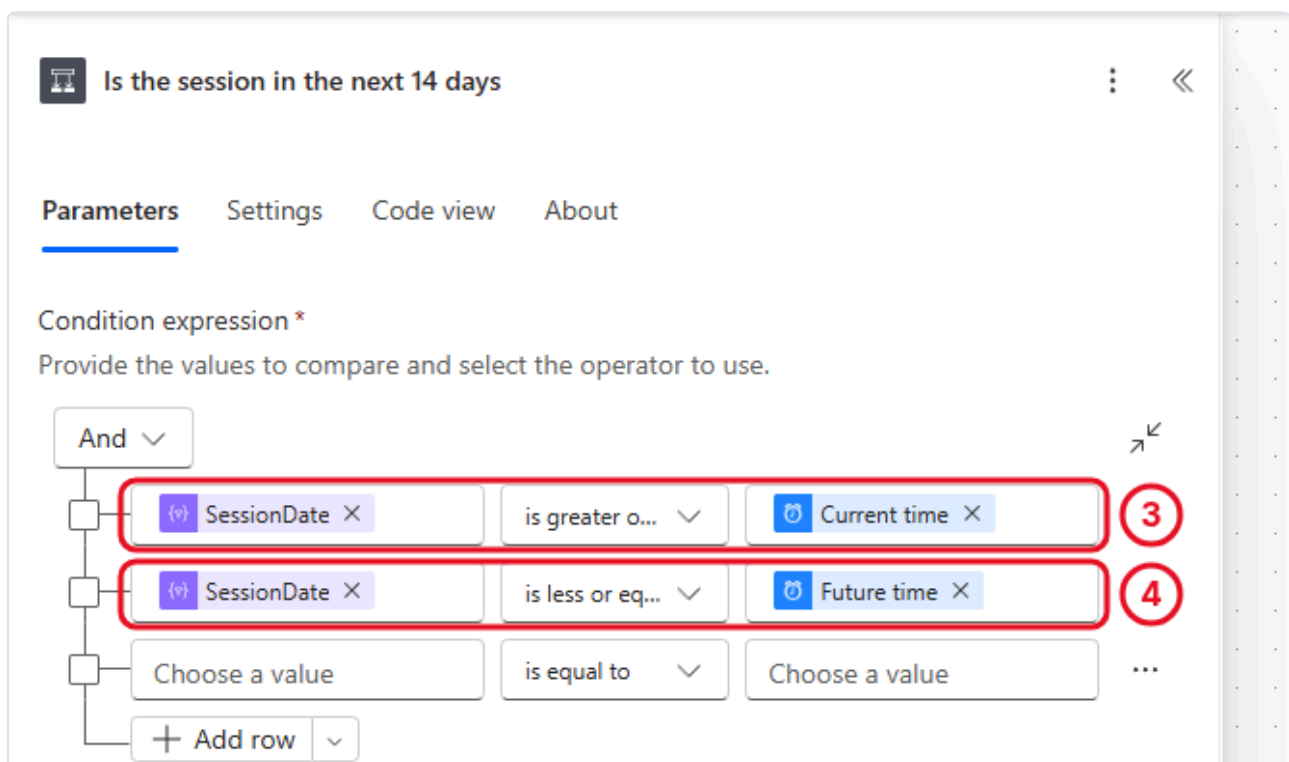
Body

The loop now only sees rows that aren't completed.

Key point: Don't leave the original Excel value here. That brings back the Chapter 5 behaviour and includes completed rows.

7.6 Check the Date Window

1. On the canvas, inside the **Apply to each** box, select the **+** just above **Send an email (V2)**, then **Add an action**. Search for **Condition** and select it under **Control**.
2. Rename it: select the title at the top of the panel and type **Is the session in the next 14 days**.
3. First row: click the left **Choose a value** box, type **/** and choose **Insert expression**. Paste `items('Apply_to_each')?['SessionDate']` and select **Add**. (See the box below for why.) The box now shows a token labelled **SessionDate**. Open the middle dropdown and select **is greater or equal to**. Click the right box, open the lightning bolt, and select **Current time** under *Current time*.
4. Select **Add row** below the rows. Second row: insert the same expression on the left, select **is less or equal to** in the middle, and on the right pick **Future time** under *Get future time*.
5. If an empty extra row appears below your two rows, select its ... and then **Delete**.
6. Check the dropdown above the rows reads **And**, so both rules must be true.
7. Drag the **Send an email (V2)** card that now sits below the condition and drop it on the **+** inside the green **True** box. (Or delete it and add it again inside True with the same settings.)



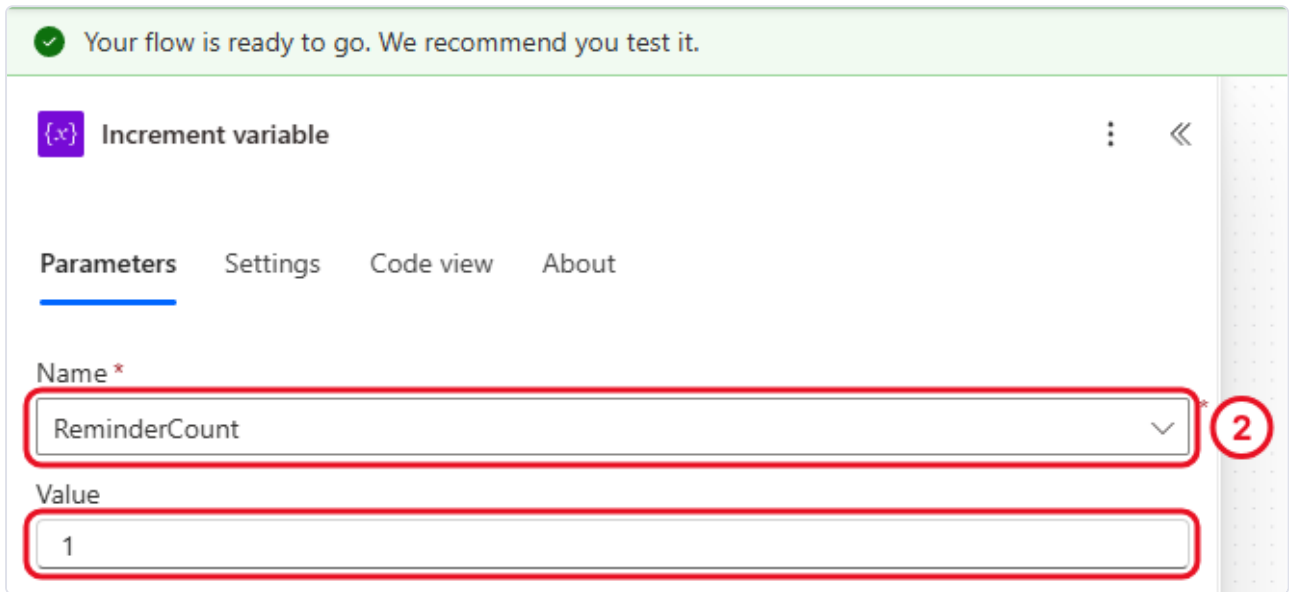
Two rules joined by AND.

Key point: why an expression here? Once the loop runs over Filter array's **Body**, the dynamic content picker no longer lists the Excel columns. If you pick **SessionDate** from *List rows present in a table* anyway, the designer wraps the condition in a second, unwanted loop. The expression `items('Apply_to_each')?['SessionDate']` simply means "the SessionDate of the row this loop is on right now". It's on the copy-paste page. The email tokens you added in Chapter 5 keep working, because they already point at the loop's current row.

This works because the Excel dates are in ISO 8601 format (set in Chapter 5), the same format as Current time and Future time. Same format means they compare correctly.

7.7 Count Each Reminder

1. On the canvas, inside the green **True** box, select the **+** below **Send an email (V2)**, then **Add an action**. Search for **Increment variable** and select it under **Variables**.
2. Open the **Name** dropdown and select **ReminderCount**. In **Value**, enter **1**.



✓ Your flow is ready to go. We recommend you test it.

Increment variable

Parameters Settings Code view About

Name *

ReminderCount

Value

1

2

Each reminder adds one to the counter.

Keep the increment *after* the email, so the count only includes reminders that reached the email step.

7.8 Decide Whether a Summary Is Needed

1. On the canvas, select the **+** below the **Apply to each** box (outside it, not one of the **+** inside the loop), then **Add an action**. Search for **Condition** and select it under **Control**.
2. Rename it **If reminders were prepared**.
3. Click the left **Choose a value** box, select the lightning bolt (or type **/** and choose **Insert dynamic content**), and pick **ReminderCount** under **Variables**.
4. Open the middle dropdown and select **is greater than**. Click the right box and type **0**.
5. If an empty extra row appears, select its **...** and then **Delete**.

✔ Your flow is ready to go. We recommend you test it.

If reminders were prepared

Parameters

Settings

Code view

About

Condition expression *

Provide the values to compare and select the operator to use.

And

ReminderCount

is greater t...

0

Choose a value

is equal to

Choose a value

+ Add row

3

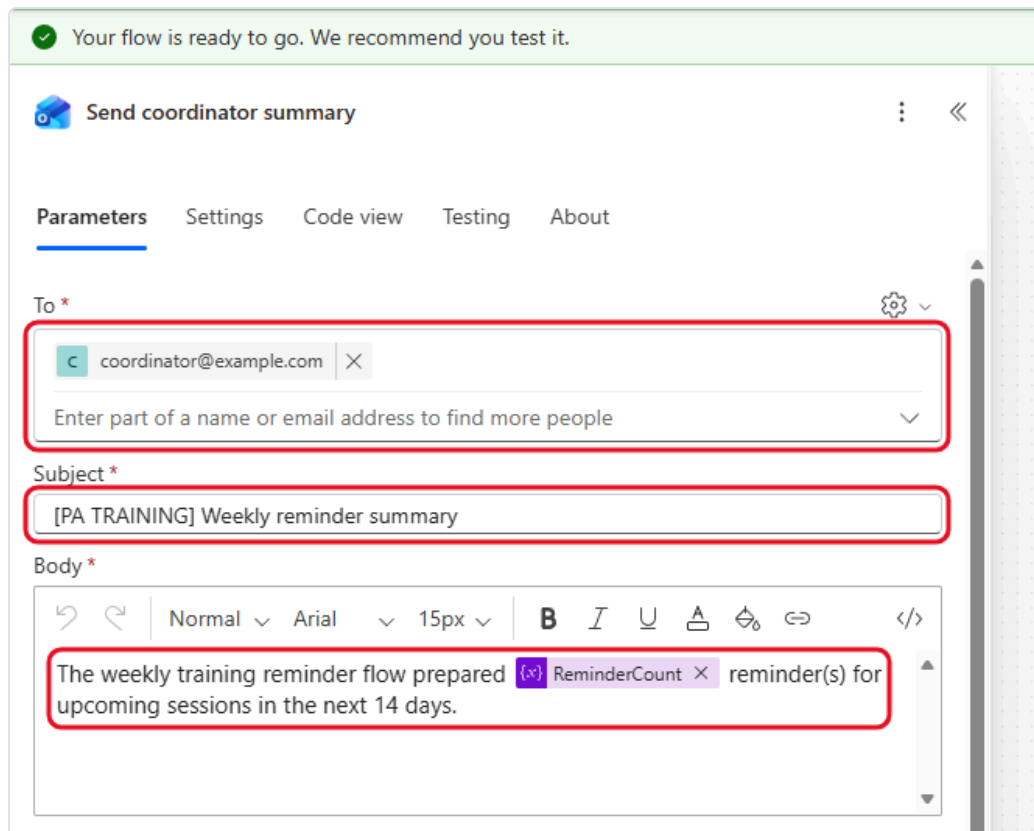
A positive count goes to True. Zero goes to False.

7.9 Prepare the Coordinator Summary

On the canvas, inside the green **True** box under **If reminders were prepared**, select the **+**, then **Add an action**. Search for `Send an email`, select **Send an email (V2)** under **Office 365 Outlook**, and rename it `Send coordinator summary`:

Field	Value
To	Type <code>coordinator@example.com</code> , then select it from the list (it shows as a custom value)
Subject	<code>[PA TRAINING] Weekly reminder summary</code>
Body	Type <code>The weekly training reminder flow prepared</code> , then select the lightning bolt (or type <code>/</code> and choose Insert dynamic content) and pick ReminderCount under <i>Variables</i> . Then type <code>reminder(s) for upcoming sessions in the next 14 days.</code>

85



The variable drops into the email like any other dynamic content.

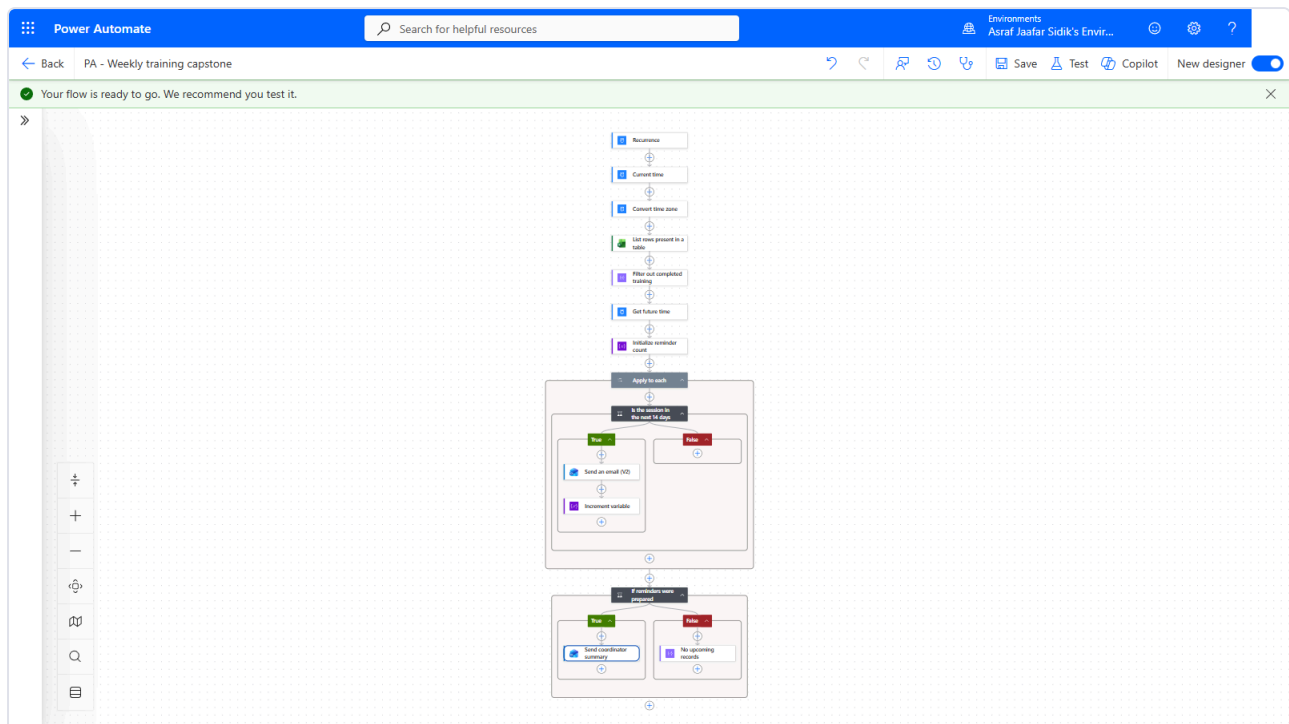
7.10 Handle a Week with No Matching Rows

On the canvas, inside the red **False** box, select the **+**, then **Add an action**. Search for **Compose** and select it under **Data Operation**. Rename it **No upcoming records**, and type **No upcoming training records were found for the next 14 days.** in **Inputs**.

That records a clear outcome in run history without sending an unnecessary email.

7.11 Save and Check the Capstone

1. Select **Save** in the toolbar at the top right. Then select **Flow checker** in the same toolbar and confirm zero errors and zero warnings.
2. Close Flow checker. In the column of buttons at the bottom left of the canvas, select **Zoom view to fit**. Check the shape: filter before the loop, date check and count inside the loop, decision after the loop.



Filter before, check and count inside, decide after.

Trainer-Controlled Test

A test can send one learner reminder per matching row plus, when the count is positive, a coordinator summary. With the sample data on 17 September 2026, four rows match (21, 22, 24 and 28 September). The trainer replaces every `example.com` address, saves the workbook, confirms the coordinator address and authorises the run first.

Afterwards, check the run history:

- Filter out completed training returned four rows
- Apply to each ran four times, and the date condition was true each time
- ReminderCount ended at 4
- The final condition took True and prepared the coordinator summary

Independent Practice

Without running the flow, change the window from 14 days to 7 by editing only **Get future time**. Which sample rows would match on 17 September 2026? Then change it back to 14.

Going further: one filter that does it all. Experienced makers often replace the filter and the date condition with a single Filter array in **advanced mode**, using an expression that also ignores capital letters and stray spaces. The expression is on the copy-paste page. You'll build flows like that in the advanced course. Optional.

Troubleshooting

Symptom	What to check
A completed row still gets a reminder	Filter array right value is exactly <code>Completed</code> , and the loop uses Filter's Body
Every date fails the condition	DateTime Format is ISO 8601 in List rows present in a table
A second Apply to each appears inside the loop	You picked SessionDate from the Excel action. Delete the extra loop and use the expression from section 7.6
No date passes the condition	The sample dates may be in the past. Ask the trainer to update SessionDate
ReminderCount stays at zero	Increment variable is inside the True branch of the date condition
Initialize variable can't be added	You're inside the loop. Variables are created at the top level
The summary runs before reminders finish	Put the final Condition below Apply to each, outside its box
Flow checker is clean but a test is unsafe	Flow checker checks structure. It doesn't approve recipients

Lesson Summary

The capstone filters out completed rows before looping, checks a date window with a two-rule condition, keeps a running count in a variable, and branches on the final count. That turns a broad weekly reminder into a controlled process with a clear result for the coordinator. Next, you'll use variables again, alongside parallel branches, in a flow that's all about you.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything in the main steps is **text**, typed or pasted straight into a field.

Flow name for the copy (text, section 7.1)

```
PA - Weekly training capstone
```

Filter array (text, section 7.2)

Action name:

```
Filter out completed training
```

Right value:

```
Completed
```

Variable (text, section 7.4)

Action name:

```
Initialize reminder count
```

Variable name:

```
ReminderCount
```

Date condition name (text, section 7.6)

```
Is the session in the next 14 days
```

Current row's session date (section 7.6)

Click the left **Choose a value** box of each condition row, type `/` and choose **Insert expression**. Paste, then select **Add**. The box shows a token labelled **SessionDate**.

```
items('Apply_to_each')['SessionDate']
```

Summary condition name (text, section 7.8)

```
If reminders were prepared
```

Coordinator summary (text, section 7.9)

Action name:

Send coordinator summary

Subject:

[PA TRAINING] Weekly reminder summary

Body: paste the first part. Select the lightning bolt (or type `/` and choose **Insert dynamic content**) and pick **ReminderCount** under *Variables*. Then paste the second part.

The weekly training reminder flow prepared

reminder(s) for upcoming sessions in the next 14 days.

No-records Compose (text, section 7.10)

Action name:

No upcoming records

Inputs:

No upcoming training records were found for the next 14 days.

Going further (optional)

One filter that does it all

Replaces both Filter array and the date condition. In Filter array, choose **Edit in advanced mode** and paste. It ignores capital letters and stray spaces in Status, and compares dates from the start of today.

```
@and(
  not(equals(toLower(trim(item()?['Status'])), 'completed')),
  greaterOrEquals(ticks(item()?['SessionDate']), ticks(startOfDay(utcNow()))),
  lessOrEquals(ticks(item()?['SessionDate']), ticks(addDays(startOfDay(utcNow()), 14)))
)
```

08

CHAPTER 8

My Morning Briefing: Variables and Parallel Branches

Every morning you check the weather, your calendar and your inbox before the day starts. This flow does all three at once and sends you one email. It only reads your own data and only emails you, so it's safe to build in a live environment.

Copy-paste values: the variable names, branch text and email body are on the copy-paste page.

Estimated time: 60 minutes

Your result: A scheduled flow that runs on weekday mornings, looks up the weather, your meetings and your important unread email in three parallel branches, and sends you one briefing email.

What You Will Learn

- Create variables with **Initialize variable** and explain why they live at the top of a flow
- Change variables with **Increment variable**, **Append to string variable** and **Set variable**
- Run independent steps side by side with **parallel branches**
- Bring parallel branches back together before a final step
- Use a variable's value in a condition and an email

How the Flow Works

```
Recurrence
> Initialize four variables
> Three parallel branches:
  Weather: Get current weather
  Meetings: Current time > Get future time > Get calendar view > loop (count + list)
  Email:    Get emails > loop (list)
> Condition: more than 4 meetings? > Set the subject
> Send me the briefing
```

The three lookups don't depend on each other. The weather doesn't need your calendar and your calendar doesn't need your inbox. When steps are independent, they can run **in parallel**, side by side, instead of waiting in a queue. The flow only moves on to the email once all three branches have finished.

Note: Your trainer has confirmed that MSN Weather, Office 365 Outlook and the Date Time actions are allowed in your environment. Nothing in this flow changes your calendar or mailbox.

8.1 Create the Scheduled Flow

1. In the left navigation, select **Create**, then **Scheduled cloud flow**.
2. In **Flow name**, enter PA - My morning briefing.
3. Set **Repeat every** to **1 Week**, tick **M T W T F**, and set the time to **8:00 AM**.
4. Select **Create**. The designer opens with a **Recurrence** card on the canvas.
5. Select the **Recurrence** card on the canvas to open its panel. If **Time zone** is blank, open the dropdown, type **Kuala** and pick **(UTC+08:00) Kuala Lumpur, Singapore**. Check that **At these hours** is **8** and the **Preview** at the bottom reads 8:00 on Monday to Friday. **Start time** and **At these minutes** are filled in for you; leave them.

Recurrence

Parameters Settings Code view About

Recurrence *

Interval * 1 Frequency * Week

Time zone (UTC+08:00) Kuala Lumpur, Singapore

Start time 2026-10-12T00:00:00.000Z

On these days Monday, Tuesday, Wednesday, Thursday, Friday

At these hours 8

At these minutes 0

Preview
Runs at 8:00 on Monday, Tuesday, Wednesday, Thursday, Friday every week

Weekday mornings at 8:00, Malaysia time.

8.2 Create the Variables

A **variable** is a named box that holds a value and can change while the flow runs. You've used one already: ReminderCount in Chapter 7. This flow uses four, of two types.

Name	Type	Starting value	What it's for
------	------	----------------	---------------

MeetingCount	Integer	0	Counts today's meetings
MeetingList	String	(leave empty)	Collects the meeting names
EmailList	String	(leave empty)	Collects the important email subjects
BriefingSubject	String	Your day ahead	The email subject. May change later

1. On the canvas, select the **+** below **Recurrence**, then **Add an action**. Search for **Initialize variable** and select it under **Variables**.
2. In **Name**, enter **MeetingCount**. Open the **Type** dropdown and select **Integer**. In **Value**, enter **0**.
3. Select the **+** below the variable you just made, then **Add an action** and search for **Initialize variable** again. Repeat until all four rows of the table are done. For the String rows, leave **Value** empty.

All four variables are created before anything else happens.

Key point: Initialize variable only works at the top level of a flow, not inside a loop, condition or branch. So every variable is created first, even ones you won't use until later.

8.3 Branch 1: The Weather

1. On the canvas, select the **+** below the last **Initialize variable** (named **Initialize variable 3**), then **Add an action**.
2. Search for **Get current weather**. Under the **MSN Weather** heading, select **Get current weather**.
3. In **Location**, type **Kuala Lumpur**. Open the **Units** dropdown and select **Metric**.

Get current weather

Parameters Settings Code view Testing About

Location *
Kuala Lumpur

Units *
Metric

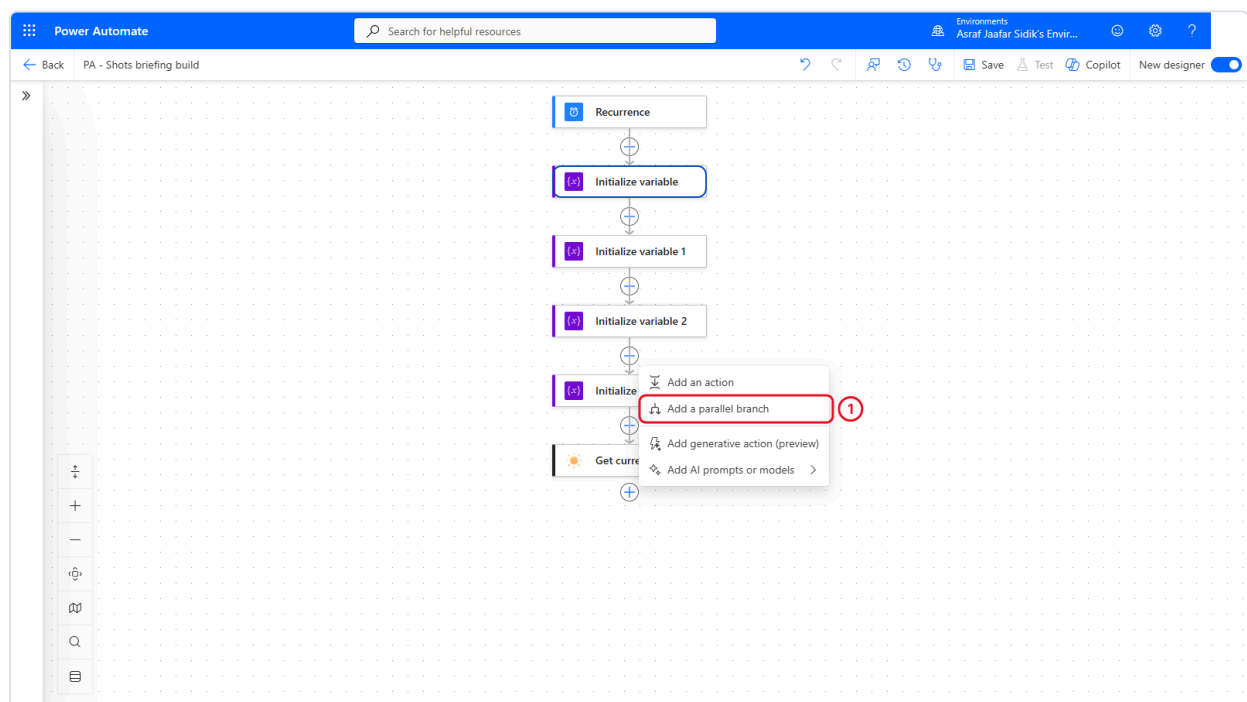
Connected to MSN Weather. [Change connection reference](#)

One action. Its values (conditions, temperature) are used in the email later.

8.4 Branch 2: Your Meetings

This branch starts *beside* the weather, not below it.

1. Select the **+** between the last **Initialize variable** and **Get current weather** (right-click it if a menu doesn't open), then choose **Add a parallel branch**.



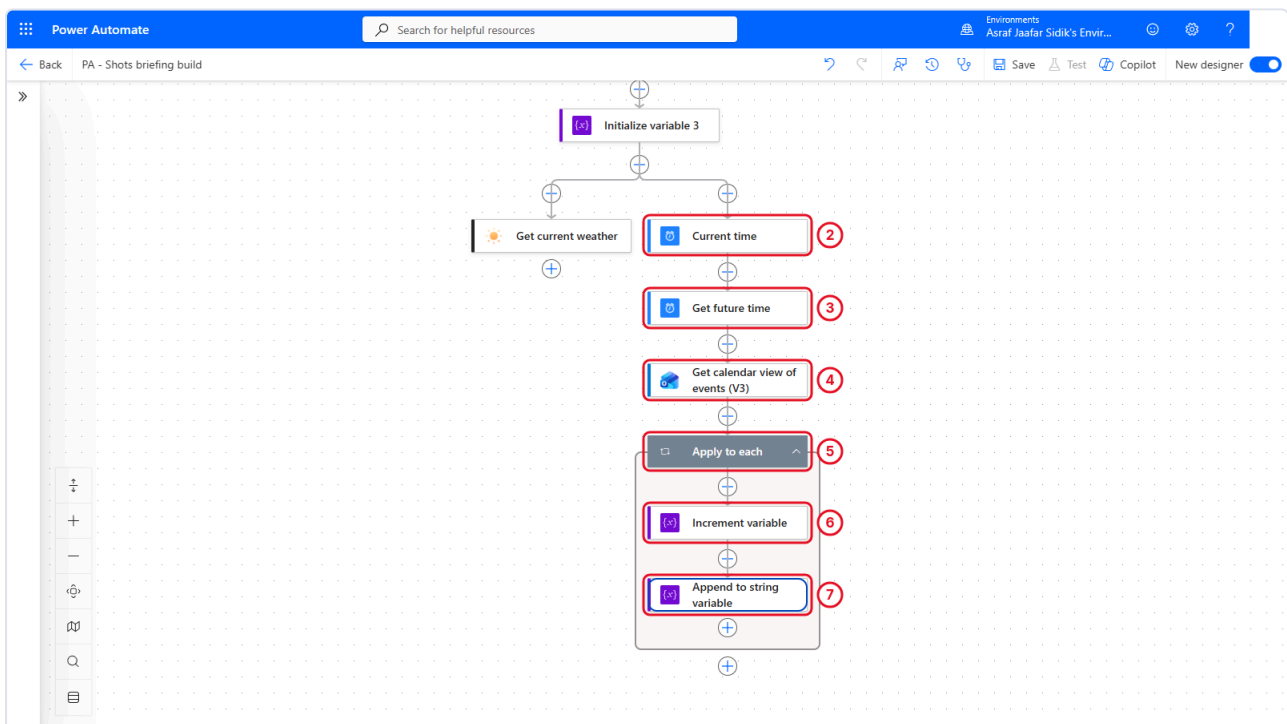
Add a parallel branch starts a new column next to the existing one.

2. The action search opens for the new branch. Search for **Current time** and select it under **Date Time**.

- On the canvas, select the **+** below **Current time**, then **Add an action**. Search for **Get future time** and select it under **Date Time**. In **Interval**, enter **12**. Open **Time unit** and select **Hour**.
- Select the **+** below **Get future time**, then **Add an action**. Search for **Get calendar view** and select **Get calendar view of events (V3)** under **Office 365 Outlook**. For **Start Time** and **End Time**, click into the box, select the lightning bolt icon beside it (or type **/** and choose **Insert dynamic content**), and pick the value:

Field	Value
Calendar Id	Open the dropdown and select Calendar
Start Time	Current time under <i>Current time</i>
End Time	Future time under <i>Get future time</i>

- Select the **+** below **Get calendar view of events (V3)**, then **Add an action**. Search for **Apply to each** and select it under **Control**. Click into **Select an output from previous steps**, open the lightning bolt, and select **body/value** under *Get calendar view of events (V3)*. That's the list of events. (It may be labelled **value**. If you can't see it, select **See more** or type **value** in the picker's search box.)
- Inside the loop, select the **+**, then **Add an action**. Search for **Increment variable** and select it under **Variables**. Open the **Name** dropdown and select **MeetingCount**. In **Value**, enter **1**.
- Inside the same loop, select the **+** below **Increment variable**, then **Add an action**. Search for **Append to string variable** and select it under **Variables**. Open the **Name** dropdown and select **MeetingList**. In **Value**, type **
•**, then open the lightning bolt and select **Subject** under *Get calendar view of events (V3)*. If Subject isn't listed, select **See more** under *Get calendar view of events (V3)*.



Each meeting adds one to the count and one line to the list.

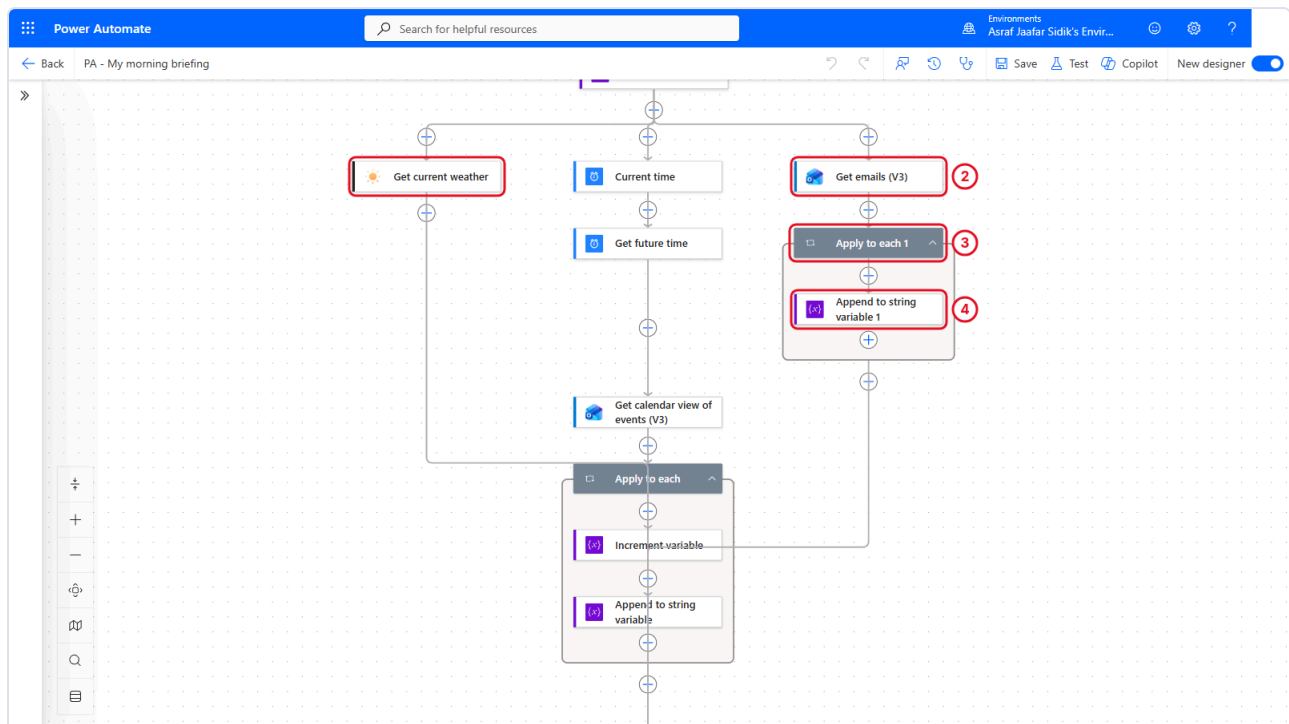
Tip: Email bodies are written in HTML, the language of web pages. `<br>` is HTML for "new line", so each meeting starts on its own line in the email.

8.5 Branch 3: Important Unread Email

1. Add a third parallel branch the same way: select the **+** on the arrow below the last **Initialize variable**, then **Add a parallel branch**.
2. Search for `Get emails` and select **Get emails (V3)** under **Office 365 Outlook**. Set the fields below. If a field isn't shown, open the **Advanced parameters** dropdown and tick it.

Field	Value
Folder	Inbox
Fetch Only Unread Messages	Yes
Importance	High
Top	5

3. On the canvas, select the **+** below **Get emails (V3)**, then **Add an action**. Search for `Apply to each` and select it under **Control**. Click into **Select an output from previous steps**, open the lightning bolt, and select **body/value** under *Get emails (V3)*.
4. Inside the loop, select the **+**, then **Add an action**. Search for `Append to string variable` and select it under **Variables**. Open the **Name** dropdown and select **EmailList**. In **Value**, type `<br>`, insert **Subject**, type `(from`, insert **From**, type `)`. Subject and From are under *Get emails (V3)* in the picker (select **See more** if they're hidden).



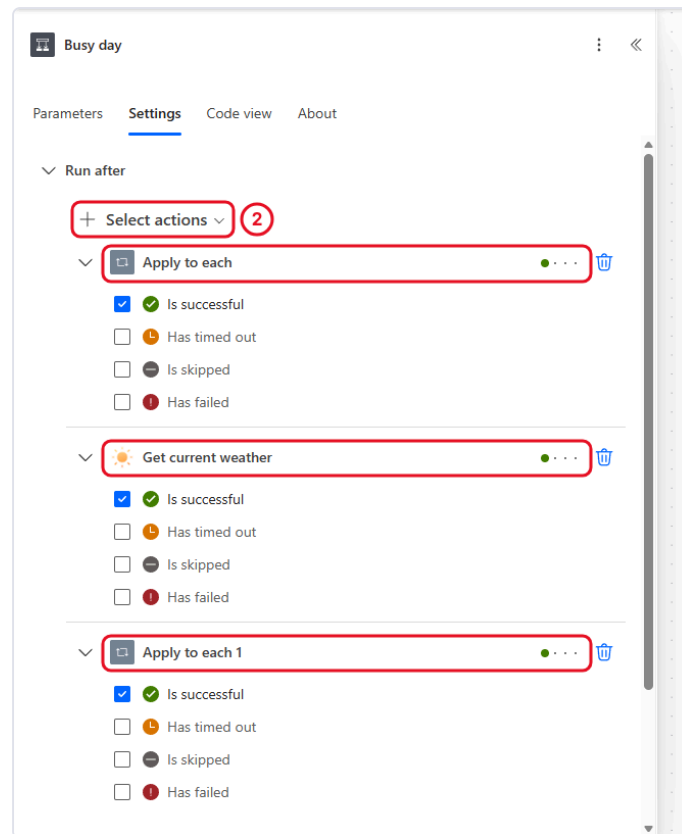
Three independent branches, running at the same time. The lines curving into the meetings loop are explained in section 8.6.

Key point: Each branch writes to its **own** variable. If all three appended to one shared variable, the sections would arrive in whatever order the branches happened to finish. Separate variables keep the email in a fixed order.

8.6 Bring the Branches Back Together

The next step must wait until all three branches have finished.

1. On the canvas, select the **+** below the meetings **Apply to each** box (outside it), then **Add an action**. Search for **Condition** and select it under **Control**. Rename it: select the title at the top of the panel and type **Busy day**.
2. Select the **Busy day** card to open its panel, then the **Settings** tab at the top of the panel. **Apply to each** is already listed under **Run after**. Open the **Select actions** dropdown and tick **Get current weather** and **Apply to each 1**.
3. Leave **Is successful** ticked for each. Check the **Run after** list shows all three actions, then select **Save**.



Run after lists one action from each branch, so Busy day waits for the slowest branch.

Note: The designer draws the lines from the weather and email branches into the side of the meetings loop, not into Busy day. That's just how it lays out the picture. The **Run after** list is what actually controls the order, so trust that, not the lines.

8.7 Change the Subject on a Busy Day

1. In the **Busy day** panel, select the **Parameters** tab. Click the left **Choose a value** box, select the lightning bolt (or type `/` and choose **Insert dynamic content**), and pick **MeetingCount** under **Variables**. Open the middle dropdown and select **is greater than**. Click the right box and type `4`. If an empty extra row appears, select its ... and then **Delete**.
2. On the canvas, inside the green **True** box, select the **+**, then **Add an action**. Search for **Set variable** and select it under **Variables**. Open the **Name** dropdown and select **BriefingSubject**. In **Value**, type `Busy day ahead:` , insert **MeetingCount** (under **Variables**), then type `meetings`.
3. Leave **False** empty. The subject keeps its starting value, *Your day ahead*.

Busy day

Parameters Settings Code view About

Condition expression *

Provide the values to compare and select the operator to use.

And ▾

{x} MeetingCount × is greater t... ▾ 4

Choose a value is equal to ▾ Choose a value

+ Add row ▾

1

Set variable replaces a variable's value. Append adds to it. Increment adds to a number.

8.8 Send the Briefing

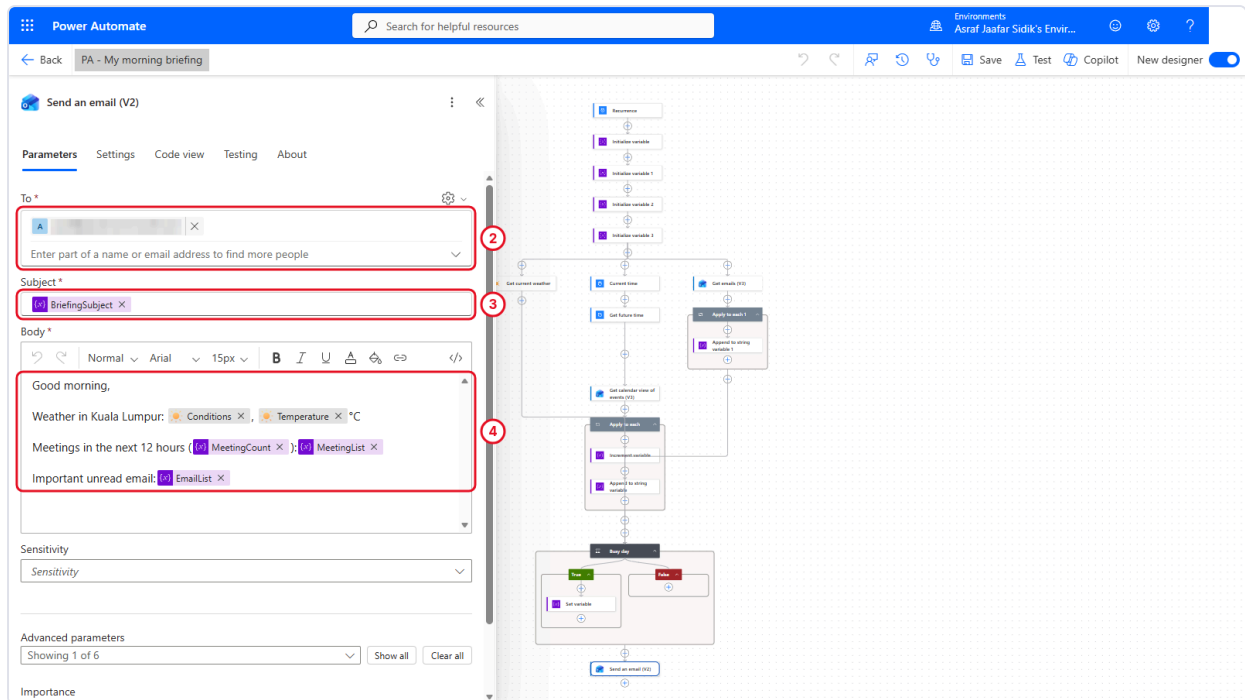
1. On the canvas, select the **+** below the **Busy day** box (outside it), then **Add an action**. Search for **Send an email**, then select **Send an email (V2)** under **Office 365 Outlook**.
2. **To:** start typing your name or email address, then select yourself from the list that appears.
3. **Subject:** click into the box, select the lightning bolt, and pick **BriefingSubject** under *Variables*.
4. **Body:** type the text below and insert the tokens shown in brackets. To insert a token, place the cursor where it goes, then select the lightning bolt (or type **/** and choose **Insert dynamic content**). **Conditions** and **Temperature** are under *Get current weather* (select **See more** if they're hidden). The rest are under *Variables*.

Good morning,

Weather in Kuala Lumpur: **[Conditions]**, **[Temperature]**°C

Meetings in the next 12 hours (**[MeetingCount]**):**[MeetingList]**

Important unread email:**[EmailList]**



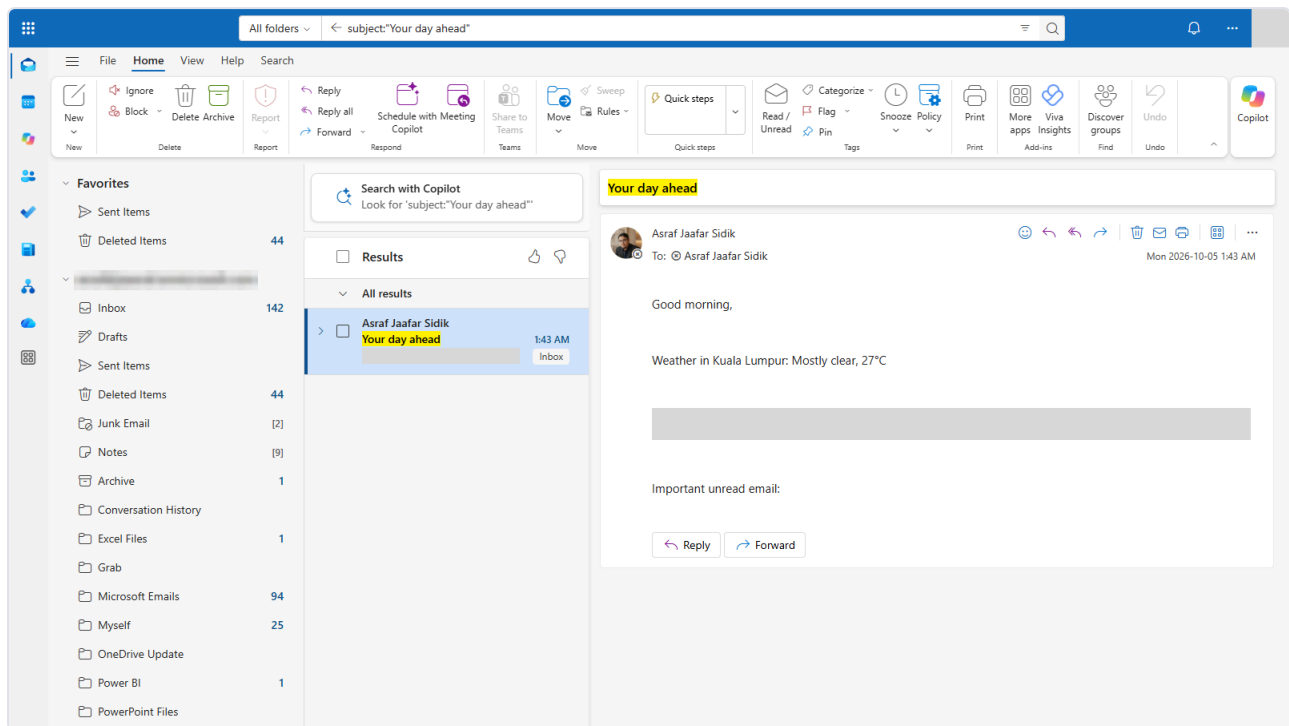
Variables drop into the email like any other dynamic content.

5. Select **Save** in the toolbar at the top right, then **Flow checker** in the same toolbar. Fix anything it reports.

8.9 Test It

This flow only emails you, so you can test it yourself.

1. In the toolbar at the top right, select **Test**, choose **Manually**, select **Test**, then **Run flow**.
2. Open the run. Notice that the three branches started at almost the same moment.
3. Select **Busy day** and check which branch it took. Select a **Set variable** or **Increment variable** step to see the value it stored.
4. In Outlook, find the email with the subject **Your day ahead** (or **Busy day ahead...**) in your Inbox. If it isn't near the top, type the subject in the search box at the top.



Your briefing. Each section came from a different branch.

If you have no meetings or no important unread email, those lists are simply empty and the count reads 0. That's correct.

Independent Practice

Add a fourth parallel branch that counts up to 25 unread emails of any importance. Create an Integer variable **UnreadCount** at the top, use **Get emails (V3)** with **Fetch Only Unread Messages** set to Yes and **Top** set to 25, and increment the variable for each email. Add a line to the briefing: *Unread email: [UnreadCount]*.

Going further: skip empty sections. Use a condition with **EmailList is equal to** (leave the right side empty) to set EmailList to `<br>Nothing urgent. Enjoy your morning. when there's no important email. Optional.`

Troubleshooting

Symptom	What to check
Initialize variable isn't offered	You're inside a branch or loop. Add it below Recurrence
Get current weather is blocked	Your environment may block MSN Weather. Ask the trainer, or skip branch 1
The meeting list is empty but you have meetings	Start Time is Current time , End Time is Future time , Calendar Id is Calendar

Meetings and email appear on one line

The `<br>` is missing at the start of the Append value

The email sends before a branch finishes

Busy day's **Run after** must include the last action of every branch

The subject never changes

Set variable is in the True branch, and the condition uses MeetingCount *is greater than 4*

Times look eight hours off

The meeting window uses UTC from Current time. For a briefing, a 12-hour window still covers your day

Lesson Summary

Variables give a flow named boxes that change as it runs. You created four at the top, then used **Increment** to count, **Append** to build lists and **Set** to replace a value. Parallel branches let independent lookups run side by side, each writing to its own variable, and a join makes the final email wait for all of them. Next, you'll ask Copilot to plan a flow for you, and review what it suggests.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

Flow name (text, section 8.1)

```
PA - My morning briefing
```

Variable names (text, section 8.2)

```
MeetingCount
```

```
MeetingList
```

```
Emaillist
```

```
BriefingSubject
```

Starting value for BriefingSubject:

```
Your day ahead
```

Weather location (text, section 8.3)

```
Kuala Lumpur
```

Start of each list line (text, sections 8.4 and 8.5)

Paste, then insert **Subject** after it.

```
<br>•
```

Condition name (text, section 8.6)

```
Busy day
```

Busy day subject (text, section 8.7)

Paste, insert **MeetingCount** (lightning bolt, under *Variables*), then type `meetings`.

```
Busy day ahead:
```

Briefing email body (text, section 8.8)

Paste, then replace each [Name] with the matching dynamic content token: delete the [Name] text, leave the cursor there, select the lightning bolt (or type / and choose **Insert dynamic content**) and pick the value. Conditions and Temperature are under *Get current weather*. The rest are under *Variables*.

```
Good morning,  
  
Weather in Kuala Lumpur: [Conditions], [Temperature]°C  
  
Meetings in the next 12 hours ([MeetingCount]):[MeetingList]  
  
Important unread email:[EmailList]
```

09

CHAPTER 9

Plan an Automation with Copilot

Copilot in Power Automate can turn a plain-language description into a suggested flow. A suggestion isn't a finished design, though. In this chapter you review what Copilot proposes, correct it precisely, and walk away without creating anything.

Copy-paste values: the three prompts for this chapter are on the copy-paste page.

Estimated time: 30 minutes

Your result: A reviewed and improved flow plan produced with Copilot, without creating or running a cloud flow.

What You Will Learn

- Describe an automation to Copilot with a clear, bounded prompt
- Review an AI-generated flow plan against the business rule
- Refine a suggestion by naming the exact change you need
- Leave the planning experience without creating a flow

The Business Problem

Copilot may leave out a calculation, add an action you don't need, or read a broad instruction differently from what you meant. You ask it to plan a weekly coordinator summary, check every proposed step, and correct it until it matches this intended plan:

Recurrence > List rows present in a table > Filter array > Compose count > Send one coordinator email

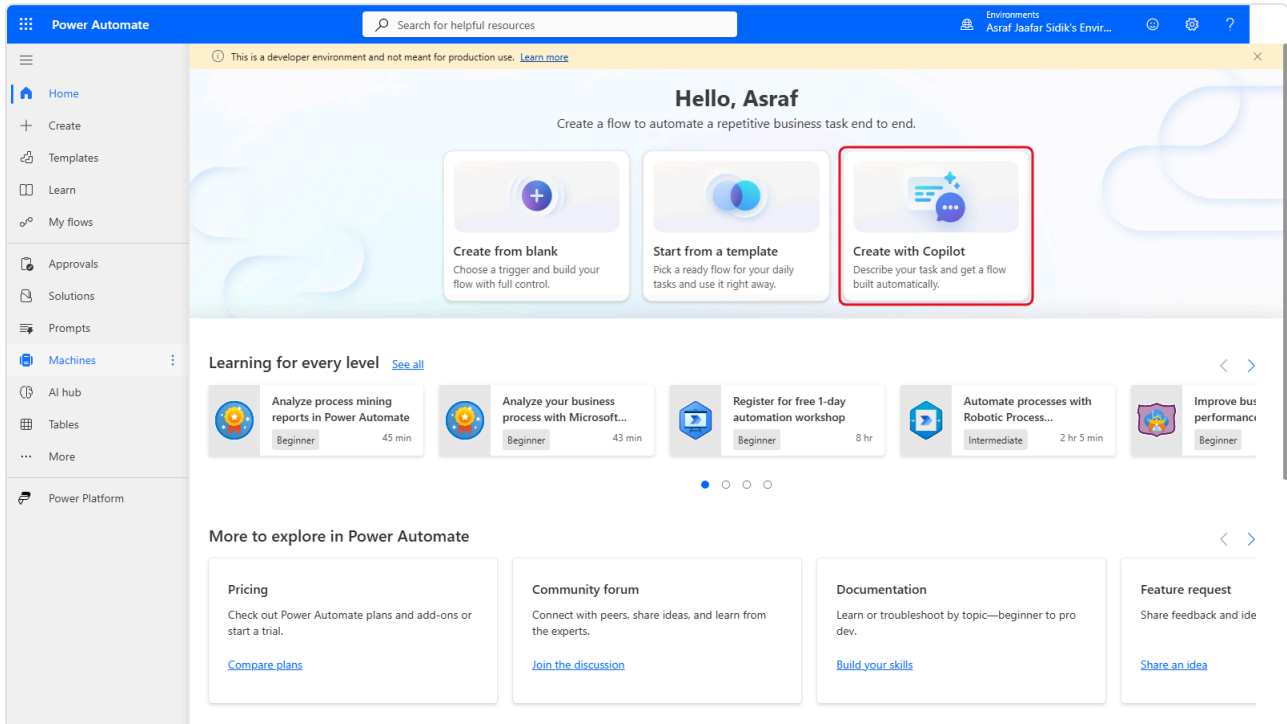
Before You Begin

Copilot availability depends on your organisation, environment, region and admin settings. If **Create with Copilot** or **Describe it to design it** isn't available, follow along with the screenshots in this chapter and answer the review questions without entering a prompt.

Important: Use only fictional details and `coordinator@example.com`. Never put confidential information, personal data, passwords or real recipients into a training prompt.

9.1 Open the Copilot Planning Experience

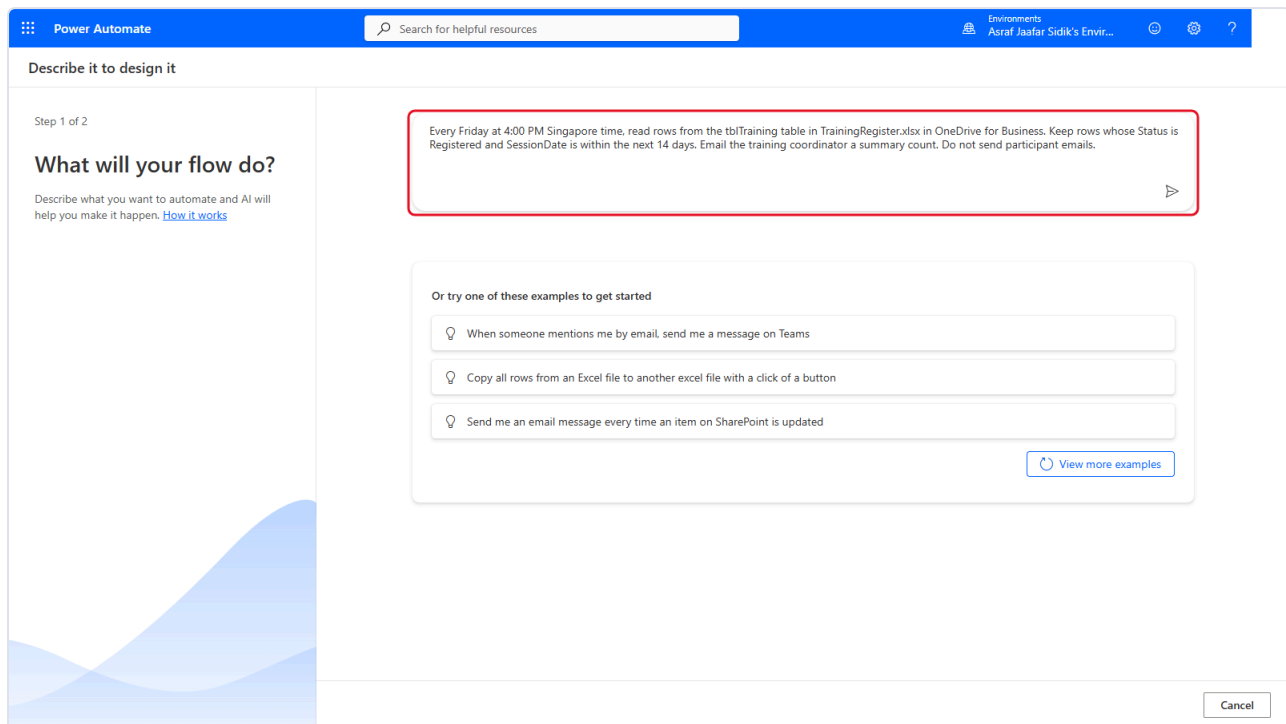
On the Power Automate home page, select **Create with Copilot**. The Create page may call the same thing **Describe it to design it**.



Create with Copilot starts from a plain-language description.

9.2 Describe the Complete Business Outcome

In **What will your flow do?**, paste prompt 1 from the copy-paste page and select the arrow icon at the bottom right of the box to send it.

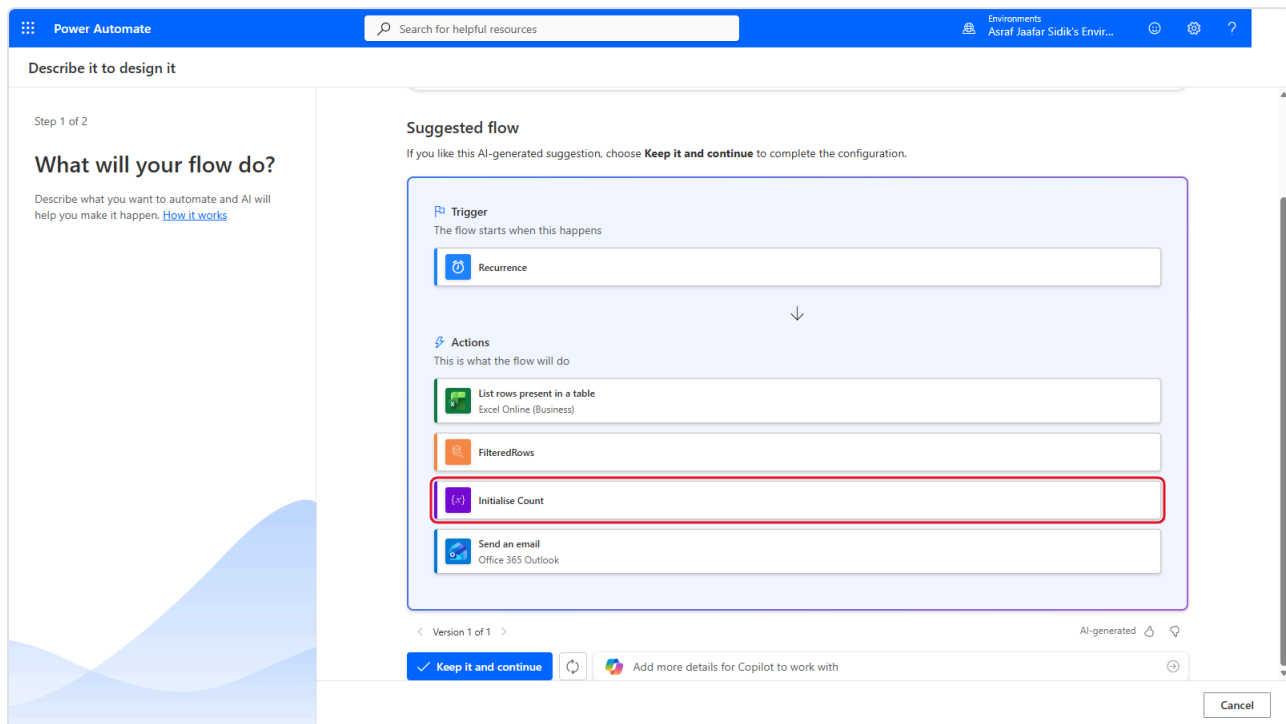


The prompt names the schedule, source, rule, output and a boundary.

A strong prompt states the event, the data source, the rule, the output and the boundaries. Even so, Copilot doesn't know your approved recipient, the exact workbook location, your organisation's policies, or how you'll test the result.

9.3 Review the First Suggestion

Copilot takes a few seconds to answer. Scroll down to **Suggested flow** and read the trigger and every action before you select anything. In the example below, the first version suggested Recurrence, List rows present in a table, FilteredRows, Initialise Count and Send an email. It looks plausible, but **Initialise Count** doesn't say how the count is calculated.



A plausible start, but the count isn't explained.

Use these questions on any AI-generated flow plan:

Review area	Question
Trigger	Does it start at the intended event, time and time zone?
Data	Does it use the correct connector, file, table, list, form or mailbox?
Logic	Are filtering, looping, conditions and calculations explicit?
Output	Is the destination approved, and could the action contact real people?
Licensing	Are all proposed connectors and actions available to participants?
Testing	Can the plan be tested safely with fictional data and approved destinations?

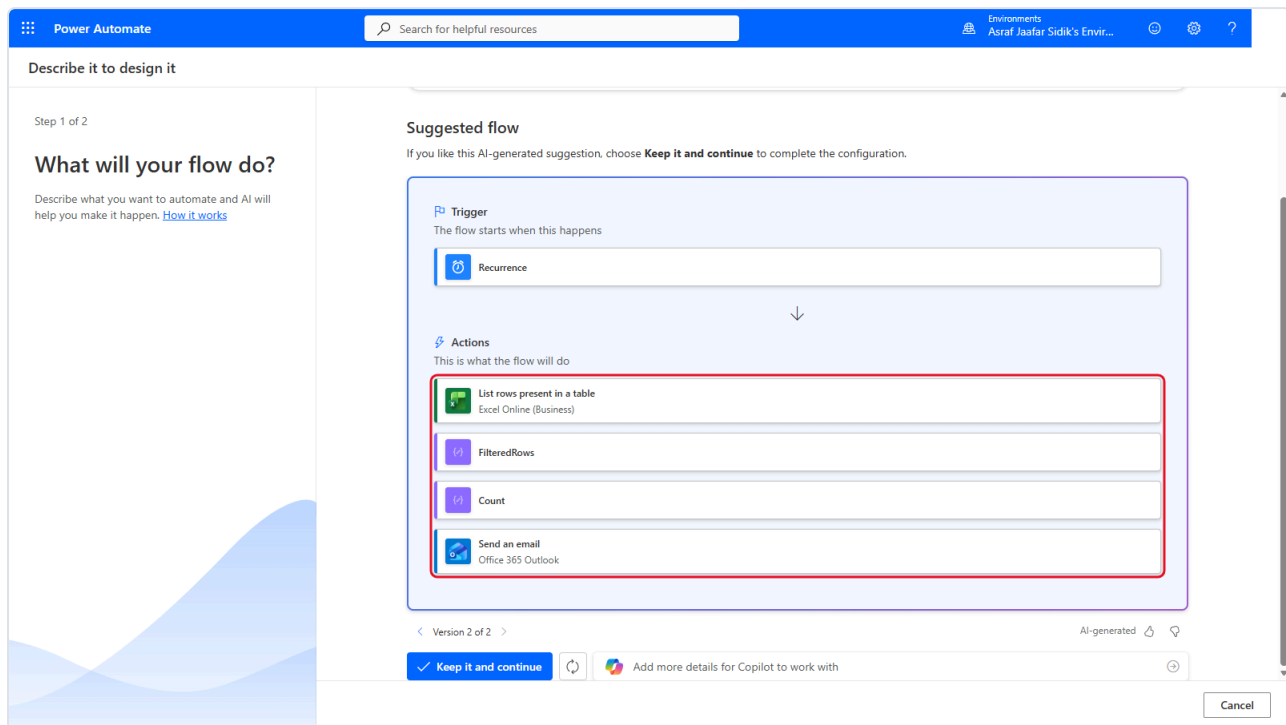
9.4 Clarify the Filter and Count

In **Add more details for Copilot to work with** (the box below the suggestion, to the right of **Keep it and continue**), paste prompt 2 and select the arrow icon at its right end to send it. It names the actions, gives the counting expression, and draws a clear line around who gets email.

9.5 Identify an Unnecessary Action

Copilot's answers vary from run to run. Check whether the second suggestion added anything you didn't ask for. A common one is **Get file metadata using path** before **List rows present in a table**. You don't need it: **List rows present in a table** can select the workbook directly, so the extra action is just another

dependency. If your suggestion is already clean, as in the screenshot below, note that and still send prompt 3: it locks in the exact structure.



In this run the second suggestion was already clean: Recurrence, List rows, FilteredRows, Count and Send an email.

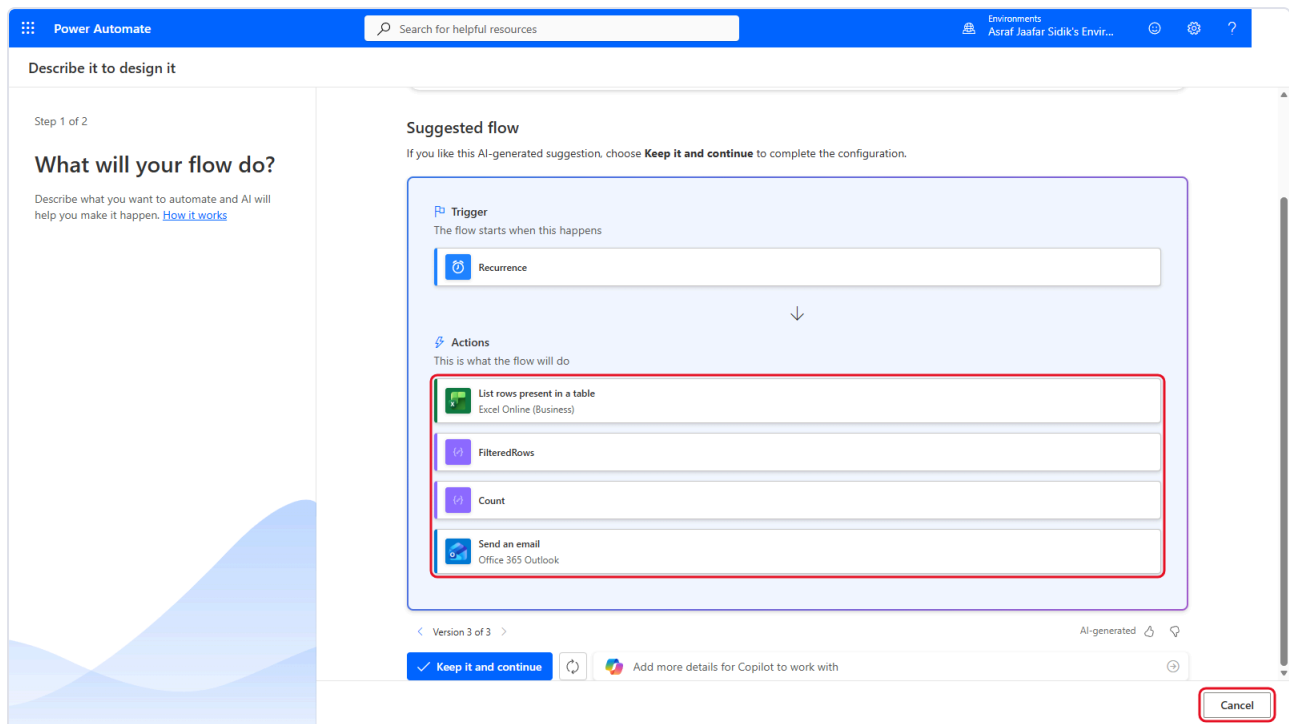
9.6 Request a Streamlined Design

Paste prompt 3 into the same **Add more details for Copilot to work with** box and select the arrow icon at its right end to send it. It says what to remove, why, and exactly which structure to keep. Copilot keeps each answer as a version. To compare them, use the small arrows beside **Version** (for example **Version 3 of 3**) below the suggestion.

9.7 Validate the Final Suggestion

The third suggestion should have exactly these steps:

Order	Proposed step
1	Recurrence
2	List rows present in a table
3	FilteredRows
4	Count (a Compose action)
5	Send an email



The outline matches the intended plan. Copilot shows the Compose action by its name, **Count**.

This only checks the outline. If you continued, you'd still need to configure the Friday schedule and time zone, the workbook and table, the filter and count expressions, the recipient, subject and body, error handling, and test data.

9.8 Exit Without Creating the Flow

Select **Cancel** at the bottom right of the page, not **Keep it and continue**. Power Automate returns to the Create page. This exercise is planning only.

Independent Practice

Write a prompt for a monthly training-capacity report. Include the schedule, data source, inclusion rule, output and one explicit boundary. Swap prompts with another participant and find one ambiguity before asking Copilot for a suggestion.

Troubleshooting

Symptom	What to check
Create with Copilot is missing	Organisation settings, environment, region, account entitlement. Follow the screenshots in this chapter instead
Copilot suggests the wrong trigger	State the exact event or schedule: frequency, day, time and time zone

The plan has a participant email loop	Say only one coordinator summary is needed and explicitly rule out participant emails
The count is unclear	Name the filtered array and ask for <code>length(body('FilteredRows'))</code>
An unnecessary connector appears	Explain which existing action already provides the data and ask Copilot to remove the extra one
The final outline looks right	Keep reviewing. Fields, expressions, connections, recipients and testing aren't configured yet

Lesson Summary

Copilot can propose a trigger and action sequence quickly, keep versions, and respond to refinements. You stay responsible for the business rule, connector choice, licensing, recipient safety, expressions, configuration and testing. The best result came from reviewing each suggestion and stating the exact correction.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner, then paste into Copilot.

These prompts use fictional details only. Never put confidential information, personal data, passwords or real recipients into a training prompt.

1. Describe the complete business outcome (section 9.2)

Paste into the box under **What will your flow do?** and select the arrow icon at the bottom right of the box.

```
Every Friday at 4:00 PM Singapore time, read rows from the tblTraining table in TrainingRegister.xlsx in OneDrive for Business. Keep rows whose Status is Registered and SessionDate is within the next 14 days. Email the training coordinator a summary count. Do not send participant emails.
```

2. Clarify the filter and count (section 9.4)

Paste into **Add more details for Copilot to work with** (below the suggested flow) and select the arrow icon at the right end of the box.

```
Use Filter array to keep rows where Status is Registered and SessionDate is between today and 14 days from today. Calculate the count with length(body('FilteredRows')). Send one summary email only to coordinator@example.com. Do not add an Apply to each or any participant email.
```

3. Request a streamlined design (section 9.6)

Paste into **Add more details for Copilot to work with** and select the arrow icon at the right end of the box.

```
Remove Get file metadata using path because List rows present in a table can select the workbook directly. Keep only Recurrence, List rows present in a table, Filter array named FilteredRows, Compose named Count using length(body('FilteredRows')), and one Send an email action to coordinator@example.com.
```

Remember: select **Cancel** at the end (section 9.8). This exercise is planning only.

10

CHAPTER 10

Move Automations with Environments and Solutions

A flow that works in one environment can still be hard to move safely. It depends on connections, tables and other components, and copying them one at a time makes it easy to miss something. Solutions package those pieces together.

Copy-paste values: the solution names are on the copy-paste page.

Estimated time: 60 minutes (trainer demonstration)

Your result: You can explain how a flow moves between environments, after watching your trainer package the capstone flow in a solution and walk through where an import starts.

What You Will Learn

- Explain what an environment is and why organisations separate development from production
- Describe what a solution can contain
- Explain how Power Automate, Power Apps and Copilot Studio can share solution components
- Follow how a solution is created and an existing cloud flow added
- Follow how an unmanaged solution is published and exported
- Identify the safe import, connection-repair, verification and testing sequence

The Migration Path

Development environment > Solution > Publish > Export package > Target environment > Import > Repair connections > Verify > Authorised test

The live demonstration stops after opening the import screen. Importing back into the same environment isn't a real migration test, and testing the capstone could send email.

Before You Begin

Important: This chapter is a **trainer demonstration**. Your class works in a live environment, and solutions create real components there (a solution, a publisher, connection references). Watch the demo and follow along on this page. Don't create, export or import solutions yourself unless your trainer gives you a separate practice environment.

The demo uses `PA - Weekly training capstone` from Chapter 7, kept turned off. Solutions also need Microsoft Dataverse in the environment, and Power Automate Free alone doesn't guarantee that.

Environments in Plain Language

An environment is a separate workspace for apps, flows, data, connections and security. The environment selector at the top of Power Automate shows which one you're in.

Environment	Typical purpose	Practical rule
Development	Build and change automations	Use fictional or approved test data
Test	Validate behaviour before release	Reconnect services and run controlled tests
Production	Run approved business processes	Restrict changes and monitor real operation

Separating these stages stops unfinished changes reaching real users, but only when permissions, connection ownership, data and release steps are controlled too.

Power Automate, Power Apps and Copilot Studio all sit on the Power Platform, so one solution can package the flows, apps, agents, tables, connection references and environment variables behind a single business process. What's available depends on the environment and licences.

10.1 Confirm the Current Environment

Check the environment selector before you create or import anything (top right of the blue bar, under the small word **Environments**), and confirm with the trainer that it's the development or training environment. If a flow or solution seems to be missing, check the environment before rebuilding it.

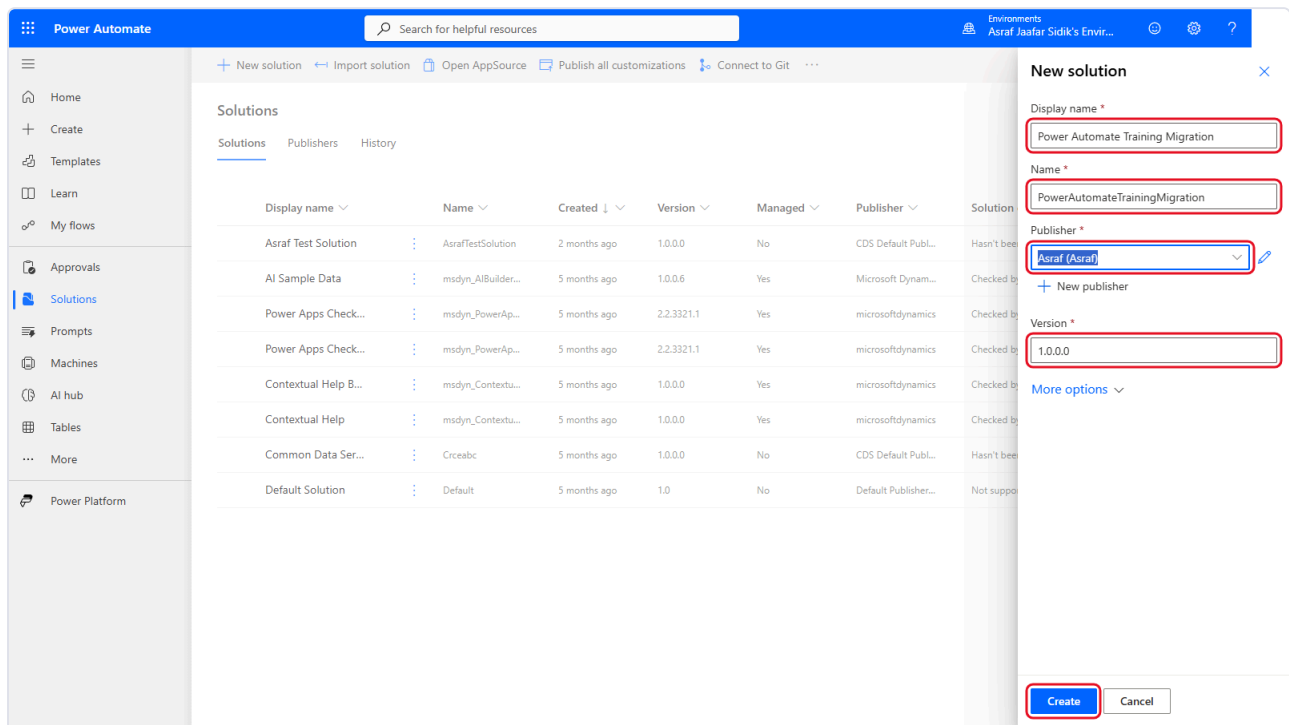
10.2 Create the Solution

In the left navigation, select **Solutions** (if you don't see it, select **More** to find it). Then select **+ New solution** at the left end of the toolbar. The **New solution** panel opens on the right. Enter:

Field	Value
Display name	Power Automate Training Migration
Name	PowerAutomateTrainingMigration

Publisher	The trainer-approved publisher
Version	1.0.0.0

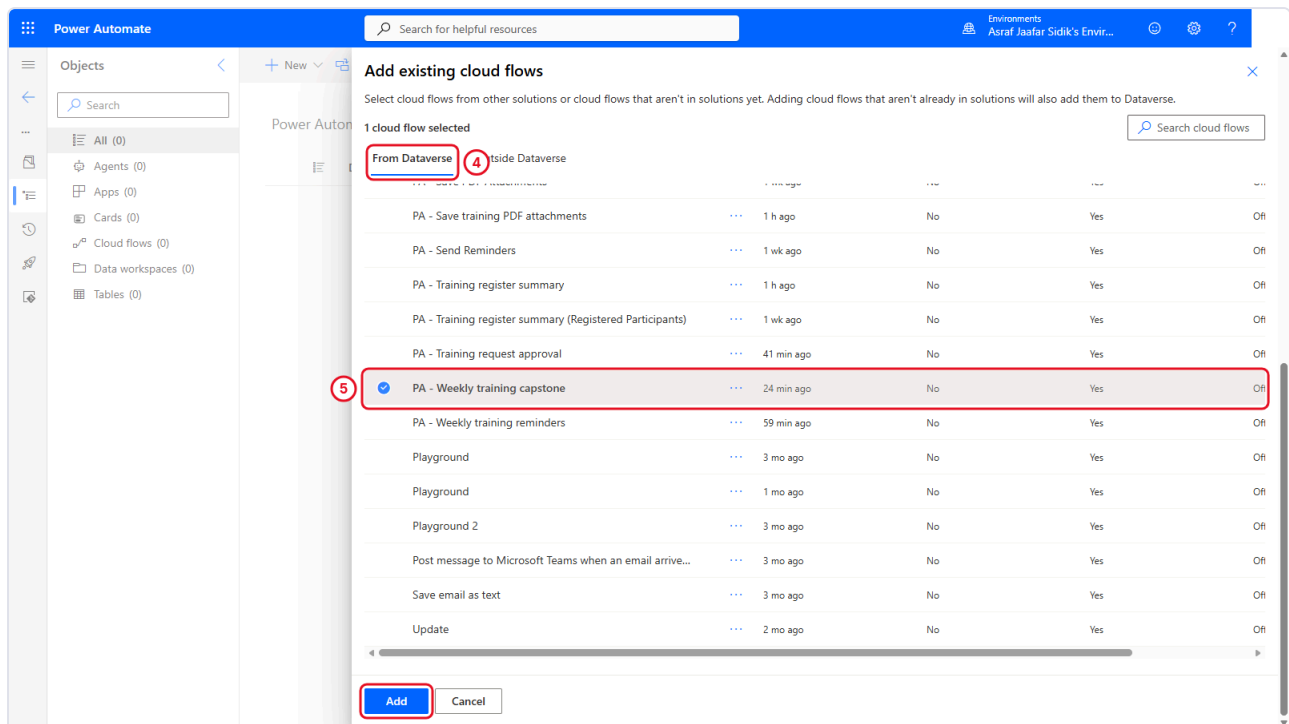
For **Publisher**, open the dropdown and pick the publisher your trainer names. Then select **Create** at the bottom of the panel. The display name is the friendly label. The unique name is used internally and normally shouldn't change later.



The display name is for people. The name is used internally.

10.3 to 10.7 Add the Capstone Flow

1. Open the new solution: in the **Solutions** list, select its display name, **Power Automate Training Migration**. Its object list is empty: a new solution is just a container.
2. In the toolbar at the top, select **Add existing** and point to **Automation**.
3. Under Automation, select **Cloud flow**.
4. Find **PA - Weekly training capstone**. Depending on how your environment stores flows, it's on the **From Dataverse** tab or the **Outside Dataverse** tab. Scroll the list, or type its name in **Search cloud flows** at the top right of the panel.
5. Select the circle to the left of the capstone's name, then select **Add** at the bottom of the panel.
6. If only the flow appears in the solution, select the circle beside it, then in the toolbar select **Advanced > Add required objects**. This adds the connection references it needs.

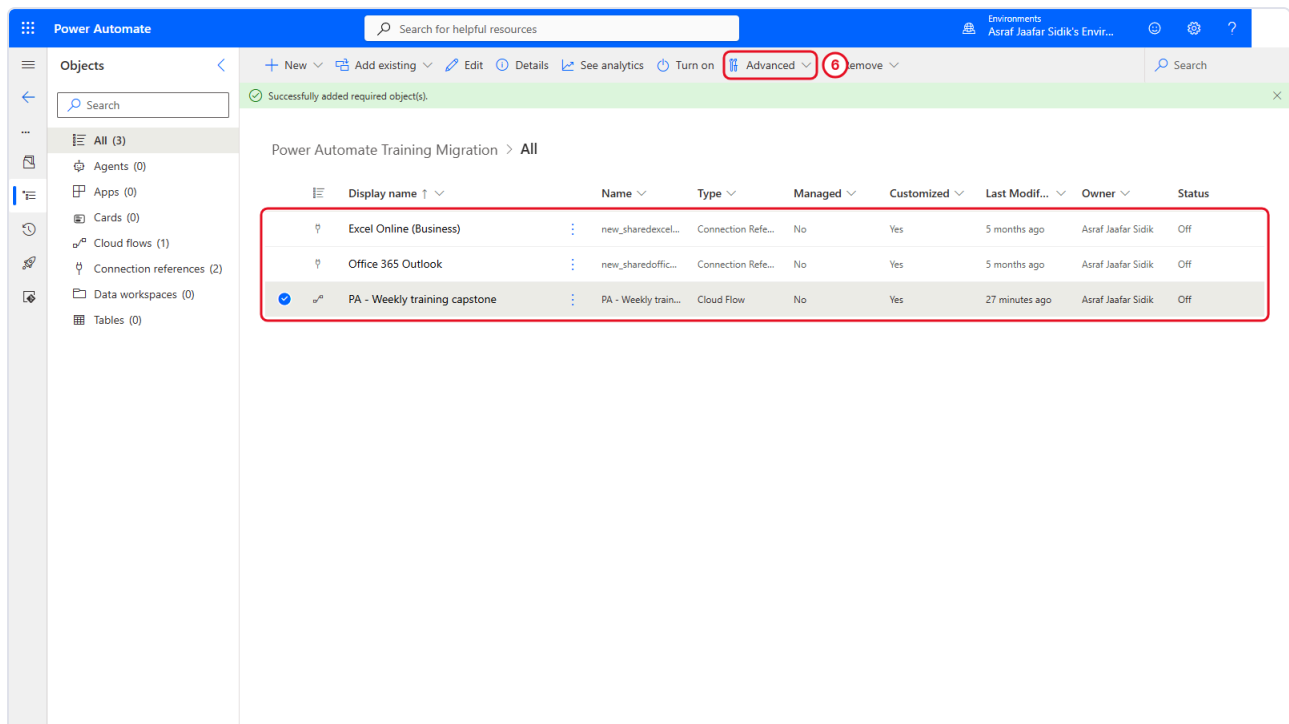


Pick the capstone from the list. Here it's on the From Dataverse tab.

Power Automate may also add **connection references**. A connection reference is a solution component that points to a connection. The target environment still needs a valid connection and an authorised account.

10.8 Verify the Solution Components

Confirm the solution has three objects: the capstone cloud flow and the Excel Online (Business) and Office 365 Outlook connection references.



Three objects: the flow and its two connection references.

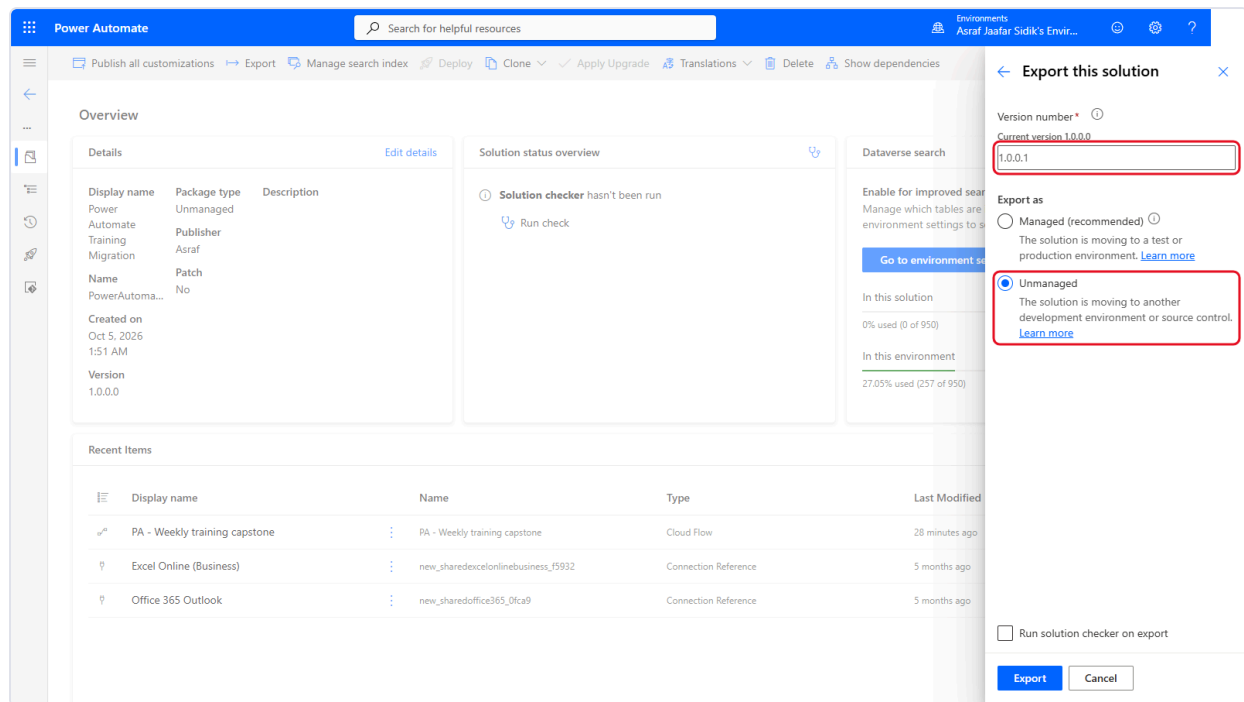
Key point: Showing up in the object list doesn't mean a dependency is ready. File locations, table names, recipients, connection accounts, environment variables and permissions can all differ in the target environment.

10.9 Review the Solution Before Export

Inside a solution, the far-left rail shows icons only. Select the **Overview** icon, the page-shaped icon just under the back arrow (hover over an icon to see its name). Check the name, version, package type, publisher and included items. **Export** is in the toolbar at the top of the Overview page.

10.10 to 10.12 Publish and Export

1. In the toolbar at the top of the Overview page, select **Export**. If a **Before you export** panel offers **Publish**, select it so the latest changes are included, and wait for **Published**.
2. Select **Next**.



Unmanaged suits a learning exercise. The version number goes up with each export.

3. Confirm the version, select **Unmanaged**, and select **Export**. Power Automate suggests the next version number (for example 1.0.0.1), so every exported package has its own version.

4. When the success message appears, select **Download** and store the ZIP in the trainer-approved location. The worked example produced `PowerAutomateTrainingMigration_1_0_0_1.zip`.

An unmanaged package suits this exercise because its components can be edited after import. Managed solutions are normally used for controlled distribution where recipients shouldn't edit the components. That's an organisational decision, so don't switch package types casually in an established process.

Tip: Don't unzip or rename the package before import. Treat it as a release artifact and record who exported it, from which environment, and why.

10.13 and 10.14 Switch Environment and Start the Import

1. Select the environment name at the top right of the blue bar and choose the trainer-approved target environment. Confirm its name before opening **Solutions** in the left navigation.
2. Select **Import solution** in the toolbar at the top of the Solutions page, then **Browse** to the exported ZIP. Select **Next** only after confirming the package name and the target environment.

The demonstration stops here. No file is uploaded and nothing is imported. Your trainer may not switch environments at all if there's no second training environment. **Never use production as a stand-in for a training target.**

10.15 Repair Connections and Settings

After the package is selected, Power Platform may ask for connection mappings or environment-specific values. Before import or first use:

1. Map each connection reference to an approved connection in the target environment.
2. Confirm the Excel workbook and `tblTraining` table exist in the intended target location.
3. Review recipients, team and channel selections, form references, schedules, time zones and environment variables.
4. Confirm the importing account and future flow owner have the required permissions.
5. Keep the imported flow turned off until verification is complete.

10.16 Verify and Test Safely

After an authorised import, open the solution and confirm the expected components are there. Run Flow checker, review every connection and setting, and use fictional data and approved destinations for the first test.

Testing is a separate release decision. An imported flow can be structurally valid and still point at the wrong workbook, mailbox, form, team, channel or recipient. Get trainer authorisation right before any test that can send messages or change shared data.

Independent Practice

Create a migration checklist for one of the earlier flows. Name its trigger, connectors, data sources, recipients, environment-specific settings, owner, safe test data, and the evidence you'd need before turning it on in production. Don't export, import or run anything.

Troubleshooting

Symptom	What to check
Solutions is missing or unavailable	Environment type, Dataverse, licence, security role and maker permissions
The flow isn't in the list	Current environment, both the From Dataverse and Outside Dataverse tabs, flow ownership, and whether it's already in another solution
Connection references are missing	Select the flow in the solution, then Advanced > Add required objects
Extra components appear	Connection references and other dependencies may be added automatically. Review each one
Export stays pending	Refresh Solutions, open export history, and confirm publishing finished
Import says the solution already exists	Check you're in the intended target, and whether this is an upgrade or an accidental same-environment import

A connection shows an error after import	Create or pick an approved target connection and remap the connection reference
The flow imports but can't find the workbook	Target OneDrive or SharePoint location, file name, table name, permissions and connector account
The imported flow is off	Leave it off while reviewing. Turn it on only through the approved release process

Lesson Summary

Environments separate workspaces, permissions, data and release stages. Solutions package related Power Platform components so a process can move in a controlled way. A safe migration covers dependency review, publishing, a versioned export, a confirmed target, connection mapping, configuration checks, Flow checker, controlled testing and a deliberate decision to switch the flow on.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

Solution display name (text, section 10.2)

```
Power Automate Training Migration
```

Solution name (text, section 10.2)

```
PowerAutomateTrainingMigration
```

Solution version (text, section 10.2)

```
1.0.0.0
```

11

CHAPTER 11

Extra Practice: Personal Automations

Four small flows for when you finish early, or for practice back at your desk. Each one only reads your own data or writes to your own OneDrive training folder, and only notifies you. None of them touches shared sites, lists, teams or other people's mailboxes.

Copy-paste values: flow names and text for every exercise are on the copy-paste page.

Estimated time: 15 to 20 minutes each, self-paced

Your result: Up to four working personal flows, each built with the skills from Chapters 3 to 8.

Which One Should I Try?

Exercise	Flow type	Practises	Connectors
A. Flagged email to To Do	Automated	Triggers and dynamic content	Office 365 Outlook, Microsoft To Do (Business)
B. Send me a note	Instant	Trigger inputs	Microsoft Teams (your own Flow bot chat)
C. Save an email to OneDrive	Automated	Trigger filters, file actions	Office 365 Outlook, OneDrive for Business
D. News digest	Automated	A non-Microsoft 365 trigger	RSS, Office 365 Outlook

Tip: Turn each flow off when you've finished testing, unless you want to keep using it. In **My flows**, select the ... beside the flow, then **Turn off**.

How the steps work: the designer basics are the same as in Chapter 3.

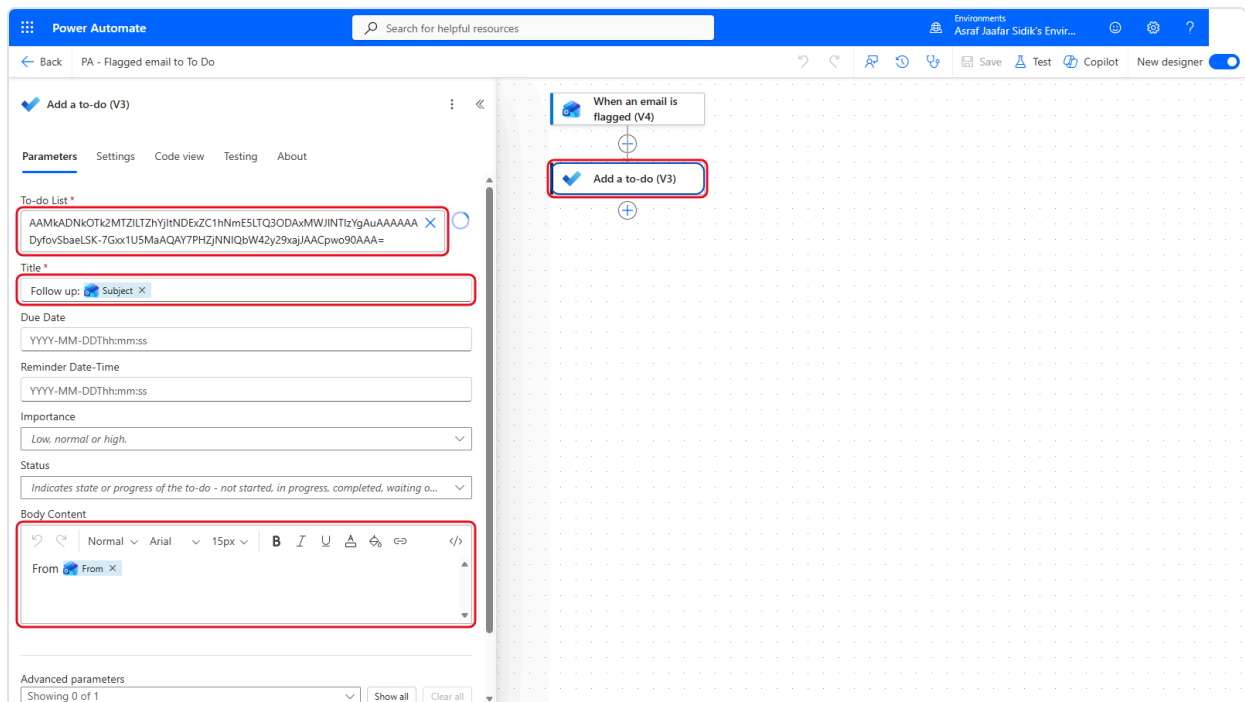
- **Add an action:** on the canvas, select the **+** under the last step, then **Add an action**. Type the action's name in the search box and pick it under the named connector.
- **Insert a value:** click into the field, then select the lightning bolt icon beside the box (or type **/** and choose **Insert dynamic content**) and pick the value. If you can't see it, select **See more** under the step's name, or type the value's name in the picker's search box.
- **Save** and **Test** are at the top right of the designer.

A. Flagged Email to To Do

The problem: You flag emails to come back to, then forget about them. This flow adds every flagged email to your personal To Do list.

1. Select **Create** in the left navigation, then the **Automated cloud flow** tile. In **Flow name**, enter **PA - Flagged email to To Do**.
2. In the trigger search box, type **flagged**, select **When an email is flagged (V4)** under **Office 365 Outlook**, and select **Create**.
3. Select the trigger card on the canvas to open its panel on the left, and set **Folder** to **Inbox**.
4. Under the trigger, select **+ > Add an action**, search **to-do**, and choose **Add a to-do (V3)** under **Microsoft To Do (Business)**. Open the **To-do List** dropdown and pick **Tasks**, then fill in the other two fields:

Field	Value
Title	Type Follow up: (with a space at the end), then insert Subject
Body Content	Type From then insert From



The flagged email's subject becomes the task title.

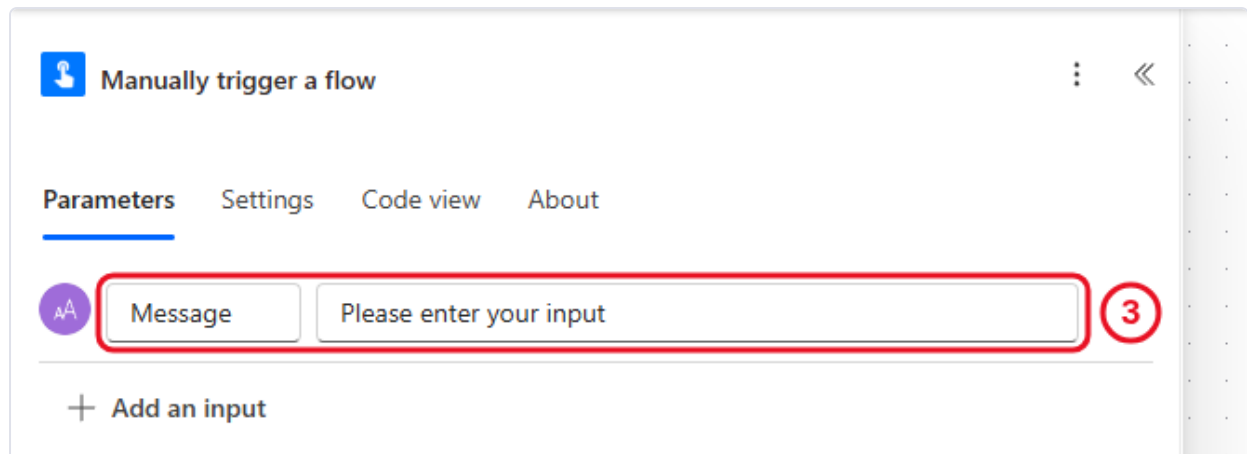
Note: After you pick **Tasks**, the To-do List field may show a long ID instead of the name. That's normal.

5. Select **Save**, then **Test** > **Manually** > **Test**. In Outlook, flag any email in your inbox. Within a minute, the task appears in Microsoft To Do.

B. Send Me a Note

The problem: You think of something and want a quick reminder where you'll see it. This instant flow asks you for a message and sends it to you as a Teams chat from Flow bot. It only posts in your own chat with Flow bot.

1. Select **Create** in the left navigation, then the **Instant cloud flow** tile. In **Flow name**, enter **PA - Send me a note**.
2. Under **Choose how to trigger this flow**, select **Manually trigger a flow**, then select **Create**.
3. Select the trigger card on the canvas. In its panel, select **+ Add an input**, then **Text**. Select the first box, which reads *Input*, delete that word and type **Message**.



A trigger input asks the person running the flow to type something first.

4. Under the trigger, select **+ > Add an action**, search `post message`, and choose **Post message in a chat or channel** under **Microsoft Teams**. Fill in the fields in this order, because **Recipient** only appears after you choose **Post in**:

Field	Value
Post as	Flow bot
Post in	Chat with Flow bot
Recipient	Your own email address
Message	Type <code>Reminder:</code> (with a space at the end), then insert Message from <i>Manually trigger a flow</i>

5. Select **Save**, then **Test > Manually > Test**. In the panel that opens, type a short message, then select **Run flow**. The message appears in Teams, in your chat with Flow bot.

Note: Earlier versions of this exercise used **Send me a mobile notification**. During testing that action reported that the Power Automate mobile app had been retired and notifications had nowhere to go, so this version uses Teams instead.

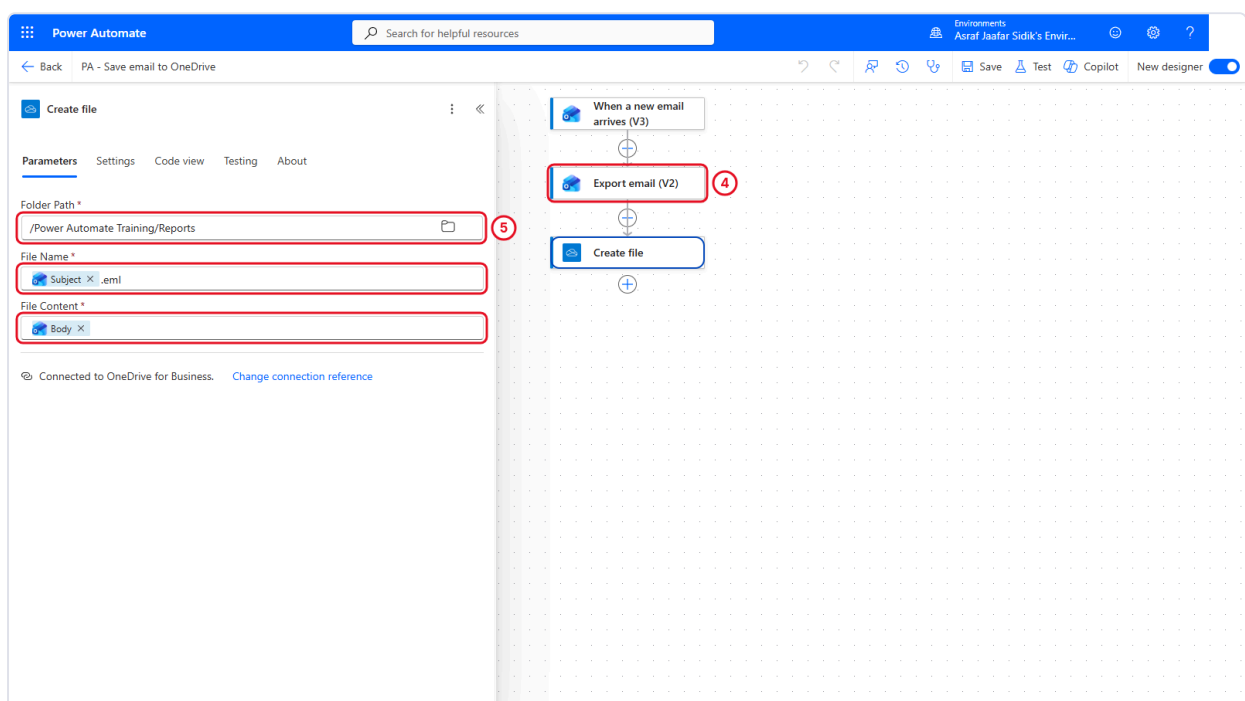
C. Save an Email to OneDrive

The problem: Some emails need keeping as files: a confirmation, a booking, an agreement. This flow saves any email you send yourself with `[PA SAVE]` in the subject as a file in your training folder.

1. Select **Create** in the left navigation, then the **Automated cloud flow** tile. In **Flow name**, enter `PA - Save email to OneDrive`.
2. In the trigger search box, type `new email`, select **When a new email arrives (V3)** under **Office 365 Outlook**, and select **Create**.
3. Select the trigger card on the canvas to open its panel, and set **Folder** to **Inbox**. Near the bottom of the panel, next to **Advanced parameters**, select **Show all**, then type `[PA SAVE]` in **Subject Filter**.

- Under the trigger, select + > **Add an action**, search `export email`, and choose **Export email (V2)** under **Office 365 Outlook**. Click into **Message Id** and insert **Message Id** from *When a new email arrives (V3)*.
- Under **Export email (V2)**, select + > **Add an action**, search `create file`, and choose **Create file** under **OneDrive for Business**:

Field	Value
Folder Path	Type <code>/Power Automate Training/Reports</code> , or select the folder icon at the right of the box and browse to Reports
File Name	Insert Subject from the trigger, then type <code>.eml</code> straight after it
File Content	Insert Body from <i>Export email (V2)</i>



Export email turns the message into a file. Create file saves it.

- Select **Save**, then **Test > Manually > Test**. In Outlook, send yourself an email with the subject `[PA SAVE] Booking confirmation`. A `.eml` file appears in Reports. Double-click it to open it in Outlook.

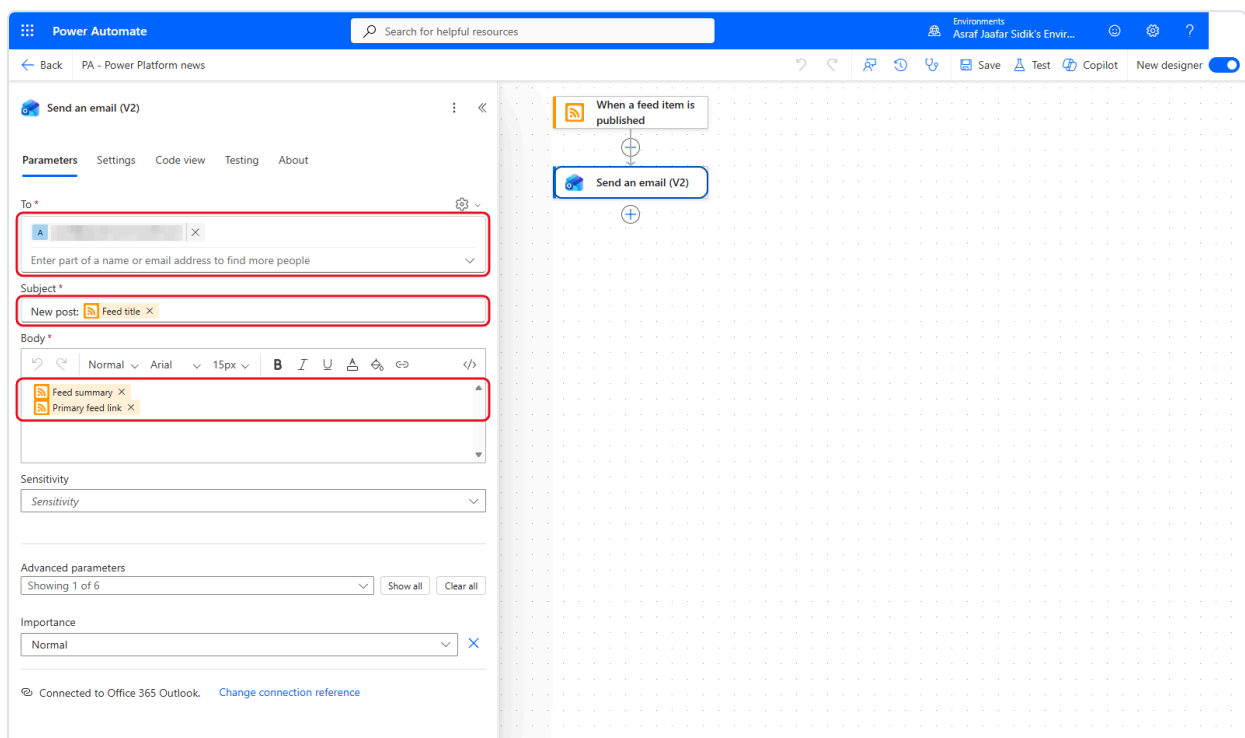
D. News Digest

The problem: You want to keep up with Power Automate news without checking the blog. This flow emails you whenever a new post is published.

- Select **Create** in the left navigation, then the **Automated cloud flow** tile. In **Flow name**, enter `PA - Power Platform news`.
- In the trigger search box, type `RSS`, select **When a feed item is published**, and select **Create**.

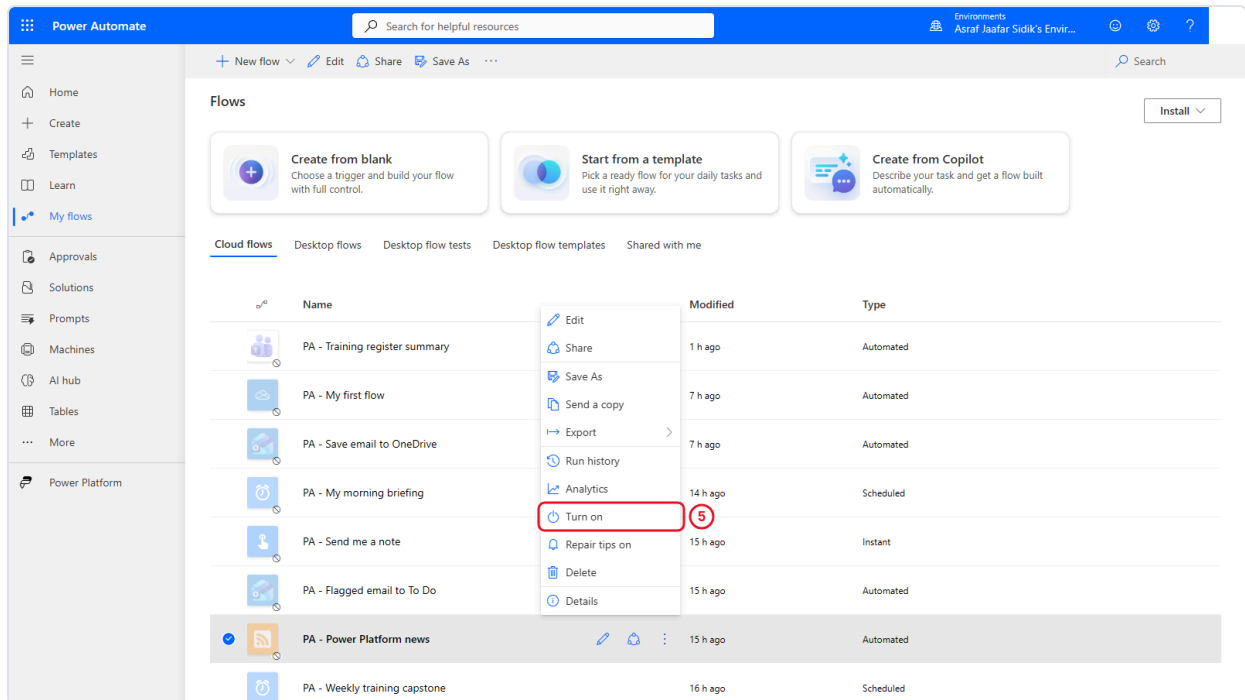
3. Select the trigger card on the canvas. In **The RSS feed URL**, paste the Microsoft Power Platform blog feed from the copy-paste page.
4. Under the trigger, select **+ > Add an action**, search `send an email`, and choose **Send an email (V2)** under **Office 365 Outlook**:

Field	Value
To	Type your own email address and select it from the suggestions
Subject	Type <code>New post: </code> (with a space at the end), then insert Feed title
Body	Insert Feed summary , press Enter, then insert Primary feed link



*The RSS connector watches a public web feed. It reads, and never writes, anything. Ignore any extra fields such as **Importance**; leave them as they are.*

5. Select **Save**, then select **My flows** on the left. Select the ... beside **PA - Power Platform news**, then **Turn on**. The next new blog post arrives in your inbox.



Turn on is in the flow's ... menu in My flows. The RSS trigger checks for new posts on a schedule, so a test may not fire straight away.

Note: If the feed URL stops working, open the blog in your browser and look for its RSS link, or try any other news site that offers an RSS feed.

Lesson Summary

Small personal flows are the best way to keep practising after the course. Each one here uses a different starting point (a flag, a button, a subject filter and a web feed) with the same building blocks: a trigger, dynamic content and an action.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

A. Flagged email to To Do

Flow name:

PA - Flagged email to To Do

Title start (insert **Subject** after it):

Follow up:

B. Send me a note

Flow name:

PA - Send me a note

Input name:

Message

Teams message start (insert **Message** after it):

Reminder:

C. Save an email to OneDrive

Flow name:

PA - Save email to OneDrive

Subject filter:

[PA SAVE]

Folder path:

/Power Automate Training/Reports

Test email subject:

[PA SAVE] Booking confirmation

D. News digest

Flow name:

PA - Power Platform news

Microsoft Power Platform blog feed:

<https://www.microsoft.com/en-us/power-platform/blog/feed/>

Subject start (insert **Feed title** after it):

New post:

Microsoft Power Automate for End Users

This book accompanies the two-day Microsoft Power Automate for End Users course. It follows one training coordinator's routine, from saving email attachments and sending an Excel summary, through weekly reminders and a Forms approval, to a filtered capstone flow and a personal morning briefing built with variables and parallel branches. Every chapter gives point-and-click steps and copy-ready values you can use during the session and back at your desk.